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Rethinking Student Housing in Berlin

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Abstract

Placeholder text is often used in design to indicate where content will eventually be placed. It serves as a visual guide for designers and developers, enabling them to lay out pages and adjust typography, spacing, and color schemes before the final content is inserted. An example of filler text is provided below for the purpose of demonstrating layout elements in a generic layout scenario. This filler content is intentionally devoid of specific meaning, meaning it contains no hidden or actionable messages, and is instead built from commonly used words and phrases to provide a textured baseline for creative exploration. It can also aid in streamlining editorial workflows, especially when translation or localization teams must preview how copy will appear in a finished design. Typical placeholder narratives may include references to common design practices, UI components, grammar conventions, and visual hierarchy principles all brought together in a fluid tone

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17 ***1.1.1 Technical University of Berlin***

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2 1. Introduction

Research aimed to explore the topic, identifying major trends, challenges, and research gaps, in order to form scientific objectives and relevance of research for Berlin Case Study.

2.1 1.1 European Student Accommodation Crisis

In order to better understand the current state of research on European Student Accommodation Crisis topic, a comprehensive literature research was conducted. *Research aimed to explore the topic, identifying major trends, challenges, and research gaps, in order to form scientific objectives and relevance of research for Berlin Case Study.*

Type of Literature Research: Bibliographical Mapping

Tools: Zotero, Python, VOSviewer

Topic: European Student Accommodation Crisis

Period of the search: 2010-2025

Area of search: European Union

Academic Database: Scopus, Emerald, Google Scholar, TU Berlin Library

Language: English

2.1.1 Literature Review Type

Bibliometric analysis is defined as “a part of scientometrics for utilising mathematical and statistical methods to analyse scientific activities in a research field” (Louarn et al., 2025). Thus, bibliographical mapping can be seen as a structured, data-driven approach for organising exciting knowledge regarding the selected topic. It can be applied to vast interdisciplinary literature, developing structured networks determined by keywords, co-citations, bibliometrics coupling, and direct citation analysis. This kind of approach matches with the overall aim to introduce innovative and structured analytical methods in urban design research. The methodology followed in review type comprises four stages: data collection, eligibility analysis, general descriptive and content analysis. (Louarn et al., 2025).

2.1.2 Limitations

In the bibliometric analyses, certain limitations were set, which should be taken into account when interpreting the results. Firstly, document type was limited to ‘journal article’ and ‘book section’ to avoid the double-collecting of similar research papers. However, during the collecting process ‘reports’ were added as a grey literature sources. Secondly, the search was limited to articles written in English. Thirdly, the search area was initially limited to Germany, which was considered an especially narrow approach. Subsequently, the search area was extended to the European Union, which provided a more comprehensive view of the problem. Last but not least, the period of search was limited to 2010-2025 years, during which the governmental student accommodation availability rate in Germany was no more than 10% (Student Housing in Germany Statistical Overview, 2023). Those limitations assist in focusing on research that is immediately connected with the Student Housing Crisis in Berlin and retain only relevant information.

2.1.3 Set Up

Import Libraries

```

import json
import pandas as pd
import geopandas as gpd
import matplotlib.pyplot as plt
import numpy as np
from shapely.geometry import Point

```

2.1.4 Data Mining

The literature collection was conducted in June 2025 using three academic databases: Scopus, Emerald, Google Scholar and TU Berlin Library as a grey literature source. The search terms was organised in five preliminary clusters:

1. Student Housing Market Terms: student housing OR student accommodation OR dormitory OR Wohnheim OR affordable housing
2. Policy: housing crisis OR rental market OR real estate investment OR financialization of housing
3. Urban Design Terms: urban displacement OR studentification OR gentrification OR social housing OR urban periphery
4. Geographic Terms: Berlin OR Germany OR EU OR European Union
5. Demographic Terms: international students OR Erasmus students OR student mobility

The search resulted in a total of 50 articles from the four data sources, which serve as an initial orientation and provide a conceptual foundation for European Student Accommodation Crisis topic.

```

file_path = '../inputs/1.1_european_student_accommodation_crisis/literature_review_list.json'

# Load the JSON data
with open(file_path) as file:
    data = json.load(file)

# Extract items (journal articles and attachments) from the JSON
items = data.get('items', [])

# Convert the items into a DataFrame
df = pd.json_normalize(items)

```

2.1.5 General Descriptive Analysis

Time Series

The Study Period (2010-2025) has seen steady growth in publication trends (Figure 1). Between 2012 and 2021, the average annual number of publications was 2. In comparison, between 2021 and 2025 average publications number of per year was 4 times higher - approximately 8 publications per year. In 2024, the number of annual publications was eight times as many as those published in 2014. The regular growth in the number of publications could be explained by the increasing acknowledgment of the importance of student housing crisis. Moreover, the number of articles related to student housing market regulations increased over the past 5 years. This could be seen as a part of the solution to the student housing crisis. Thus, providing new approaches to student housing, regulating the market, and recognising the right of students to social housing be taken into account in forming a theoretical framework.

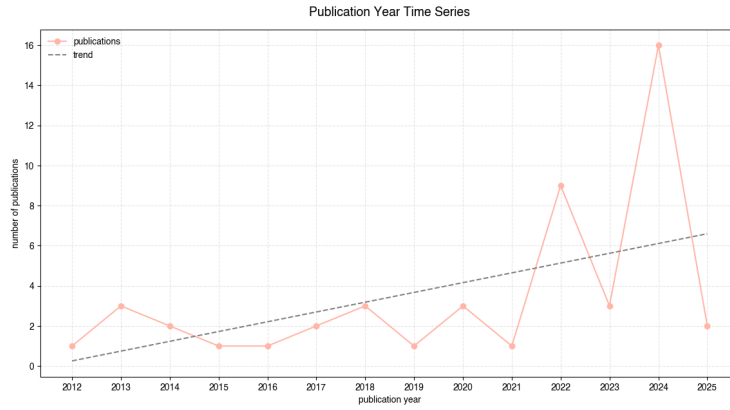


Figure 1. Publication Year Time Series

Publication Type Distribution

Publication Type analysis (Figure 2) depicts the dominance of peer-reviewed journal articles (74%) and book chapters (16%) with the strong academic foundation of the research field on student housing market and urban design. In contrast, the reports (10%), representing grey literature, offer valuable insights from policy institutions and housing organisations. Although non-academic, these sources provide empirical evidence and real-world perspectives that bridge the gap between theory and practice. The combination of academic and grey literature thus ensures a balanced and multidimensional understanding of the student housing crisis, where data-driven analysis is informed both by academic research and applied policy contexts.

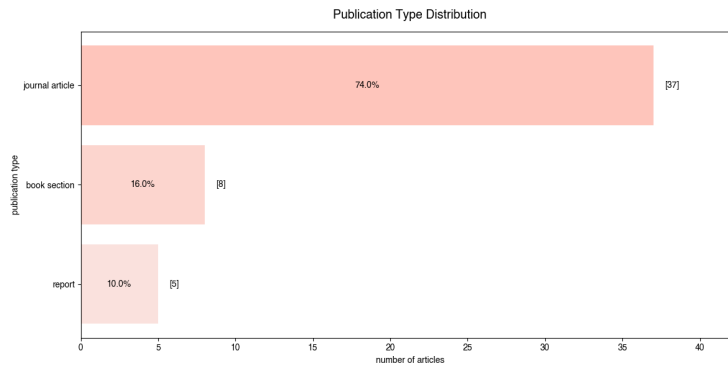
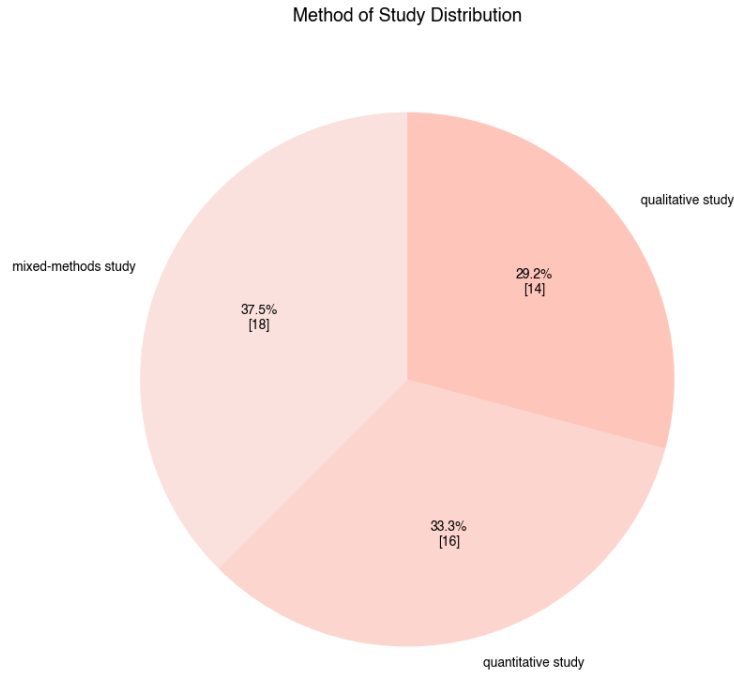


Figure 2. Publication Type Distribution

Method of Study Distribution

Methods of Study (Figure 3) refer to the techniques and tools used in research to collect, analyse, and interpret data. Such methods can be broadly categorised into qualitative, quantitative, and mixed-method approaches. Qualitative study methods (29.2%) have a strong emphasis on understanding the social dynamics and subjective experiences, highlighting topics such as cultural diversity, moving abroad, and community living. Quantitative studies (33.3%) indicate a desire for evidence-based insights related to housing costs, student satisfaction, or demographic trends. However, the majority of studies are using mixed-method approaches with an emphasis on understanding the social processes within the data-driven framework. Thus, merging qualitative methods with quantitative ones is the most relevant approach to access the Student Housing Crisis topic.



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Figure 3. Method of Study Distribution

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Data Source Distribution

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During the data source distribution analysis (Figure 4), four primary data source categories were identified: numerical data (30.3%), social data (27.5%), literature review (25.7%), and other data (16.5%). The balance between four categories reflects the interdisciplinary nature of the Student Crisis topic. Statistics, institutional records, market data provide measurable insights into housing, while social data (surveys, interviews, case studies, street observations) capture students' lived experiences and perceptions. The strong representation of literature sources ensures theoretical grounding in urban design, economics and sociology.

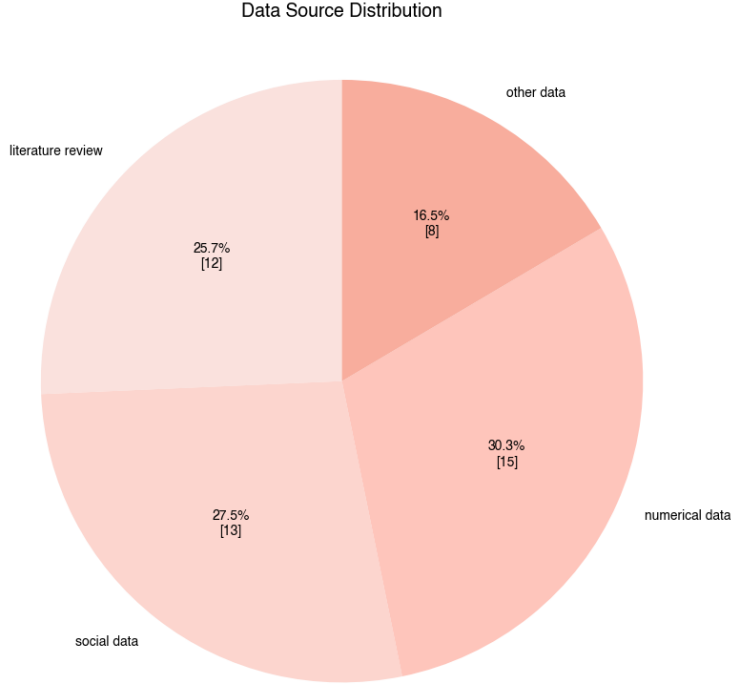


Figure 4. Data Source Distribution

Studies Areas

In selected literature, 46 cities were mentioned as case studies. The most frequently mentioned cities are Berlin, Hamburg, Stockholm and Vienna (Table 1), all of which have populations of over 1.5 million. This can be explained by the fact that the more people live in a city, the easier it is to obtain a representative sample for social research. Large cities also tend to have more open statistics and data.

```
city.head(4)
```

	city_name	count	longitude	latitude	geometry
0	Berlin	24	13.404954	52.520008	POINT (391776.158 5820073.118)
1	Hamburg	6	9.993682	53.551086	POINT (168441.945 5945241.585)
2	Stockholm	3	18.068581	59.329323	POINT (674570.668 6580745.518)
3	Vienna	3	16.373819	48.208174	POINT (602066.671 5340350.73)

Table 1. Literature Review Cities

Spatial Distribution of Case Studies (Figure 5) shows geographical origin of the academic and empirical datasets used in the literature, highlighting that topic of European Student Accommodation Crisis is researched mainly in Western Europe (e.g., Germany, Austria, Netherlands) and Northern Europe (e.g., the UK, Scandinavia), while East Europe (e.g., Poland, Slovakia, Hungary) remains relatively under-researched. It is important to note that, since geographic search terms include “Germany”, the highest data density correlates with the frequency of mentions of Germany in search terms.

In the context of the student housing crisis, Germany serves not only as a subject of analysis, but also as a model for comparative European studies. For example, Berlin is compared with Vienna (Holm, Andrej et al., 2024), Prague, Stockholm, and Amsterdam (Wijburg and Aalbers, 2017), as well as other German cities such as Munich, Hamburg, Stuttgart, Hanover, Wiesbaden, and Saarbrücken (Mitze et al., 2015).



Figure 5. Spatial Distribution of Case Studies Across Europe

This spatial distribution of Case Studies across Germany (Figure 6) underlines that Berlin’s housing challenge is embedded in broader German dynamics of socio-economic crisis. By choosing Berlin as a case study, research will not only analyse local challenges, but also contributes to the German discourse on urban housing justice.



Figure 6. Spatial Distribution of Case Studies Across Germany

Key Words Analysis

Network analysis (Figure 7), created using VOSviewer, visualises the bibliographic relationships between recurring keywords in the student housing literature from 2010 to 2025. The central cluster — formed by searching keywords **student housing**, **students**, **housing** and **Germany** — anchors the field. Peripheral nodes such as **gentrification**, **financialisation**, **studentification**, **displacement** show the expanding theoretical dimensions. In the same time social dimensions represented by **international students**, **discrimination**, **student needs**, **affordable housing**, keywords, highlighting the importance of students experience. Last but not least, student market represented by **urban housing policy**, **strategy**, **rent**, **housing prices**, **rent control**, **subsidised housing** keywords, emphasising that student housing is a trade objective.

The colour gradient indicates the temporal evolution of the literature: earlier studies (2010–2015) focused on affordability and social equity, while more recent works (2020–2025) increasingly address financialisation, rent control, urban mobility and housing satisfaction in relation to international students.

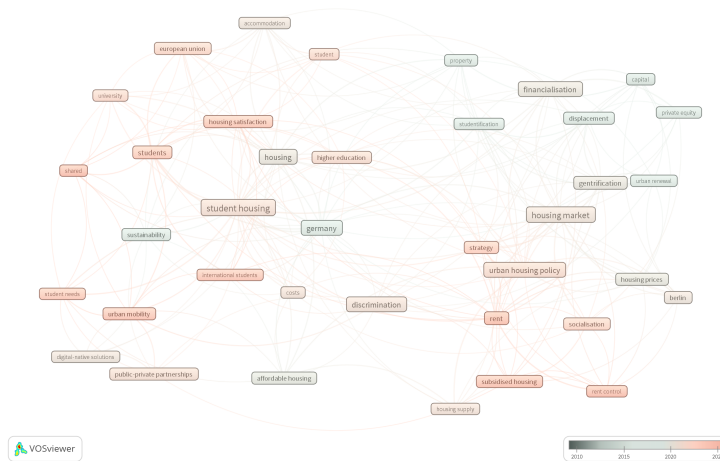


Figure 7. Time Overlay Network Visualisation

Bibliographic mapping demonstrates that the research area has evolved from descriptive housing studies toward data-driven, policy-oriented analyses. It situates your work within an emerging scholarly effort to link spatial analytics, social data, and governance in tackling urban housing challenges.

2.1.6 Trends

Contemporary literature agrees that the housing affordability crisis in Europe is linked not only to demographic growth, but also to the financialisation of the housing sector and the declining role of the public sector.

In the case of Germany, it has been shown conclusively that, despite the myth of a ‘stable, non-financialised’ market, the country has followed its own path of financialisation through the privatisation of social housing and the growth of listed real estate investment trusts (Wijburg & Aalbers, 2017). The authors describe three waves of financialisation: from the failed liberalisation of housing loans to the mass transfer of municipal housing stock to private equity funds and the subsequent concentration of stock in the hands of large privat corporations (Vonovia, Deutsche Wohnen, etc.) (Wijburg & Aalbers, 2017). As a result, the German market has become the next frontline for global investment.

These trends are particularly noticeable in Berlin. According to the IBB Wohnungsmarktbericht 2024, the average cold rent offered in the city in 2024 reached EUR 15/m², which is more than double the comparative rent according to Mietspiegel (EUR 7.21/m²) and represents a gap of around 118% — the largest among major German cities (Investitionsbank Berlin, 2024). At the same time, the rate of new housing construction is falling behind population growth, and the shortage is particularly severe in central areas.

In this context, students, young adults and migrants are the least protected. Student halls of residence cover only about 8% of students in Germany (Deutsches Studierendenwerk, 2024). European-wide data shows that one in four students spends more than 40% of their budget on housing (EUROSTUDENT, 2024).

Thus, international students are most vulnerable to the housing crisis: they are being pushed out of central areas, concentrated in the most marginal segments of the market (shared flats with 6-8 people per room, short-term contracts, illegal or semi-legal rentals) and have a disproportionately high rental burden.

2.1.7 Challenges

Despite notable progress, the literature faces a number of limitations.

First, there is a lack of data and weak integration of sources. Many studies rely on local or one-off cases (Miessner, 2021; von Wittenhorst zu Sonsfeld & Beusker, 2024). Nationwide and Europe-wide studies (EUROSTUDENT, 2024) provide a valuable overview (e.g., the proportion of students spending >40% of their budget on housing), but rarely differentiate data by type of housing (dormitories vs. WGs vs. private rentals) and status (international vs. local students).

Secondly, a significant portion of the studies provide qualitative assessments without quantitative verification. For example, the housing affordability crisis is described through an array of questionnaire complaints and narratives (stress, fraud, language barriers), but is rarely accompanied by spatial modelling or econometric estimates of the impact on mobility participation. Even where quantitative analysis is available, patterns are not always systematised. Miessner (2021) and Avogo et al. (2022) identify important behavioural patterns (private landlords’ focus on students), but do not link them to urban structure or patterns of student housing distribution.

Finally, most studies highlight the role of private actors (investors, landlords, platforms), but provide much less analysis of how the state and universities reproduce or moderate inequality through their decisions.

2.1.8 *Gaps*

Based on the literature reviewed, several key gaps can be identified:

Lack of a comprehensive approach to universities and student housing. Universities and Studentenwerke (student unions) are often viewed merely as background players (providers of dormitories) rather than active actors in the housing field. For example, the IBB report analyses the Berlin market in detail, but hardly highlights universities as a separate policy area (Investitionsbank Berlin, 2024).

Insufficient consideration of the role of the public sector in the current housing crisis for students and migrants. Although some authors emphasise that the abandonment of housing subsidies and the privatisation of municipal housing were key triggers of the crisis (Wijburg & Aalbers, 2017), there are still few studies that directly link the reduction in student accommodation/state programmes with the increased vulnerability of the student group in particular. Paradoxically, even in the discussion of social commons (Stoll, 2022), student housing is hardly mentioned, although logically it is a classic candidate for public ownership and democratic governance.

Most studies are either macro-level (national/European surveys) or micro-cases of individual student cities and campuses. There are no studies that systematically examine the spatial distribution of student and migrant housing in large cities (e.g., Berlin) at the planning district level.

These gaps justify the need for research that examines student housing in conjunction with university structures and urban housing policy, uses spatial analysis (at the city and district level), and assesses how different management models (state/municipal, university, private, cooperative) affect the daily lives of students in different ways.

2.2 1.2 Berlin Housing Crisis History

The history of housing in Berlin is a story of constant crisis and adaptation. Throughout its history, the city has undergone a series of cycles of destruction, reconstruction and transformation. This chapter tells the recent history of Berlin from the post-war period to the present day.

2.2.1 1945-1989. *Post-World War II Developments*

When World War II ended, Berlin was heavily damaged and destroyed. All political parties perceived the housing shortage as the most pressing problem of the time and supported state investment into housing and strong regulation of the market. In this respect, estates built under socialism in East Berlin and under capitalism in West Berlin share many commonalities, despite the political differences of their respective regimes during the division (1949–1990) and particularly during the times of the Berlin Wall (1961–1989) (Urban, 2018). During this period, the authorities in East and West Berlin viewed housing construction as a key instrument of social policy.

West Berlin Housing

In the West, most multifamily buildings—both small and large estates—were built by state-sponsored institutions, owned by municipalities or sometimes other public bodies such as trade unions (Urban, 2018). Between 1952 and 1970, over 85% of new housing construction was publicly subsidized, and frequently constructed by state-owned housing companies operating under the principle of common public interest (Fields & Uffer, 2016)

Interventionist housing policy, which funneled large amounts of public disproportionate funding into housing, was set up in the early 1950s, and at the time approved by the West German Federal Government, ultimately aimed at sustaining the city’s population levels (Lindenthal and Eichholtz, 2012). Despite government support, by the 1970s these neighbourhoods were criticised for their social isolation and standardised architecture, becoming a symbol of the bureaucratic approach to housing.

East Berlin Housing

In East Germany, the paradigm change towards large estates was far more comprehensive (Urban, 2018). After the Building Conference in 1955, the GDR began the ‘industrialisation of the construction industry’, which led to the mass construction of houses from pre-fabricated reinforced concrete panels (Plattenbau). Private landownership was almost entirely abandoned and state-led housing associations provided mass housing, becoming the most conspicuous urban form (Fields and Uffer, 2016). Although the new mass construction embodied the idea of socialist equality, over time it became associated with uniformity and social control by the state.

2.2.2 1989-1991. *Political and Spatial Reunion*

The fall of the Berlin Wall meant not only political but also spatial unification of two separate housing systems. The unification of the city provided for a massive shock to housing supply.

After the opening up of the East, households could live in East Berlin or in surrounding Brandenburg instead of West Berlin. This increased competition is hypothesized to lower prices in peripheral locations of West Berlin. In the same time, the central quarters became “more central”, since they are now in the middle of an overall larger metropolitan area (Lindenthal and Eichholtz, 2012).

Municipal policy towards Berlin’s housing estates was somewhat contradictory. On the one hand, in the early 1990s, both the local authority and the national government

poured generous subsidies into the housing stock supporting renovation. On the other hand, along with increasing international popularity of market-centred ideas, market was turning against social housing, and thus against the every institution from which these estates derived (Urban, 2018).

2.2.3 1992-2012. Privatisation, Financialisation and Gentrification

Berlin's fiscal instability motivated privatisation of its state-owned housing stock and abandonment of social housing subsidies. In 1991, the city owned approximately 480,000 housing units (28% of the entire housing stock), but by the early 2000s, most of this housing had been sold to private and institutional investors (Fields and Uffer, 2016). While privatisation created expectations that private investors would contribute their financial resources to modernize the housing stock, Berlin's housing demand was low.

Between 2000 and 2006, Berlin sold 100,000 dwelling units to international investors, thus sacrificing a system working with long-term success for short-term profit (Fields and Uffer, 2016). Private investors were speculating on a rising market seeing an opportunity to increase rents through modernisation and luxurious upgrading (Urban, 2018).

Research by Fields and Uffer (2016) shows that private investment has exacerbated inequality in access to housing, and global financialisation mechanisms have come to determine urban policy more strongly than social priorities. High-yield investors have renovated residential buildings, increased rents and removed flats from the social rental system. This process has been accompanied by the gentrification of central areas and the social displacement of low-income residents to the periphery.

2.2.4 2012 - 2024. New Actors, Old Problems

Since the early 2010s, Berlin has been experiencing a new wave of housing crisis.

Compared to other major German cities, Berlin still has potential for land use both in the city and in the surrounding area (Berlin Strategy, 2018). Which makes it unique for urban planning research.

At the same time, a variety of challenges are already looming that the city will have to overcome in the coming years. Above all, these include tasks associated with urban growth. According to the Senate Department for Urban Development, Building and Housing (2022), Berlin already reached a population of 3.75 million. The population forecast assumes that the number of residents will rise to around 3.93 million by 2030. The growth is primarily driven by immigration from abroad (CBRE GmbH, 2025). Such residents accounted for 24.4 percent of Berlin's population in 2024. The number of newcomers who moved to Berlin from abroad was twice as high as the number of people who moved to the capital from other parts of Germany.

Opportunities accompanying this growth include the potential for more affordable housing. However, these developments often come with specific spatial requirements and property policies (Berlin Strategy, 2018). Moreover, the application of planning law become particularly challenging in light of rising costs, inflation, and high interest rates, which carry the risk of social inequality, spatial polarisation, and housing crisis.

2.2.5 Conclusion

Berlin Housing History is a story of recurring crises and compromises. The contradiction between housing as a social right and housing as a market commodity remains unresolved.

2.3 1.3 Theoretical Framework

Theoretical framework can be seen as a point of view of to research question. In order to make research interdisciplinary it is a need to consider urban and social theories. While there are many theories that could help to solve the student housing crisis in Berlin, this chapter examines only three main theories that respond to literature gaps and influenced the forming of research interest.

2.3.1 *Urban Theories*

Central Place Theory

Central Place Theory helps to construct spatial research at the level of a large city such as Berlin, as well as at the level of its planning districts, defining urban centrality:

For many economic activities located in a city it is important that they be central to some larger market area that extends beyond the boundaries of the urban place itself. This is especially so for activities that serve the community surrounding the urban place (King, 1986).

This theory can be adapted to the student housing crisis by considering universities as central points that establish spatial economic relations in the student housing market.

Central Place Theory, considering universities as central places, helps to unite the existing Berlin student real estate market and the universities that form the demand for this market. Depending on the ownership of student accommodation, the theory helps to define the roles of both the private and public construction sectors.

Thus, the economic interrelationships between higher education universities and student accommodation come sharply into focus when, for example, policy issues concerning metropolitan and regional government reorganisation are raised.

2.3.2 *Social Theories*

Right to the City Theory

Right to the City Theory is a necessary response to the financialisation that has taken place in Berlin over the past decades.

The concept of Right to the City, developed by French philosopher and sociologist Henri Lefebvre, is a critical approach to understanding social relations in the modern city (Lefebvre, 1996). Lefebvre argues that the city is not just a physical space, but a product of social relations where residents have an unconditional collective right to urban life. Lefebvre's concept allows us to rethink the student housing crisis from the perspective of social justice, accessibility, and commercialisation.

The future of the city must be shaped by its inhabitants, not by the interests of capital. (Lefebvre, 1968).

Berlin's urban environment must take into account the needs of all inhabitants, not just the economically privileged groups. Students are an important part of the urban population, providing a driver of change in the city. Every student has the right to social justice. Factors influencing students' housing choices should be taken into account in Berlin's long-term urban planning.

The city is an oeuvre, closer to a work of art than a simple material product. (Lefebvre, 1968).

The housing market in Berlin should be shaped by social demands and not by commercial interests. The city is not an object of speculation. Prioritising private development over the creation of new affordable housing contradicts the student's right to the city.

In Berlin, this theory is particularly relevant and has actually been written into the city constitution:

Everyone has the right to adequate housing. The state promotes the creation and provision of adequate housing, especially for people on low incomes, and the cultivation of condominium ownership – Article 28 (1) of the Berlin Constitution.

This theory aims to protect students' rights to affordable housing, counteract the displacement of students from the central urban fabric, and halt the privatisation of social housing.

Affordability Theory Affordability Theory complements spatial and social analysis with an economic dimension of housing affordability. The main problem with student housing in Berlin is not only a shortage of supply, but also the fact that prices are out of line with students' actual financial capabilities.

The Affordability Theory in Housing, developed by Michael Stone (2006), defines affordability beyond rental prices, incorporating income, basic needs expenses, transport costs, and structural limitations of the housing market. Stone's theory allows us to define affordable housing in a broader economic and social context.

Housing affordability is not just about the cost of housing, but about the balance between housing costs and the ability to meet other basic needs. (Stone, 2006).

The assessment of housing affordability for students should include not only the cost of housing itself and its proportion in relation to the student's income but also the cost of studying at Berlin's universities, as well as basic financial needs such as transport, student insurance, and groceries. In addition to basic financial needs, the term affordability also includes basic social needs such as the quality of housing, the state of the infrastructure, and the possibility for students to participate in urban life. Student housing affordability is the ability to cover housing costs without compromising basic financial and social needs.

Affordability Theory serves as a conceptual basis for analysing student housing needs and preferences and requires a behavioural and qualitative approach. Thus, it complements the quantitative spatial research of Central Place Theory with a qualitative assessment of the actual living conditions of students, revealing the gap between formal housing supply and actual affordability.

Together, Central Place Theory, Right to the City Theory, and Affordability Theory form a comprehensive analytical framework that allows us to view the student housing crisis in Berlin as a spatial, social, and economic problem simultaneously. These theories provide a theoretical basis for data-driven analysis aimed at reducing the gap between the opportunities offered by the student housing market and the real needs of students in the context of the ongoing housing crisis.

2.4 1.4 Rethinking Student Housing in Berlin

2.4.1 Problem Statement

Housing is a basic human need and a fundamental right, comparable in importance to access to water or clean air (Lefebvre, 1968). Stable, affordable and safe housing is a necessary condition for a dignified life, social integration and full participation in urban and educational spaces. In this context, housing should be viewed not only as physical infrastructure, but also as a socio-political tool that shapes access to opportunities, resources and rights.

As Roberta Burghardt (2025) notes, architects, urban planners and designers can influence not only the form of living spaces, but also models of ownership, participation schemes, and mechanisms for managing and allocating housing. Thus, the issue of student housing goes beyond the design of buildings and becomes part of a broader discussion about urban justice and governance.

2.4.2 Context

Historically, in the post-war period, students in Berlin mainly lived in shared rental accommodation. However, over the last few decades, the structure of student housing has changed significantly. Purpose-Built Student Accommodations (PBSA) — specially built and managed residential complexes for students — have begun to play a special role, which is particularly noticeable in the international student segment.

Nevertheless, despite the growth in the number of such facilities, only about 8% of students in Berlin live in PBSA (Deutsches Studierendenwerk, 2024). Most students are still forced to look for accommodation on the general rental market — in shared flats, sublets or corporate models — due to the general rental pattern of the German housing market. As a result, students compete with other population groups for a limited number of available flats, leading to rising prices, housing instability and housing vulnerability.

The process of financialisation of student housing reinforces this imbalance: PBSAs are viewed by investors as stable investment products with high returns, due to the lack of sufficient alternatives in the public sector. As a result, the social function of student housing is taking a step back, making way for market logic (Annas Ejaz, Anika Gercke, 2025).

2.4.3 Limitations

At the same time, a variety of challenges are already looming that the city will have to overcome in the coming years. Above all, these include tasks associated with urban growth. According to the Senate Department for Urban Development, Building and Housing (2022), Berlin already reached a population of 3.75 million. The population forecast assumes that the number of residents will rise to around 3.93 million by 2030. The growth is primarily driven by immigration from abroad (Housing Market Report, 2024). Such residents accounted for 24.4 percent of Berlin's population in 2024. The number of newcomers who moved to Berlin from abroad was twice as high as the number of people who moved to the capital from other parts of Germany.

Berlin's housing policy stipulates that at least 30% of new housing must be social housing, with the aim of preserving social diversity and housing affordability in all areas of the city (Senate Department for Urban Development, Building and Housing, 2023). However, this target has not yet been sufficiently achieved in the area of student housing.

As ESN and ESU (2023) emphasise, expanding public student housing should be a priority across Europe, especially in countries and cities where it is underdeveloped. Such housing plays a key role not only in ensuring affordability, but also in supporting the internationalisation and mobility of students. Higher education institutions, as important urban actors, should engage more actively with local authorities and the private sector, as many of the challenges faced by students are not only market-related but also involve contractual and administrative barriers, where the public sector can play a decisive role.

The study focuses exclusively on international students and purpose-built student accommodation, which allows for a deeper understanding of the current structural vulnerabilities of this model, but limits the ability to generalise the results to the entire student population.

2.4.4 Research Question

How can data-driven decision-making reshape Berlin's student housing, to bridge the gaps between student housing market capacities and students demands, addressing basic financial and social needs, during the city's ongoing housing crisis?

2.4.5 Scientific Objectives

In accordance with the question posed, the study has the following scientific objectives:

1. Student Housing Market: Capacities, Demand, and Spatial Distribution **Where are student accommodations and universities located?**

How are they spatially distributed across the city?
[resume]

1. Student Housing Market Imbalance **How do student university demand compare to student housing market housing capacities?**

Where do spatial patterns of demand and supply generate elevated market pressure?
[resume]

1. Students Behind the Numbers **Who are the users of student accommodations?**

What experience they have in ongoing housing crisis? What are common needs, preferences, difficulties and problems?

The study aims to combine spatial, quantitative and qualitative analysis to go beyond abstract market indicators. The use of geodata and statistics allows us to identify areas of increased pressure and structural imbalances, while surveys and interviews reveal the everyday experiences of students that are hidden behind aggregated data. This combined approach not only identifies housing shortages, but also understands how they translate into stress, unstable housing strategies, discriminatory practices and social vulnerability, especially among international students. Ultimately, the goal of the study is to form a basis for more equitable and informed decisions in the field of student housing, taking into account the real needs of users and the constraints of the urban context.

3 2. Capacities, Demand & Spatial Distribution

3.1 2.1 Overview

Chapter Capacities, Demand & Spatial Distribution gives an overview of Berlin's student housing market. It focuses on evaluation of initial market capacities and demand and how they spread across the city.

3.1.1 *What? Student Housing Market: Capacities, Demand, and Spatial Distribution*

The student housing market can be understood as the intersection between demand — represented by the number of international students and their housing needs — and capacity, defined by the number of available purpose build student accommodations.

By definition given in MLP Studentenwohnreport (2024):

Demand - mainly depends on the number of students, which often changes by location. Students also compete with young workers, trainees, or commuters, so student numbers only partly show demand. For this research, the target group is international students enrolled in Berlin's higher education, at any level, regardless of current residency.

Capacity- is determined by the number of advertised apartments in the corresponding segments. Alternative housing options, such as student union dormitories, are also available. During the research process, all categories will be merged in a broader term - **purpose-built student accommodation**, categorised by type of ownership.

For the simplicity of data collection, student housing market will be understood as the intersection between demand — represented by the number of international students and their housing needs — and capacity, defined by the number of available purpose build student accommodations.

In a broader perspective, both demand and capacity are dynamic parameters influenced by university services, urban policy and social drivers. A more detailed discussion of their interrelation will be presented in the Discussion Chapter.

3.1.2 *Why? Initial Student Housing Market Description*

The aim of this chapter is to describe the current state of Berlin's student housing market by answering the following research questions:

Where are student accommodations and universities located?

How are they spatially distributed across the city?

3.1.3 *How? Spatial Data Analysis and Spatial Econometrics*

The descriptive process combines Urban Data Mining, Statistical Analysis, and Spatial Autocorrelation to examine market characteristics and explore spatial relationships.

Urban Data Mining

Urban Data Mining is a data collection method that aims to gain new insights by identifying patterns, exceptions, and anomalies in the data (Giseke et al., 2021). It involves selecting, exploring, cleaning, interpreting, and visualising datasets. Data Mining also involves transforming data from non-spatial to spatial. Therefore, it can be seen as the first step of Spatial Data Analysis (SDA). Due to the need to work with large amounts of data. Data Mining is primarily achieved through software, with Python Programming and Artificial Intelligence methods used to further develop urban environments and improve residents' daily lives (Giseke et al., 2021).

Urban Data Mining is used - one of the most time-consuming stages. Since there is no pre-made data set that combines data on all student dormitories and educational institutions in Berlin. Therefore, the data had to be collected in two stages: first manually, then automatical preparation. The quality of manual collection depends directly on the time spent on it and carries the risk of incomplete data, as different data sources indicate different sets of attributes. Meanwhile, automatic data preparation allows you to identify inaccuracies in the data and fill in missing attributes.

Descriptive Statistics

Descriptive Statistics is a set of statistical procedures that summarize the essential characteristics of a distribution through calculating and plotting. Statistical methods are employed to analyse spatial data, quantify a spatial process, discover hidden patterns or unexpected trends and model these data in a geographic context (Giseke et al., 2021).

Descriptive statistics in the context of this work is considered as the initial stage of data analysis, which allows a quantitative assessment of the characteristics of student housing and higher educational institutions, as well as the identification of initial quantitative patterns.

Spatial Autocorrelation

Spatial Autocorrelation is the degree of spatial dependency, association, or correlation between the value of an observation of a spatial entity and the values of neighboring observations of the same variable. It is unlikely that the spatial distribution of Student Accommodations and Higher Education Universities will be homogeneous in the entire Berlin due to the nonuniformity and noncontinuity of space. The magnitude of spatial autocorrelation may vary from one neighborhood to another due to spatial heterogeneity. To estimate spatial autocorrelation at the local level, there is a need to use local measures of spatial autocorrelation, like Local Moran's I index (Grekousis, 2020).

Unlike Urban Data Mining and Descriptive Statistics, Spatial Autocorrelation works directly with spatial dimensions and allows you to identify patterns in the spatial distribution of student accommodations and higher education institutions, as well as evaluate the spatial differences between them.

Through the combination of quantitative, spatial, and computational approaches, this chapter establishes a foundation for the empirical analysis of Berlin's student housing system. It links theoretical concepts of demand and capacity with practical spatial evidence, creating a bridge between urban data science and housing policy research.

3.2 2.2 Urban Data Mining

Urban Data Mining, or the process of collecting and preparing data for analysis, consisted of two stages, which can be divided into manual collection and automated data preparation. The purpose of the manual stage was to collect data and convert it into tabular form, while the automated process was aimed to convert the tabular data into geospatial data.

3.3 Manual Data Collection

The main task of manual data collection is searching purpose-built student accommodation and higher education universities in Berlin and collecting capacity and demand data, respectively.

3.3.1 Student Accommodation

The Studentenwerk (StW, 2025) website was used as the initial source of data for the purpose of building student accommodation. StW is a public, non-profit organisation that provides essential services to students in Germany, including affordable housing, operating student dormitories in collaboration with universities. The StW website provides not only detailed information about addresses, prices, accommodation types, and living space in student accommodation, but also references to other publicly funded companies that provide social housing for students, such as Howoge, Gewobag, Berlinvo, and others. Data on the capacity of public student dormitories was collected from the Study in Germany website, which is supported by the Federal Ministry of Education and Research (BMBF, 2025) and is part of the ‘Germany – Land of Ideas’ campaign.

The missing addresses of private companies were collected from online platforms of companies specialising in student housing and property rental in Berlin. While public companies publish data openly, private companies have less transparent policy, which limits the research process and complicates the manual data mining process.

A complete list of online sources used for manual data collection on student accommodation is presented in Table 2.

	source	comment ¹	link
0	Study in Germany	An official informational platform aimed at in...	https://www.study-in-germany.com/de/
1	HOWOGE Wohnungsbaugesellschaft mbH	Municipal housing company with dedicated stude...	https://www.howoge.de/immobiliensuche/neu
2	GESOBAU AG	Public housing provider offering projects like...	https://www.gesobau.de/wohnungsbau/neu
3	GEWO BAG	New-build and modernized housing projects, som...	https://www.gewobag.de/
4	WBM	Municipal housing firm offering apartments in ...	https://www.wbm.de/neubau-berlin/
5	Berlinovo	Offers student apartments across Berlin. Filte...	https://www.berlinovo.de/de/apartments
6	BGG Berlin	Private housing group with student-suitable ap...	https://www.bgg-berlin.com/de
7	STW Berlin	Official student dorm provider in Berlin; list...	https://www.stw.berlin/en/
8	Studentendorf Berlin	Community-oriented residences in Adlershof and...	https://www.studentendorf.berlin/
9	The FIZZ	International student housing brand; premium h...	https://www.the-fizz.com/en/
10	Neon Wood	Private student housing provider; fully furnis...	https://neonwood.com/de/
11	Home in Berlin	Central Berlin housing platform targeting stud...	https://home-in-berlin.de/
12	Insider Apartments	Furnished rental apartments for students and p...	https://insiderapartments.com/projects
13	Studentenhotel Hubertusallee	Hotel-style short- and long-term housing optio...	https://studentenhotel-hubertusallee.hotel-in-...
14	Studentenwohnheim	External platform listing some Berlin student ...	http://www.studentenwohnheim.ch/berlin/
15	Felixx Student	Premium student housing in Berlin-Charlottenburg.	https://www.felixx-student.de/
16	Smartments Student	Fully furnished student housing units in centr...	https://smartments-student.com/de/
17	MyGermanUniversity	Informational site for international	https://www.mygermanuniversity.com/

Table 2. Manual Data Collection Sources for Student Accommodation

The address is a critical parameter for conducting spatial econometric analysis. However, many student accommodations have more than one building. For example, WH Spandauer Damm comprises five buildings located at Spandauer Damm 134–142 (Studierendenwerk Berlin, 2025). During manual data collection, one of the tasks was to collect unique addresses related to the student housing market in order to calculate centroids afterwards, during automated data preparation.

For student housing, the data collected included the address (**street**, **house_nummer**, **plz**, **city**), the name of the building complex (**name**), the ownership company name (**ownership**), the ownership model (**ownership model**), number of residential units (**!units**), their capacity (**!students**) and type (**units type**), maximum and minimum price (**price_min**, **price_max**), maximum and minimum living space (**area_min**, **area_max**), link to the source (**link**).

As a result of manual data collection, data was collected on 94 proposed student accommodation buildings located at 171 unique addresses.

```
fp = r"../inputs/2.2_urban_data_mining/input_pbsa.xlsx"

pbsa = pd.read_excel(fp, sheet_name='SA', dtype={'street': str, 'house_nummer': str, 'plz': str})

pbsa.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 171 entries, 0 to 170
Data columns (total 16 columns):
#   Column                Non -Null Count  Dtype
--  --
0   name                  171 non -null    object
1   street                171 non -null    object
2   house_nummer          171 non -null    object
3   plz                   171 non -null    object
4   city                  171 non -null    object
5   model                 171 non -null    object
6   ownership              171 non -null    object
7   ownership_model        171 non -null    object
8   !units                93 non -null     float64
9   !students             148 non -null    float64
10  units type            168 non -null    object
11  price_min              148 non -null    float64
12  price_max              96 non -null     float64
13  area_min              154 non -null    float64
14  area_max              144 non -null    float64
15  link                  168 non -null    object
dtypes: float64(6), object(10)
memory usage: 21.5+ KB
```

3.3.2 Higher Education Universities

In order to indicate Berlin's student demand, it was necessary to collect data on all of Berlin's higher education institutions and the number of students enrolled in them. Although the initial plan was to limit the search to public educational institutions only. Nevertheless during the data collection process it became evident that the list and statistics for private universities were also publicly available.

The dataset provided by the Statistisches Bundesamt (DESTATIS, 2025) serves as the most comprehensive source, capturing detailed information on student enrollments, origin, and gender at public and private German higher education institutions. The data provided was so detailed that it included information about branches of Berlin universities in other regions of the country. For example, Media Design School Berlin also has a campus in Munich. These universities were excluded from the statistics as they do not put pressure on the Berlin Student Housing Market.

After cleaning, the Higher Education Institution list from the Federal Statistical Office was double-checked with the official Berlin city directory of higher education institutions (Berlin City Portal, 2025) and the official registry maintained by the German Rectors' Conference (Hochschul Kompass, 2025).

A complete list of online sources used for manual data collection on higher education institutions is provided in Table 3.

```
sources[[
  "source",
  "comment1",
  "link",
]].iloc[23:]
```

	source	comment ¹	link
23	STW	Official public map of university-related faci...	https://www.stw.berlin
24	Hochschul Kompass	Official registry of higher education institut...	https://www.hochschulkompass.de/hochschu
25	Berlin City Portal	Official Berlin city directory of higher educa...	https://www.berlin.de/sen/wissenschaft/einr
26	Federal Statistical Office of Germany (Destatis)	Statistik der Studenten (Code: 21311-0002) is ...	https://www.openstreetmap.org/#map=12/

Table 3. Manual Data Collection Sources for Higher Education Universities

After secondary verification of the list of educational institutions, the next task was to collect unique addresses, as large universities have more than one campus. For example, Charité - Universitätsmedizin Berlin has four campuses in Berlin, which are located at a total of 41 unique addresses. The manual search for addresses was carried out using two services: Open Street Map and Google Maps.

For Higher Education Universities, the data collected included the code according to the Federal Statistical Office, the abbreviation (**abr**), the name (**name**) and the full name (**fname**), the address (**adr**), the ownership model (**own**), type of educational institution (**stud**), number of German students (**!s_de**), number of foreign students (**!s_aus**) and total number of students (**!s_all**).

As a result of manual data collection, data was collected on 70 higher education institutions in Berlin, located at 357 unique addresses.

```

753 op = r"../inputs/2.2_urban_data_mining/input_uni.xlsx"
754
755 uni = pd.read_excel(op, sheet_name='Uni', dtype={'adr': str})
756
757 uni.info()
758
759 <class 'pandas.core.frame.DataFrame'>
760 RangeIndex: 357 entries, 0 to 356
761 Data columns (total 11 columns):
762 #   Column  Non -Null Count  Dtype
763 ---  ---
764 0    code    357 non -null    object
765 1    abr      70 non -null    object
766 2    name     70 non -null    object
767 3    fname    70 non -null    object
768 4    adr      357 non -null    object
769 5    own      70 non -null    object
770 6    stud     71 non -null    object
771 7    !s_de    53 non -null    float64
772 8    !s_aus   52 non -null    float64
773 9    !s_all   54 non -null    float64
774 10   url      42 non -null    object
775 dtypes: float64(3), object(8)
776 memory usage: 30.8+ KB

```

775 3.4 Automated Data Preparation

776 During Manual Data Collection, one of the biggest challenges of data-driven research
777 - finding and selecting data - was successfully overcome.
778 The next challenge within Urban Data Mining is data cleaning, formatting, and stan-
779 dardalising. Data preparation using the Python programming language not only allows
780 you to identify errors in manual input data, but also automates problem-solving during
781 data import, cleaning, and export, as well as during geocoding.

782 3.4.1 Set Up

783 Import Libraries

```

784 import pandas as pd
785 import numpy as np
786 from tqdm import tqdm
787 import geopandas as gpd
788 from geopy.geocoders import Nominatim, Photon
789 from geopy.extra.rate_limiter import RateLimiter
790 import matplotlib.pyplot as plt
791 from matplotlib.colors import LinearSegmentedColormap
792 from shapely.geometry import MultiPoint
793 from sklearn.cluster import DBSCAN
794 import time

```

795 Scripts

796 In order to access spatial analysis and spatial economics methods and Tools, it is
797 necessary to convert addresses to geographical coordinates, which consist of longitude

and latitude. Geocoding converts addresses into geographical coordinates to be placed on a map.

Two services, Nominatim and Photon, were used for geocoding. Despite the use of several services, about 10% of addresses were not recognised or were misinterpreted, indicating errors in the manually collected data.

To improve the accuracy of the coordinate search, the names of the city (Berlin) and country (Germany) were added to each address, which made it possible to uniquely identify objects in the German capital. In addition, to account for the particularities of the German language and possible spelling errors, an alternative spelling mechanism was implemented: special characters such as ‘ β ’ were automatically replaced with ‘ss’, which increased the likelihood of successfully matching the address with the geographic database.

Since the addresses for student accommodation and higher education universities were collected in different formats, two different scripts were also used for geocoding, which differed from each other in their query (`df['`query']`). At the same time, it would be advisable to avoid writing two scripts by collecting data on student accommodation and higher education universities in a single format (for example, `df['query'] = df['street'] + ' ' + df['house_nummer'] + ', ' + df['plz'] + ', ' + df['city'] + ', Germany'`).

```
def geocode_pbsa(df):
    # Initialize geocoders
    nom = RateLimiter(Nominatim(user_agent="berlin_housing", timeout=10).geocode,
                      min_delay_seconds=1)
    photon = RateLimiter(Photon(user_agent="berlin_housing", timeout=10).geocode,
                         min_delay_seconds=0.5)

    # Create address strings
    df['query'] = df['adr'] + ', Germany'
    df['query'] = df['street'] + ' ' + df['house_nummer'] + ', ' + df['plz'] + ', ' + df['city']
    df['query_alt'] = df['query'].str.replace(' \beta', 'ss') # Alternative spelling

    # Geocoding function with retries
    def geocode(address, geocoder, retries=2):
        for _ in range(retries):
            try:
                loc = geocoder(address)
                return (loc.latitude, loc.longitude) if loc else (None, None)
            except:
                time.sleep(1)
        return (None, None)

    # Primary geocoding (Nominatim)
    tqdm.pandas(desc="Primary geocoding", disable=True)
    df[['lat', 'lon']] = df['query'].progress_apply(
        lambda x: pd.Series(geocode(x, nom))
    )

    # Fallback (Photon for failed addresses)
    missing = df[df['lat'].isna()]
    if not missing.empty:
        tqdm.pandas(desc="Fallback geocoding", disable=True)
        missing[['lat', 'lon']] = missing['query_alt'].progress_apply(
            lambda x: pd.Series(geocode(x, photon)))
        df.update(missing)
```

```

850
851     # Convert to GeoDataFrame (Berlin CRS)
852     return gpd.GeoDataFrame(
853         df.drop(['query', 'query_alt'], axis=1, errors='ignore'),
854         geometry=gpd.points_from_xy(df['lon'], df['lat']),
855         crs="EPSG:4326"
856     ).to_crs("EPSG:25833")

857 def geocode_uni(df):
858     # Initialize geocoders
859     nom = RateLimiter(Nominatim(user_agent="berlin_housing", timeout=10).geocode,
860                       min_delay_seconds=1)
861     photon = RateLimiter(Photon(user_agent="berlin_housing", timeout=10).geocode,
862                          min_delay_seconds=0.5)
863
864     # Create address strings
865     df['query'] = df['adr'] + ', Germany'
866     df['query_alt'] = df['query'].str.replace(' \beta', 'ss') # Alternative spelling
867
868     # Geocoding function with retries
869     def geocode(address, geocoder, retries=2):
870         for _ in range(retries):
871             try:
872                 loc = geocoder(address)
873                 return (loc.latitude, loc.longitude) if loc else (None, None)
874             except:
875                 time.sleep(1)
876         return (None, None)
877
878     # Primary geocoding (Nominatim)
879     tqdm.pandas(desc="Primary geocoding", disable=True)
880     df[['lat', 'lon']] = df['query'].progress_apply(
881         lambda x: pd.Series(geocode(x, nom))
882     )
883
884     # Fallback (Photon for failed addresses)
885     missing = df[df['lat'].isna()]
886     if not missing.empty:
887         tqdm.pandas(desc="Fallback geocoding", disable=True)
888         missing[['lat', 'lon']] = missing['query_alt'].progress_apply(
889             lambda x: pd.Series(geocode(x, photon)))
890         df.update(missing)
891
892     # Convert to GeoDataFrame (Berlin CRS)
893     return gpd.GeoDataFrame(
894         df.drop(['query', 'query_alt'], axis=1, errors='ignore'),
895         geometry=gpd.points_from_xy(df['lon'], df['lat']),
896         crs="EPSG:4326"
897     ).to_crs("EPSG:25833")

898     Berlin Spatial Boundaries

899 pf = r"../inputs/2.2_urban_data_mining/lor_pgr.gpkg"
900 pgr = gpd.read_file(pf)
901 pgr.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

```

```

bf = r"..inputs/2.2_urban_data_mining/adm_bz.gpkg"
bz = gpd.read_file(bf)
bz.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

of = r"..inputs/2.2_urban_data_mining/adm_out.gpkg"
out = gpd.read_file(of)
out.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

```

3.4.2 Data Import

The data obtained as a result of manual data collection served as input data for the automated stage.

```

fp = r"..inputs/2.2_urban_data_mining/input_pbsa.xlsx"

pbsa = pd.read_excel(fp, sheet_name='SA', dtype={'street': str, 'house_nummer': str, 'plz': str})

op = r"..inputs/2.2_urban_data_mining/input_uni.xlsx"

uni = pd.read_excel(op, sheet_name='Uni', dtype={'adr': str})

```

3.4.3 Geocoding

As part of the automated data preparation stage, the addresses of student accommodation and higher education institutions were geocoded to obtain accurate geographical coordinates for each unique address. The data was then converted into a geospatial format (GeoDataFrame) and reprojected into the official Berlin coordinate system (EPSG:25833).

Student Accommodation

```
pbsa = geocode_pbsa(pbsa)
```

```
pbsa.info()
```

```

<class 'geopandas.geodataframe.GeoDataFrame'>
RangeIndex: 171 entries, 0 to 170
Data columns (total 19 columns):

```

#	Column	Non -Null Count	Dtype
0	name	171 non -null	object
1	street	171 non -null	object
2	house_nummer	171 non -null	object
3	plz	171 non -null	object
4	city	171 non -null	object
5	model	171 non -null	object
6	ownership	171 non -null	object
7	ownership_model	171 non -null	object
8	!units	93 non -null	float64
9	!students	148 non -null	float64
10	units type	168 non -null	object
11	price_min	148 non -null	float64
12	price_max	96 non -null	float64
13	area_min	154 non -null	float64
14	area_max	144 non -null	float64

```

948     15 link                168 non -null    object
949     16 lat                 171 non -null    float64
950     17 lon                 171 non -null    float64
951     18 geometry            171 non -null    geometry
952 dtypes: float64(8), geometry(1), object(10)
953 memory usage: 25.5+ KB

```

954 Higher Education Universities

```
955 uni = geocode_uni(uni)
```

```
956 uni.info()
```

```

957 <class 'geopandas.geodataframe.GeoDataFrame'>
958 RangeIndex: 357 entries, 0 to 356
959 Data columns (total 14 columns):
960  #   Column      Non -Null Count  Dtype
961  --  --
962  0   code        357 non -null    object
963  1   abr          70 non -null    object
964  2   name        70 non -null    object
965  3   fname       70 non -null    object
966  4   adr         357 non -null    object
967  5   own         70 non -null    object
968  6   stud        71 non -null    object
969  7   !s_de       53 non -null    float64
970  8   !s_aus      52 non -null    float64
971  9   !s_all      54 non -null    float64
972  10  url         42 non -null    object
973  11  lat         357 non -null    float64
974  12  lon         357 non -null    float64
975  13  geometry    357 non -null    geometry
976 dtypes: float64(5), geometry(1), object(8)
977 memory usage: 39.2+ KB

```

978 This results in a shift from tabular data to geospatial data with a new attribute
979 geometry.

980 3.4.4 Data Cleaning

981 Student Accommodation

982 Student Housing capacity is calculated as a single capacity indicator (!cap) accord-
983 ing to the following rule: if the number of students living in student accommodation (students)
984 is known, it is taken; if not, the number of housing units (units) is taken, while purposed
985 build student accommodation, where both values are unknown, is excluded.

```

986 pbsa['!cap'] = pbsa['!students'].where(
987     pbsa['!students'].notnull(),
988     pbsa['!units']
989 )
990
991 pbsa = pbsa.dropna(subset=['!cap'])

```

After that, there is a transition from unique addresses to centroids or geographical centers (Figure XX), for which grouping by name is performed (`pbsa.groupby('~name')`). Thus, student dormitories with more than one building are assigned a single point, and the capacity of all buildings is summed (`!cap': ``sum''`).

```
pbsa_c = (
    pbsa.groupby('name')
    .agg({
        'geometry': lambda x: MultiPoint([pt for pt in x]).centroid,
        'plz': 'first',
        'city': 'first',
        'model': 'first',
        'ownership': 'first',
        'ownership_model': 'first',
        '!cap': 'sum',
        'units type': 'first',
        'price_min': 'first',
        'price_max': 'first',
        'area_min': 'first',
        'area_max': 'first',
        'link': 'first'
    })
    .reset_index()
)

pbsa_c = gpd.GeoDataFrame(pbsa_c, geometry='geometry', crs=pbsa.crs)
```

This results in a transition from 168 points to 88 centroids, which simplifies and speeds up further use of data in the data analysis (Figure 8).

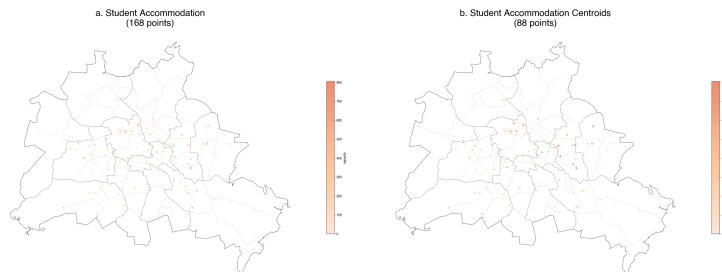


Figure 8. Visual Comparison Initial Unique Student Accommodation Addresses with Final Centroids

It should be noted that unique addresses were collected specifically for the correct calculation of centroids. Meanwhile, for student communities, the calculation of centroids is relatively simple due to the unique names for each community and the proximity of its buildings. For universities, however, the task is not so straightforward, as grouping by name will not give the desired result, since a university with the same name may be located in different parts of the city, for example, Humboldt-Universität, which has three campuses: Campus Mitte, Campus Adlershof, and Campus Nord.

Higher Education Universities

Higher Education Universities' demand for each educational institution is calculated as the number of international students enrolled in it. Universities for which there is no data on students are excluded.


```
1033 uni = uni.dropna(subset=['!s_all'])
```

1034 Afterwards, there is a transition from unique addresses to centroids (Figure XX).
 1035 For this purpose, grouping by code (`uni[['code']]`) is performed, while demand indi-
 1036 cators: the number of German students (`!s_de`), the number of foreign students (`!s_aus`)
 1037 and the total number of students (`!s_all`) are distributed evenly among unique addresses
 1038 — buildings — with a single unique code (`for code in uni[['code']].unique()`). This
 1039 allows us to obtain an ‘average’ representation of the demand indicators for each indi-
 1040 vidual building, while these data for different buildings are initially combined.

```
1041 copy_cols = ['abr', 'name', 'fname', 'own', 'stud', 'url']
1042 split_cols = ['!s_de', '!s_aus', '!s_all']
1043
1044 # Loop over unique codes
1045 for code in uni['code'].unique():
1046     mask = uni['code'] == code
1047     group = uni[mask]
1048
1049     # Copy values from first row in group
1050     for col in copy_cols:
1051         uni.loc[mask, col] = group.iloc[0][col]
1052
1053     # Distribute numeric values
1054     for col in split_cols:
1055         if pd.notnull(group.iloc[0][col]):
1056             uni.loc[mask, col] = round(group.iloc[0][col] / len(group), 2)
1057         else:
1058             uni.loc[mask, col] = pd.NA
```

1059 To further transition from unique addresses to a single address for a complex of uni-
 1060 versity buildings, the data were grouped based on a distance threshold. The threshold
 1061 parameter was responsible for separating unique addresses belonging to the same uni-
 1062 versity campus based on the spatial factor. The threshold value was obtained through
 1063 multiple manual tests. At the same time, the demand values for each unique address were
 1064 summed up.

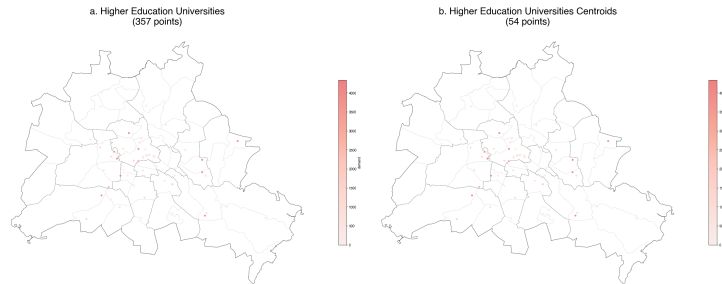
```
1065 distance_threshold = 650 # in meters
1066 records = []
1067
1068 for code, group in uni.groupby("code"):
1069     coords = np.array([(geom.x, geom.y) for geom in group.geometry])
1070     if len(coords) == 0:
1071         continue
1072     # Cluster points that are close to each other
1073     clustering = DBSCAN(eps=distance_threshold, min_samples=1).fit(coords)
1074     group = group.copy()
1075     group['cluster'] = clustering.labels_
1076     for cluster_id, cluster_points in group.groupby('cluster'):
1077         mp = MultiPoint(cluster_points.geometry.tolist())
1078         # Prepare a record with all columns
1079         record = cluster_points.iloc[0].to_dict()
1080         record['geometry'] = mp.centroid
1081         # Sum the specified columns
1082         for col in ['!s_de', '!s_aus', '!s_all']:
1083             record[col] = cluster_points[col].sum()
```

```

1084     record['cluster'] = f"{code}_{cluster_id+1}"
1085     records.append(record)
1086
1087 # Create the GeoDataFrame
1088 uni_c = gpd.GeoDataFrame(records, geometry='geometry', crs=uni.crs)

```

Thus, unique addresses for individual buildings were grouped not only by code but also by university campus affiliation. Next, geometric centers (centroids) were calculated for each group. This resulted in a transition from 357 points to 54 centroids (Figure 9).



1092

1093 *Figure 9. Visual Comparison Initial Unique Higher Education Universities Addresses*
 1094 *with Final Centroids*

1095 The verification method was a manual visual comparison of the final centroids with
 1096 the original unique addresses in QGIS using the Measure Tool.

1097 **3.4.5 Data Export**

1098 The processed data for analysing the capacity and demand for student housing is
 1099 stored in geographic packages (.gpkg) — an open format for storing geospatial data, which
 1100 differs from the tabular format (.excel) in that it contains coordinates.

```

1101 ef = r"../outputs/2.2_urban_data_mining/pbsa.gpkg"
1102 pbsa.to_file(ef, driver="GPKG")
1103
1104 ef = r"../outputs/2.2_urban_data_mining/pbsa_c.gpkg"
1105 pbsa_c.to_file(ef, driver="GPKG")
1106
1107 ef = r"../outputs/2.2_urban_data_mining/uni.gpkg"
1108 uni_copy.to_file(ef, driver="GPKG")
1109
1110 cf = r"../outputs/2.2_urban_data_mining/uni_c.gpkg"
1111 uni_c.to_file(cf, driver="GPKG")

```

3.5 2.3 Descriptive Market Statistics

In the 2024/25 winter semester, a total of 49,801 students with non-German citizenship were enrolled. This is more than in the previous winter semester (+1,439). The proportion of international students thus increased from 24.1% to 24.8% (Statistik Berlin Brandenburg, 2025). According to data collected in the Urban Data Mining chapter, the total capacity of planned student accommodation is 25,817 students. In other words, only 51.8% of international students can be accommodated in halls of residence if there is no competition from local German students.

While describing a dataset is usually the first task in any analysis, in order to rethink the student housing crisis, it is necessary to use descriptive statistics (also called ‘summary statistics’). Descriptive statistics quickly provide an understanding of data variability and allow for the identification of possible patterns (Grekousis, 2020).

3.5.1 Set Up

Import Libraries

```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.path_effects as path_effects
import geopandas as gpd
from shapely.geometry import Point, MultiPoint
from pyproj import CRS
from scipy.stats import norm
```

Berlin Spatial Boundaries

```
pf = r"../inputs/2.3_descriptive_market_statistics/lor_pgr.gpkg"
pgr = gpd.read_file(pf)
pgr.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

bf = r"../inputs/2.3_descriptive_market_statistics/adm_bz.gpkg"
bz = gpd.read_file(bf)
bz.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

of = r"../inputs/2.3_descriptive_market_statistics/adm_out.gpkg"
out = gpd.read_file(of)
out.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)
```

3.6 Student Accommodation Capacity

The geospatial student accommodation data, obtained as a result of Urban Data Mining, served as a base for statistical research.

```
ef = r"../inputs/2.3_descriptive_market_statistics/pbsa_c.gpkg"
pbsa_c = gpd.read_file(ef)
```

3.6.1 Common Statistics

The numeric fields are the basis for collecting statistics, for which it is possible to calculate the sum (`sum`), mean (`mean`), median (`median`) and standard deviation (`std`),

1156 minimum (`min`) and maximum (`max`). In the case of Student Accommodation, such data
 1157 includes: capacity (`!cap`) maximum and minimum price (`price_min`, `price_max`), max-
 1158 imum and minimum living space (`area_min`, `area_max`).

```
1159 cols_numeric = ['!cap', 'price_min', 'price_max', 'area_min', 'area_max']
```

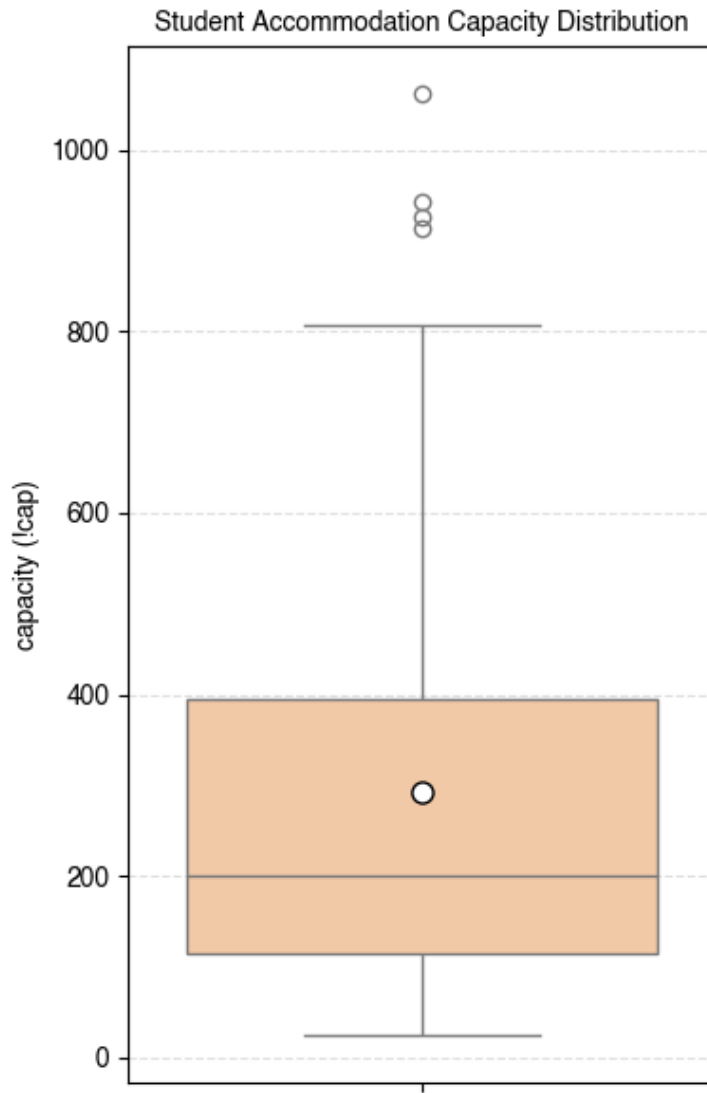
1160 Summary Statistics (Table 4) shows considerable diversity in student accommo-
 1161 dation data, as evidenced by significant deviations from the mean value for the main nu-
 1162 meric fields (`std`).

```
1163 summary = pbsa_c[cols_numeric].agg(['mean', 'median', 'std', 'min', 'max'])
1164 summary = summary.T.round(2)
1165
1166
1167 summary
```

	mean	median	std	min	max
!cap	293.38	200.0	248.18	24.0	1062.0
price_min	447.33	400.0	136.85	256.0	720.0
price_max	615.94	469.5	310.19	352.0	1499.0
area_min	15.11	16.0	4.05	9.0	25.0
area_max	30.79	27.0	17.08	10.0	149.0

1169 *Table 4. Student Accommodation Numeric Fields Summary*

1170 Capacity indicators (`!cap`) show significant dispersion (Figure 10), where the me-
 1171 dian value is equal to the standard deviation (`mean` $\approx$ `std`), reflecting the polarised na-
 1172 ture of housing supply — from small private dormitories, for example, Reichsstrasse Dor-
 1173 mitory designed for 61 students, to large public-owned residential complexes, such as WH
 1174 Siegmunds Hof from Studierendenwerk Berlin with more than 500 places.

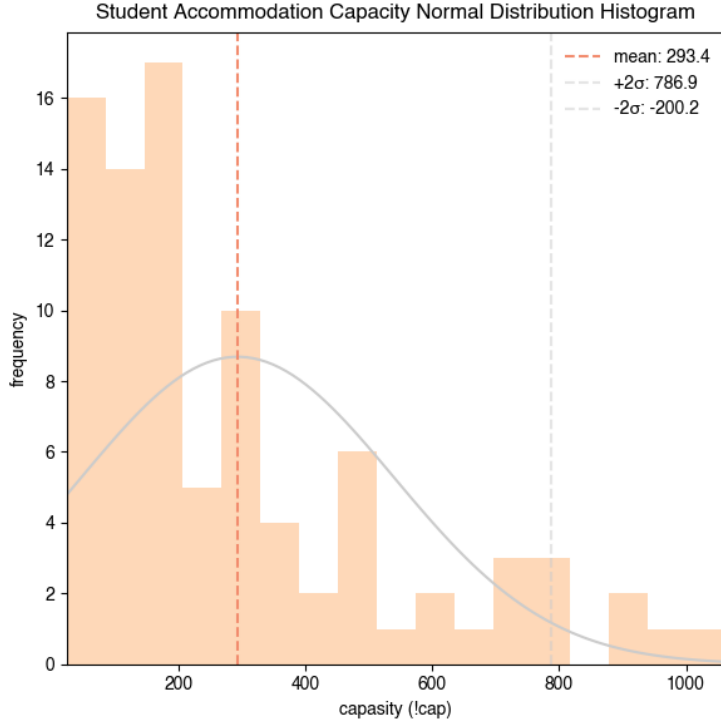


1175

1176 *Figure 10. Student Accommodation Capacity Distribution*

1177 The Student Accommodation Capacity Normal Distribution Histogram (Figure 11)
 1178 shows that student accommodation consists of small halls of residence with fewer than
 1179 300 students, indicating fragmentation and a small scale of supply within Berlin's ex-
 1180 isting urban structure.

1181 Only a few large complexes exceed 600–800 places, reflecting isolated high-density
 1182 developments, often located on the periphery or on redeveloped sites.



1183

1184

Figure 11. Student Accommodation Capacity Normal Distribution Histogram

1185

1186

1187

1188

1189

1190

1191

Price Distribution (Figure 12) shows a clear disparity between minimum and maximum student accommodation prices in Berlin, illustrating both baseline affordability constraints and significant upper-end price volatility. Minimum prices on average are $447\text{€} \pm 136\text{€}$, which aligns with Berlin's Mietspiegel, suggesting a relatively narrow—but still high—cost floor for most students. In contrast, maximum prices have multiple high-value outliers, reaching above 1,400€, indicating the presence of high-cost or highly commodified units within the market.

1192

1193

1194

1195

1196

1197

When discussing student housing prices, it is important to remember that housing is a human right. (Lefebvre, 1968), not an object of trade. The Berlin Constitution clearly state: **Everyone has the right to adequate housing. The state promotes the creation and provision of adequate housing, especially for people on low incomes, and the cultivation of condominium ownership.** Students can be seen as people of low income, who can not afford such high-cost options.

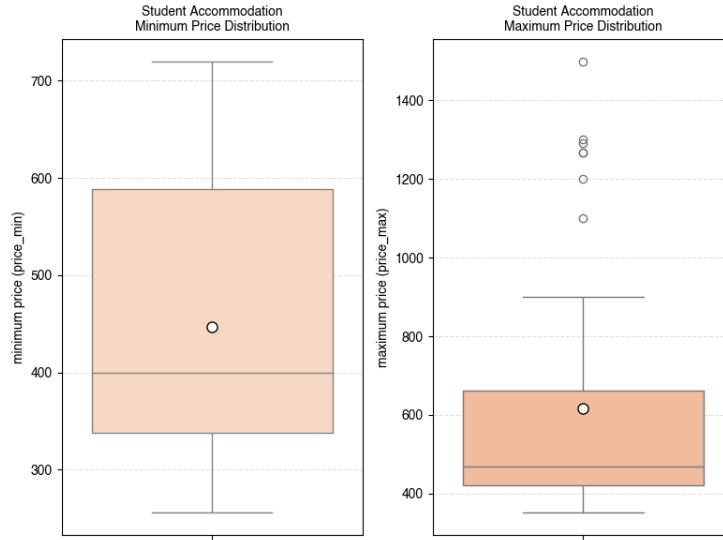


Figure 12. Student Accommodation Price Distribution

Spatial characteristics (`area_min`, `area_max`) demonstrate additional heterogeneity — living spaces range from compact studios ($<12 \text{ m}^2$, $\sigma=\pm 4.7$) to spacious shared flats ($>40 \text{ m}^2$, $\sigma=\pm 11.2$), illustrating differences in the morphology of student housing (Figure 13).

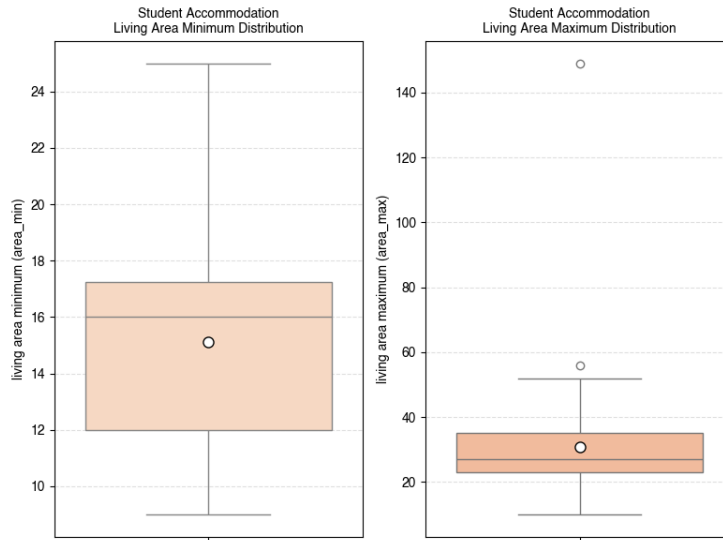


Figure 13. Student Accommodation Living Area Distribution

The significant variability (Table XX) identified at the beginning of the chapter requires additional statistical processing. The heterogeneity essentially reflects the fragmented housing market in Berlin, where supply imbalances lead to differences in housing typology across different urban areas.

For a more detailed analysis of the housing market and evaluation of numeric data, student accommodations were divided into groups according to ownership model (`own`).

3.6.2 Ownership Model

Key Student Accommodation Ownership Companies (Table 5) are the state-owned companies Studierendenwerk Berlin, Berlinovo, Degewo and the private companies Home in Berlin GmbH and Campus Viva GmbH. Together, these five companies own 68.9% of the student accommodation market.

```
summary_ownership = (pbsa_c.groupby('ownership', dropna=False)[cols_numeric].agg(['sum', 'mean',
summary_ownership.sort_values(by=('!cap', 'sum'), ascending=False, inplace=True)
summary_ownership.head(5)
```

	!cap	price_min	price_max	area_min	area_max							
	sum	mean	median	std	sum	mean	median	std				
ownership												
Studienwerk Berlin	394230.668312	285.9010325	322.4344355	237.162269	8394.02712	0.363903	29.128	0.803				
Berlinovo	378734.2732	282.394043	710.07870	0.0	NaN	NaN	NaN	NaN	145.06116	0.060162	20.220	0.276
Home in Berlin GmbH	890305.078315	371560.659	565.565679	0.0790	NaN	91.01517	3.042210	25.526	0.574			
Campus Viva GmbH	108554.6543	59.21905	95.695076	789910	991099	NaN	38.01900	9.014172	0.36036	0.1414		
Degewo	406532.5325	57.910	NaN	NaN	NaN	0.0	NaN	NaN	39.01950	9.535461	0.30530	0.578

Table 5. Key Student Accommodation Ownership Companies

When dividing student accommodation into state and privately owned (Table 6), the gap between the minimum and maximum prices (**price_min** and **price_max**) for monthly rent for different ownership models become evident. Table XX analyses the numerical fields for student accommodation according to different ownership types.

The capacity of state and private accommodation differs slightly. Nevertheless, prices for state-owned accommodation vary from €368 to €439, while privately owned accommodation costs significantly more: from €590 to €968.

At the same time, the minimum income for students set by the Federal Foreign Office in 2025 for foreign students to obtain a visa is €992 per month. This makes private accommodation unaffordable for most international students.

Reducing the price gap between state and private accommodation through price regulation could be part of the solution to the student housing crisis.

```
summary_ownership_model = (
    pbsa_c
    .groupby('ownership_model', dropna=False)[cols_numeric]
    .agg(['mean', 'median', 'std']).round(2)
    .sort_index()
```


1241
1242
1243

)

summary_ownership_model

1244

!cap	price_min	price_max	area_min	area_max
mean median std	mean median std	mean median std	mean median std	mean median std
ownership_model				
Private-Owned	295.50 245.0 205.93	90.52 600.0 96.00	968.94 887.5 307.59	16.00 17.0 3.18
State-Owned	292.28 62.5 269.13	68.43 52.0 79.89	439.44 22.0 60.25	14.72 15.0 4.36
				31.29 27.0 19.65

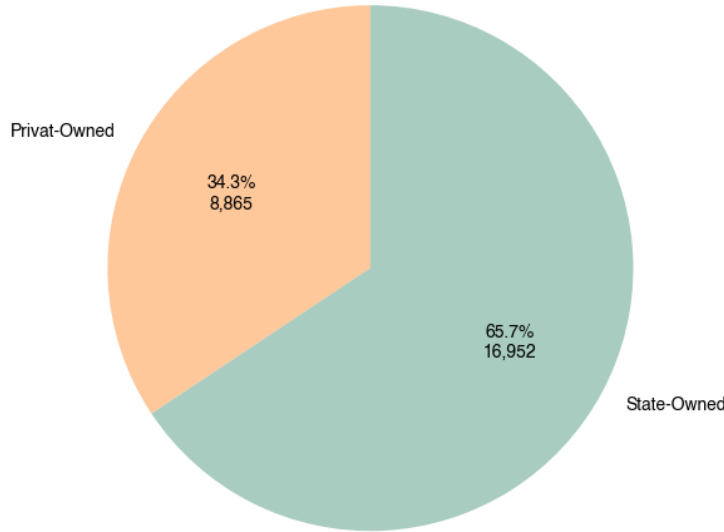
1245

Table 6. Student Accommodation Ownership Model Numeric Fields Summary

1246
1247
1248
1249

Despite their inaccessibility in terms of cost, private student accommodation accounts for 34.3% of the market supply (Figure 14). Private accommodation is not uncommon – it houses about one-fifth of all international students. This suggests that the private market should be regulated systematically, rather than on a case-by-case basis.

Student Accommodation Capacity Distribution by Ownership Model



1250

1251

Figure 14. Student Accommodation Capacity Distribution by Ownership Model

1252
1253
1254

The map (Figure 15) illustrates the spatial distribution of student housing in Berlin by ownership model, revealing clear structural patterns that contribute to the ongoing student housing crisis in the city.

1255
1256
1257

Public residences are more evenly distributed across the periphery, reflecting their role as affordable, welfare-oriented infrastructure, but their limited density indicates insufficient public capacity relative to demand.

Private facilities are more concentrated in central areas and high-demand areas, particularly in Mitte, Friedrichshain-Kreuzberg and Charlottenburg-Wilmersdorf, reflecting market logic that prioritises profitable locations with high rents over equal access.

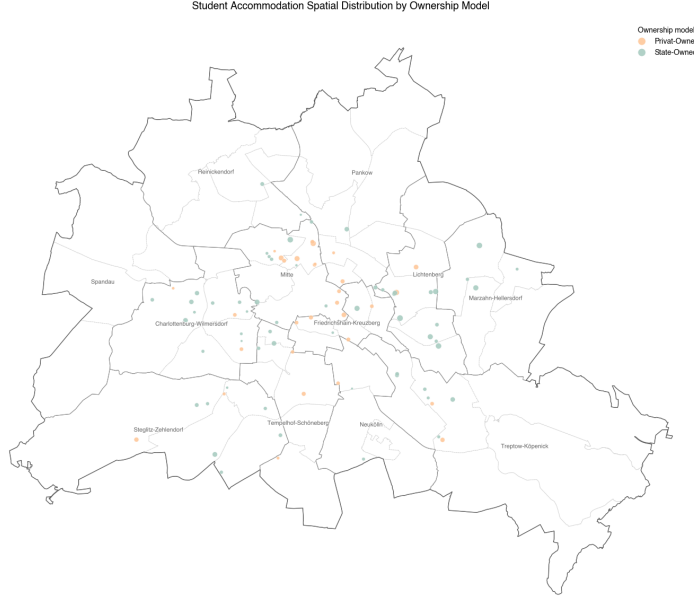


Figure 15. Student Accommodation Spatial Distribution by Ownership Model

Two histograms (Figure 16) show that public and private student residences in Berlin have almost identical average capacities (≈ 290 – 300 places), but their distribution differs in aspects that are relevant to the housing crisis.

Public facilities have a wider range, including several large complexes with more than 900 places (e.g. WH Coppelstraße, Iranische Straße, Ludwig-Renn-Straße Hellersdorf, WH Aristotelessteig), reflecting their role in providing significant long-term public capacity.

In private student accommodation, by contrast, small and medium-sized facilities predominate, with fewer large complexes, indicating a market that tends to prioritise scattered, small investments over large-scale projects.

The public sector is at least attempting to implement large projects, while private supply remains fragmented.

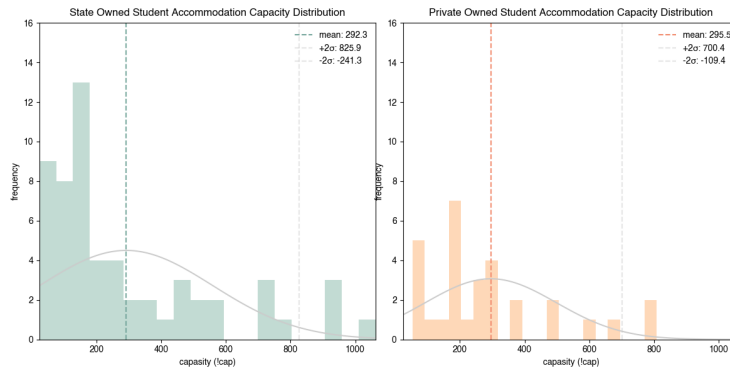


Figure 16. State and Private Owned Student Accommodation Normal Capacity Distribution Comparison

This division highlights a fundamental problem: public student housing is too infrequent and located on the outskirts to offset affordability pressures, while private housing is concentrated in the most expensive areas. As a result, students face geographical and financial mismatches, reinforcing the need for data-driven planning strategies that rebalance supply, expand capacity, and redirect development to underserved areas.

Full descriptive statistics on student accommodation are available for review in the Materials & Data chapter.

```
output_path = r"../outputs/2.3_descriptive_market_statistics/student_acomodation_statisticcs.xls"

# Write DataFrames to separate sheets in one Excel file
with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
    summary.to_excel(writer, sheet_name='common_statistics')
    summary_ownership.to_excel(writer, sheet_name='by_ownership')
    summary_ownership_model.to_excel(writer, sheet_name='by_ownership_model')
    summary_unit_type.to_excel(writer, sheet_name='by_unit_type')
```

3.7 Higher Education Universities Students Demand

The higher education universities' geospatial data, obtained as a result of Urban Data Mining, served as a base for statistical research.

```
cf = r"../inputs/2.3_descriptive_market_statistics/uni_c.gpkg"
uni_c = gpd.read_file(cf)
```

3.7.1 Common Statistics

In the case of Higher Education Universities, the numerical data are the number of German students (!s_de), the number of foreign students (!s_aus) and the total number of students (!s_all).

```
cols_numeric = ['!s_all', '!s_de', '!s_aus']
```

Higher Education Universities Data (Table 7) shows a sharp difference in student numbers between universities in Berlin. The average university enrolment is 3,720 students, but a typical university (median 842 places) is 4.4 times smaller. This means that several large universities take on the main part of the load (such as Free University, Humboldt University and Technical University of Berlin).

Foreign students make up 24.1% of the total, meaning that one in five students in Berlin is a foreigner. Comparing the standard deviations for German and foreign students, one can see segmentation based on nationality, which manifests itself in a 3.3-times higher dispersion of German students than foreign students, revealing a cluster model of foreign student placement in large state universities. It is noteworthy that Free University (8,280 foreign students), Humboldt University (6,415 foreign students) and Technical University of Berlin (10,262 foreign students) account for 24,957 foreign students, or 51.5% of total international student demand.

```
summary = uni_c[cols_numeric].agg(['mean', 'median', 'std', 'sum'])
summary = summary.T.round(2)

summary
```

	mean	median	std	sum
!s_all	3720.15	842.0	8498.32	200888.0
!s_de	2822.80	489.0	6604.27	152431.0
!s_aus	897.35	172.5	1999.27	48457.0

Table 7. Higher Education Universities Numeric Fields Summary

Free University, Humboldt University and Technical University Berlin are exceptions to the rule (or so-called ‘statistical outliers’). Most universities have up to 600 students. Thus, Higher Education Universities Students Demand Distribution (Figure 17) is far from normal.

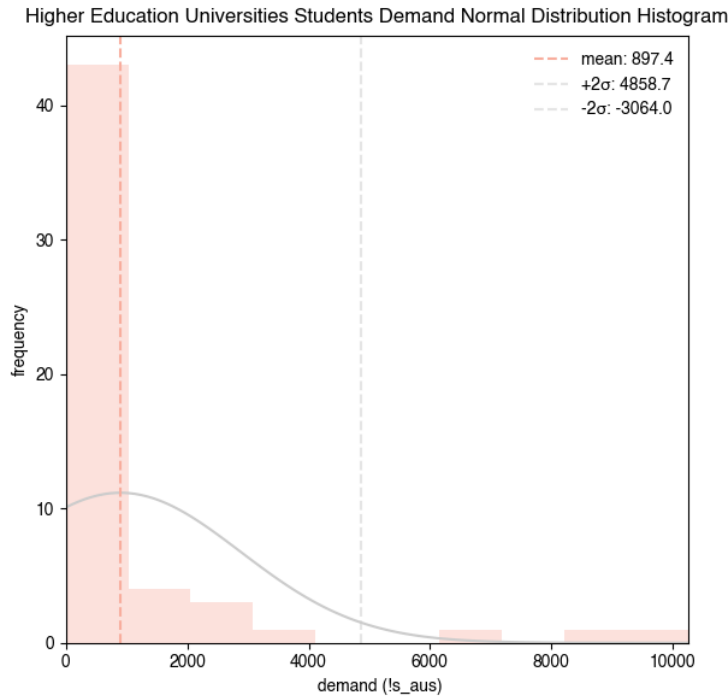


Figure 17. Higher Education Universities Students Demand Normal Distribution Histogram

In order to gain a deeper understanding of the imbalance in supply, by analogy with student accommodations, higher education universities can be divided into two groups based on their ownership model (**own**): Public and Private Universities.

3.7.2 Ownership Model

Higher Education Universities Ownership Model Numeric Fields Summary (Table 8) shows a dramatic structural difference between public and private higher education institutions in Berlin. Public universities dominate the academic landscape, accepting an average of more than 9,000 students per institution, while private universities accept fewer than 1,000 students. This gap persists among both German and foreign students: public universities enrol the majority of German students (an average of $\approx 7,100$) and attract significantly more foreign students (an average of $\approx 2,000$) than private institutions.

The much higher overall figures and variability in the public sector highlight the concentrated demand pressure on public universities, which are the main drivers of the student housing market.

```
summary_ownership_model = (
    uni_c
    .groupby('own', dropna=False)[cols_numeric]
    .agg(['mean', 'median', 'std', 'sum']).round(2)
    .sort_index()
)
summary_ownership_model
```

	!s_all			!s_de			!s_aus					
	mean	median	std	sum	mean	median	std	sum	mean	median	std	sum
own												
Privat Uni- ver- sity	804.89	597.0	727.67	28171.04	498.49	267.0	584.14	17447.08	606.40	102.0	493.02	10724.0
Public Uni- ver- sity	9090.37	4226.0	12832.96	572717.71	104.42	805.0	9895.76	134984.19	85.95	400.0	3062.69	7733.0

Table 8. Higher Education Universities Ownership Model Numeric Fields Summary

Private universities contribute less significantly to the total number of students (Figure 18). Nevertheless, 22.1% of foreign students choose private universities for their education. Of all students at private universities, 38% are foreigners (vs. 22% at state universities).

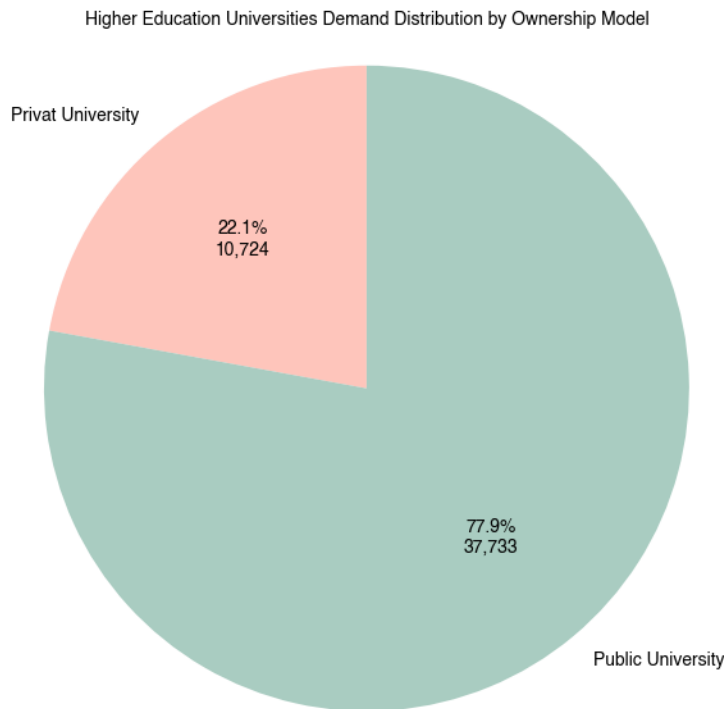


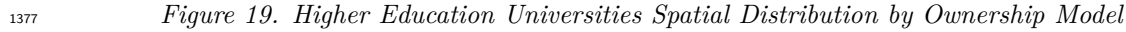
Figure 18. Higher Education Universities Demand Distribution by Ownership Model

The spatial distribution of higher education institutions in Berlin (Figure 19) shows a strong central concentration of private universities, in contrast to the more dispersed location of public universities throughout the city.

Private universities are mainly concentrated in Mitte, Charlottenburg-Wilmersdorf and Friedrichshain-Kreuzberg – areas that are already characterised by high demand for housing – while public universities occupy both central areas and more peripheral areas such as Lichtenberg, Steglitz-Zehlendorf and Treptow-Köpenick. This uneven geography increases spatial inequality in the demand for student housing: large public universities create high pressure on the housing market across the city, while private universities add local spikes in demand in already highly densely populated central areas.

For large public universities, state-owned dormitories remain a priority, while the private sector requires integration into the urban housing ecosystem and lower prices as a market driver that is developing faster than the public sector.

The map highlights a broader structural problem: the spatial logic of Berlin's university system does not correspond to the distribution of student housing, exacerbating the city's ongoing student housing crisis.



Public universities have a wider range, including several large complexes with more than 8,000 students. (e.g., Technical University of Berlin and Free University of Berlin), indicating a high concentration of international demand in a limited number of campuses.

The contrast between the two distributions confirms that the housing demand of international students in Berlin is unevenly distributed, creating local areas of market overload.



Full descriptive statistics on student accommodation are available for review in the
Materials and Data Chapter.

–51–

```
1396
1397 # Write DataFrames to separate sheets in one Excel file
1398 with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
1399     summary.to_excel(writer, sheet_name='common_statistics')
1400     summary_ownership_model.to_excel(writer, sheet_name='by_ownership_model')
1401     summary_uni_type.to_excel(writer, sheet_name='by_uni_type')
```


3.8 2.4 Global Spatial Autocorrelation

In the Descriptive Market Statistics Chapter, a descriptive analysis of two datasets was conducted: student accommodations and higher education universities. However, in order to access spatial distribution across cities, it is necessary to measure spatial dependency.

Spatial autocorrelation measures spatial dependency. In other words, how much the value of a variable in a specific location is related to the values of the same variable at neighboring locations. Spatial autocorrelation (either positive or negative) is a key concept in geographic analysis. Spatial autocorrelation can be explained by the first law of geography. According to the law, objects in a neighborhood tend to have more similarities and interactions than those lying further away (Grekousis, 2020).

3.8.0.1 Set Up Import Libraries

```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib as mpl
import seaborn as sns
from matplotlib.colors import LinearSegmentedColormap
import matplotlib.patches as mpatches
import matplotlib.path_effects as path_effects
import geopandas as gpd
from shapely.geometry import Point, MultiPoint
from pyproj import CRS
from scipy.stats import norm
import libpysal
from esda import Moran
from libpysal.weights.spatial_lag import lag_spatial
from libpysal.weights import KNN
from libpysal.weights import weights
```

Berlin Spatial Boundaries

```
pf = r"../inputs/2.4_global_spatial_autocorrelation/lor_pgr.gpkg"
pgr = gpd.read_file(pf)
pgr.replace({'-': np.nan, 0: np.nan, ':': np.nan}, inplace=True)

bf = r"../inputs/2.4_global_spatial_autocorrelation/adm_bz.gpkg"
bz = gpd.read_file(bf)
bz.replace({'-': np.nan, 0: np.nan, ':': np.nan}, inplace=True)

of = r"../inputs/2.4_global_spatial_autocorrelation/adm_out.gpkg"
out = gpd.read_file(of)
out.replace({'-': np.nan, 0: np.nan, ':': np.nan}, inplace=True)

mf = r"../inputs/2.4_global_spatial_autocorrelation/berlin_mauer.gpkg"
mauer = gpd.read_file(mf)
mauer.replace({'-': np.nan, 0: np.nan, ':': np.nan}, inplace=True)
```

3.9 Student Accommodation Capacity

3.9.1 Data Import

The data obtained as a result of Urban Data Mining was used as input data for the analysis (since no changes were made to the data during Descriptive Market Statistics).

```
pf = r"../inputs/2.4_global_spatial_autocorrelation/pbsa_c.gpkg"
pbsa_c = gpd.read_file(pf)
```

Before calculating global spatial autocorrelation, it is also necessary to perform tests: Moran's I Scatterplots Test and K-Nearest Neighbor Test, as well as calculate the Significance Threshold test.

Since the research focused on market parameters, global autocorrelation will focus on **capacity**, which measures the market supply.

3.9.2 Standardisation

Data standardisation is used to assess the correlation of neighbouring values relative to the overall average and to eliminate the outliers identified in the previous chapter. Standardisation converts data into dimensionless values, which allows for accurate comparison of clustering for different student accommodations. Positive standardised capacity values (!cap_std) indicate above-average objects, while negative values indicate below-average objects. This will help to clearly identify student accommodation clusters in the future.

```
pbsa_c["!cap_std"] = pbsa_c["!cap"] - pbsa_c["!cap"].mean()
```

3.9.3 Moran's I Scatterplots Test & K-Nearest Neighbor Test

Several diagnostic measures can be used to identify spatial autocorrelation. Those that estimate spatial autocorrelation by a single value for the entire study area are named global spatial autocorrelation measures. The most commonly used is Moran's I (Grekousis, 2020).

Moran's I computes global spatial autocorrelation by taking into account feature locations and attribute values (of a single attribute) simultaneously. It is calculated by the following formula:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (1)$$

where:

- n — number of spatial features
- x — attribute value of feature i (a variable in a spatial analysis context)
- x — attribute value of feature j
- $\bar{x}$ — mean of this attribute
- w — spatial weight between feature i and j
- w — aggregation of all spatial weights

In order to calculate spatial weights between features, K-nearest neighbor weights are used. K-nearest neighbor weights are distance-based weights defines the neighbor set of a particular observation as containing its nearest observations, where the user spec-

ifies the number of nearest neighbors (k). If we do not test k , Moran's I Index might have values beyond the $[-1,1]$ boundaries (Grekousis, 2020).

To understand the intuition behind its math, it is useful to begin with a graphical interpretation: the Moran Plot for different numbers of neighbors (Figure XX). The Moran Plot is a way of visualizing a spatial dataset to explore the nature and strength of spatial autocorrelation. It is essentially a traditional scatter plot in which the variable of interest is displayed against its spatial lag (Rey et al., 2023).

The Moran's I scatterplots (Figure XX) across increasing k -values (`k_values = list(range(1, 21))`) reveal that student accommodation capacity in Berlin exhibits a weak but consistently positive spatial autocorrelation, which strengthens as the neighborhood definition expands. Each point represents a student accommodation and its relationship to the capacities of its surrounding neighbors. The four quadrants provide insight into the spatial logic of student housing distribution:

Q1 – High–High (HH)

These locations have high student accommodation capacity and are surrounded by other areas with similarly high capacities. These clusters signal where the market has already developed a strong supply.

Q2 – Low–High (LH)

These student accommodations have low capacity themselves, but are neighboring high-capacity ones. They often represent transition zones—places that have the potential to absorb future student housing development because they border well-served districts. These locations highlight opportunity areas where relatively small additions could significantly improve spatial balance.

Q3 – Low–Low (LL)

These locations exhibit low capacity and are surrounded by other low-capacity areas, marking systemic undersupply zones. For policy-making, these locations indicate structurally disadvantaged parts of the city, where students face limited options.

Q4 – High–Low (HL)

These student accommodations have high capacity but are surrounded by low-capacity neighbors. Such areas represent isolated clusters—individual dorms or private student residences located in otherwise undersupplied districts. They highlight mismatches where supply exists but lacks integration into broader student housing networks, potentially limiting accessibility.

In conclusion, the Moran's I scatterplots illustrate that Berlin's student housing supply is spatially uneven, featuring concentrated high-capacity cores, isolated developments, transitional opportunity zones, and persistent undersupply areas.

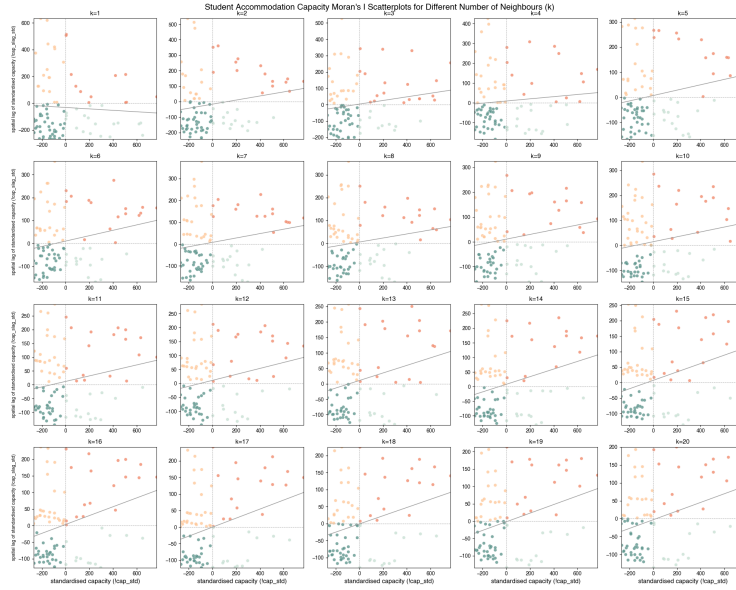


Figure 21. Student Accommodation Capacity Moran's I Scatterplots for Different Number of Neighbors (k)

The difference between the scatterplots (Figure 21) must be evaluated based on Moran's I index and P-value. Through these metrics, we assess whether spatial autocorrelation is statistically significant. To do this, we need to calculate the parameters for each value of the number of nearest neighbors (k) (Table 9).

Moran's I index and P-value can be interpreted as follows. A Moran's I index score higher than 0.3 is an indication of relatively strong positive autocorrelation, while a score lower than -0.3 is an indication of relatively strong negative autocorrelation. If the p-value is large (usually $p > 0.05$), the results are not statistically significant, and we cannot reject spatial dependency (Grekousis, 2020).

```

np.random.seed(55)

k_values = [ 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
morans_I = []
p_values = []

for k in k_values:
    w = libpysal.weights.KNN.from_dataframe(pbsa_c, k=k)
    w.transform = "R"
    moran = Moran(pbsa_c['!cap'], w)
    morans_I.append(moran.I)
    p_values.append(moran.p_sim)

# Create DataFrame
pbsa_k_test = pd.DataFrame({
    'k': k_values,
    'Moran's I': morans_I,
    'P -value': p_values
})

pbsa_k_test

```

	k	Moran's I	P-value
0	2	0.131920	0.076
1	3	0.110553	0.057
2	4	0.054687	0.163
3	5	0.107291	0.030
4	6	0.117724	0.017
5	7	0.100457	0.018
6	8	0.088330	0.039
7	9	0.095008	0.016
8	10	0.097889	0.012
9	11	0.099631	0.013
10	12	0.106905	0.008
11	13	0.122254	0.005
12	14	0.130466	0.002
13	15	0.135846	0.003
14	16	0.133675	0.001
15	17	0.136209	0.001
16	18	0.120816	0.002
17	19	0.124204	0.001
18	20	0.121062	0.002

Table 9. Student Accommodation Capacity Moran's I index and P-value for Different Number of Neighbors (k)

The task of finding the desired value of the number of nearest neighbors (k) can be solved by plotting a graph of the dependence of Moran's I index and P-value on k (Figure 22). On this graph, it is necessary to find the maximum values of the Moran's I index modulus and then compare the P-value.

The maximum values of Moran's I index are achieved at k from 15 to 17. Since there are only 88 student dormitories, the minimum value of 15 neighbors ($k = 15$) was chosen. That is, each student dormitory is compared with its 15 closest neighbors in order to identify spatial dependence.

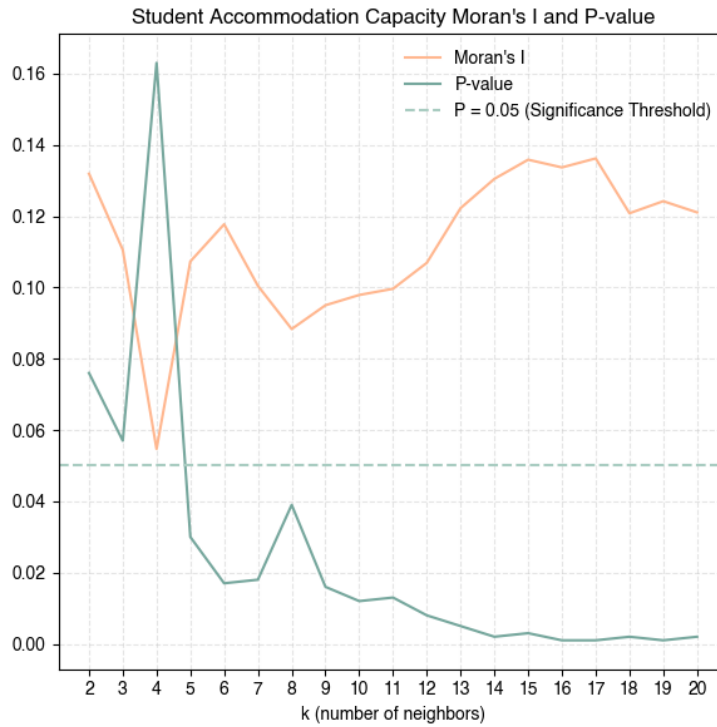


Figure 22. Student Accommodation Capacity Moran's I index and P-value for Different Number of Neighbors (k)

3.9.4 Significance Threshold Test

In the original Moran's I Scatterplots for Different Number of Neighbors (Figure XX), the points are distributed across four quadrants: HH, HL, LH, LL. However, points near zero may accidentally fall into false categories, even though they do not demonstrate real spatial dependence. This can lead to distorted results and the identification of false points of spatial autocorrelation.

The Significance Threshold Test sets a significance threshold (`threshold`) for standardized capacity (`!cap_std`) and spatial lag standardized capacity (`!cap_slag_std`), and filters out not significant points where both variables take neutral values.

Option 1 - Percentile -based threshold: 234.375
(75th percentile of absolute values)
Option 2 - Standard deviation -based threshold: 124.090
(0.5 * standard deviation)
Option 3 - Fixed threshold: 50.000
(Fixed value for data range)
Option 4 - Median -based threshold: 78.750
(0.5 * median of absolute values)

Test 1: threshold = 234.375
NS: 66 (75.0%)
HH: 8 (9.1%)
LH: 6 (6.8%)
LL: 6 (6.8%)
HL: 2 (2.3%)

Test 2: threshold = 124.090

```

1596     NS: 25 (28.4%)
1597     HH: 15 (17.0%)
1598     LH: 10 (11.4%)
1599     LL: 26 (29.5%)
1600     HL: 12 (13.6%)
1601
1602 Test 3: threshold = 50.000
1603     NS: 6 (6.8%)
1604     HH: 16 (18.2%)
1605     LH: 12 (13.6%)
1606     LL: 35 (39.8%)
1607     HL: 19 (21.6%)
1608
1609 Test 4: threshold = 78.750
1610     NS: 11 (12.5%)
1611     HH: 16 (18.2%)
1612     LH: 12 (13.6%)
1613     LL: 35 (39.8%)
1614     HL: 14 (15.9%)

```

1615 In order to correctly interpret the Significance Threshold Test, it is necessary to
1616 pay attention to the distribution of points across four squares - spatial clusters.

1617 **Option 1 - Percentile-based threshold: 234.375**

1618 With a high threshold value, 75% of the points became neutral, and the global autocor-
1619 relation analysis lost its meaning.

1620 **Option 2 - Standard deviation-based threshold: 124.090**

1621 It will eliminate the outliers that were described in the Descriptive Market Statistics chap-
1622 ter, as it is calculated according to the standard deviation.

1623 **Option 3 - Fixed threshold: 50.000**

1624 A threshold value that is too low results in the percentage of neutral points being too
1625 low at 6.8%.

1626 **Option 4 - Median-based threshold: 78.750**

1627 Median-based threshold does not react to abnormally large values. Therefore, this thresh-
1628 old creates a threshold that reflects the real “central” distribution structure (`threshold`
1629 `= threshold_median`).

1630 Using three tests: Moran’s I Scatterplots Test, K-Nearest Neighbour Test, and Sig-
1631 nificance Threshold Test, it can be confirmed that the results of local spatial autocor-
1632 relation are stable.

1633 **3.9.5 Spatial Weights & Moran’s I & P-value**

1634 After performing all necessary tests, it is possible to proceed to the direct calcu-
1635 lation of Spatial Weights, Moran’s I, and P-value for the selected value of numbers of
1636 nearest neighbors (`k=15`). To do this, the method of spatial weights standardization must
1637 be defined. Row-standardization (`w.transform = 'R'`) makes Moran’s I and other
1638 spatial indices comparable between different areas, even if the number of neighbors for
1639 objects is different. For each row of the weight matrix (each object), the sum of all neigh-
1640 bor weights is normalized to 1.

$$w_{ij}^{(R)} = \frac{w_{ij}}{\sum_j w_{ij}} \quad (2)$$

```

1641 w = KNN.from_dataframe(pbsa_c, k=15)
1642
1643 w.transform = "R"
1644
1645 values = pbsa_c['!cap']
1646
1647 moran = Moran(values, w)
1648 print(f"Moran's I: {moran.I:.3f}")
1649 print(f"P -value: {moran.p_sim:.3f}")
1650
1650 Moran's I: 0.136
1651 P -value: 0.001

```

1652 Moran's I (Moran's I: 0.136) indicates positive spatial autocorrelation. In the
 1653 context of urban planning, this indicates that student accommodations have a weak but
 1654 stable tendency to form clusters.

1655 P-value (P-value: 0.001) indicates that the probability of obtaining such a re-
 1656 sult by chance is 0.1%. In the context of urban planning, this shows that the observed
 1657 spatial clusters actually exist and are not random.

1658 *3.9.6 Spatial Lag & Standardised Spatial Lag*

1659 After calculating spatial weights, it is necessary to compute the spatial lag. The
 1660 spatial lag operator is one of the most common and direct applications of spatial weights
 1661 matrices (called $\mathbf{W}$ formally) in spatial analysis. Conceptually, the spatial lag captures
 1662 the behavior of a variable in the immediate surroundings of each location; in that respect,
 1663 it is akin to a local smoother of a variable (Rey et al., 2023).

1664 We can formally express it in matrix notation as:

$$Y_{sl} = \mathbf{W}Y \quad (3)$$

1665 or, in individual notation as:

$$y_{sl-i} = \sum_j w_{ij} y_j \quad (4)$$

1666 where:

1667 w_{ij} is the cell in $\mathbf{W}$ on the i -th row and j -th column, thus capturing the spatial
 1668 relationship between observations i and j , in other words y_{sl-i} is the weighted average
 1669 values in a specified neighborhood.
 1670 y_{sl-i} thus captures the product of the values and weights of each observation other than
 1671 i in the dataset.

1672 Spatial lag is a key element of many spatial analysis techniques including spatial
 1673 autocorrelation (Rey et al., 2023).

```

1674 pbsa_c["!cap_slag"] = lag_spatial(w, pbsa_c["!cap"])
1675 pbsa_c["!cap_std"] = pbsa_c["!cap"] - pbsa_c["!cap"].mean()
1676 pbsa_c["!cap_slag_std"] = lag_spatial(w, pbsa_c["!cap_std"])

```


The maps (Figure 23) illustrate the various stages of data preparation for global autocorrelation — from actual capacity values (!cap) and their spatial relationships (!cap_slag) to their standardized values (!cap_std and !cap_slag_std).

a. Student Housing Capacity (!cap)

The first map shows the absolute capacity of student housing. Despite the presence of large facilities in central areas, the overall structure remains fragmented, with significant heterogeneity between districts. This map reflects the actual market and serves as a starting point for analysis.

b. Student Housing Capacity Influence (!cap_slag)

The second map displays the spatial lag of capacity — the average capacity in neighboring locations. It captures the influence of neighboring facilities, smoothing out local peaks and revealing latent clusters that are not visible in the absolute data. This shows not only where large dormitories are located, but also where their surroundings are saturated or deficient.

c. Standardized Capacity (!cap_std)

The third map shows standardised values (centered around the mean). Here you can see which areas have above- or below-average capacity for the city, rather than simply “large/small.” This is a key step for Moran’s I, as it allows you to identify deviations from the overall trend rather than absolute values.

d. Standardized Capacity Influence (!cap_slag_std)

The fourth map displays standardized spatial lag—how much neighboring areas are above or below average. In combination with map (c), it shows where High–High (HH) and Low–Low (LL) clusters form and where anomalies (High–Low / Low–High) occur. These patterns form the basis for interpreting the Moran Scatterplot.

Together, maps (a–d) demonstrate that the distribution of student housing in Berlin is highly heterogeneous and structurally dependent on the spatial environment. The transition from absolute values (!cap) to standardized spatial lags (!cap_slag_std) allows us to identify hidden clusters and areas of shortage — key information for rethinking student housing placement strategies and developing spatially sensitive solutions.

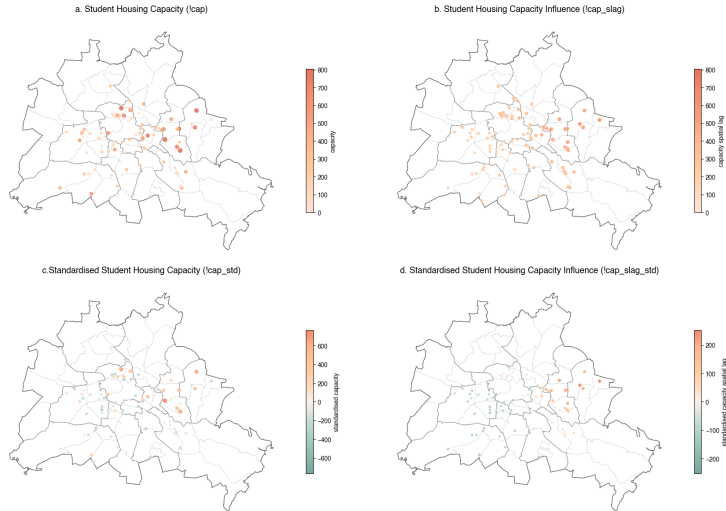


Figure 23. Stages of Preparing Student Accommodation Capacity Data

Moran’s graph (Figure 24) shows the spatial relationship between the standardised capacity of student housing (!cap_std) and its spatial lag (!cap_slag_std) — that

is, the average capacity of neighbouring locations. Dividing the graph into four quadrants allows us to identify types of spatial patterns:

Q1 – High–High (HH)

These are high-capacity clusters where several large student residences are concentrated close to each other. For example, Studentenwohnheim Alfred-Jung-Straße, with a capacity of 800, is located less than 200 metres from another large student accommodation facility, WH Storkower Straße, with a capacity of 508 students, forming a stable centre for foreign students. At the same time, 12 of the 17 student facilities in the HH quadrant are state-owned. This leads to the conclusion that these centres of attraction are formed by state-owned facilities.

Q2 – Low–High (LH)

These are local pockets near developed clusters where a particular facility has low capacity compared to its saturated surroundings. For example, four facilities located in Wed: Norderoog, Studentenwohnheim Paul-Hertz-Haus, Süderoog, WH Augustenburger Platz with a total capacity of 471 students. Such areas are particularly important for analysis, as they may represent missed opportunities for growth.

Q3 – Low–Low (LL)

These are supply shortage clusters where both the property and its surroundings have low capacity. For example, student accommodation in the direct vicinity of the Technical University of Berlin (Insider Apartments Leibniz Straße, WH Fraunhoferstraße, WH Hardenbergstraße). Such areas indicate structurally underserved areas – potential territories for development.

Q4 – High–Low (HL)

These are spatial anomalies where one large property appears in an environment where neighbouring areas have low supply. For example, Student Village Berlin Adlershof, is an early indicator of the development of new student areas.

The graph shows that the distribution of student housing capacity in Berlin includes both stable high-density clusters (HH) and structural gaps (LL), as well as a number of spatial anomalies (HL, LH), indicating uneven market development. These patterns describe indicators of the student housing crisis.

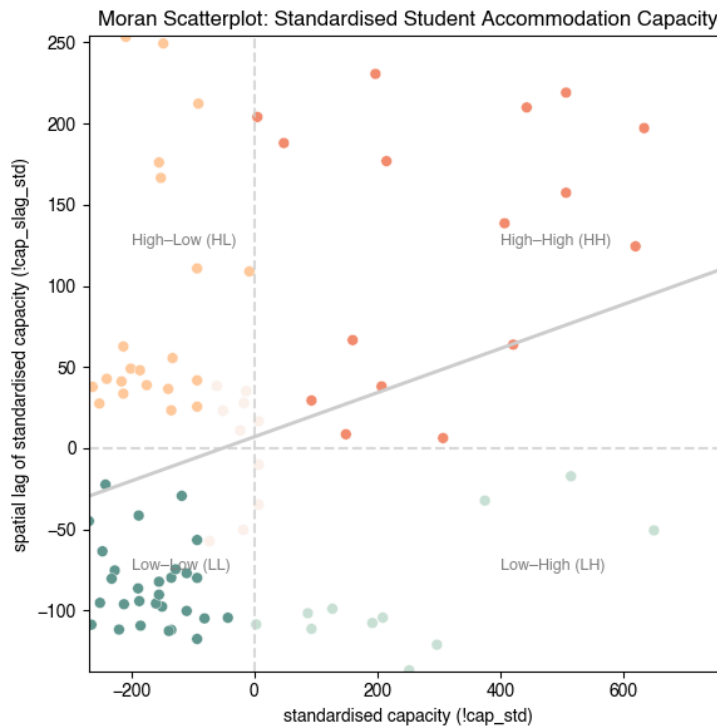


Figure 24. Student Accommodation Capacity Moran Scatterplot

The map (Figure 25) visualises the results of global spatial autocorrelation, which allows spatial clusters of student housing capacity to be identified.

In addition to identifying point clusters, the map clearly demonstrates the structural difference between East and West Berlin, which persists decades after the city's reunification. The map reveals clear spatial segregation.

East Berlin shows a high concentration of large student complexes and compact clusters. In the districts of Lichtenberg, Marzahn-Hellersdorf, and parts of the central districts, large halls of residence (HH) are spatially clustered, indicating strategically strong market areas. Individual high-capacity facilities (HL) are located in areas with low supply (e.g., in the south-east of the city). This is partly due to the historical legacy of the former East Berlin, where student residences were actively developed as part of the social infrastructure.

Western Berlin, on the other hand, is characterised by a predominance of stable areas with low capacity and a shortage of supply. Areas of systemic shortage are concentrated mainly in Charlottenburg-Wilmersdorf, Tempelhof-Schöneberg, Neukölln, and Steglitz-Zehlendorf. Properties with low capacity are surrounded by large student complexes, most often in areas where local conditions or urban planning restrictions hinder expansion, such as in Mitte. This reflects a market that is more oriented towards the private rental sector, where student accommodation has developed more slowly.

The spatial distribution of student housing in Berlin remains uneven and historically conditioned: the east has concentrated large, state-owned housing clusters, while the west faces a systemic supply shortage. This asymmetry increases pressure on rental markets in western districts and serves as a key signal of the ongoing student housing crisis.

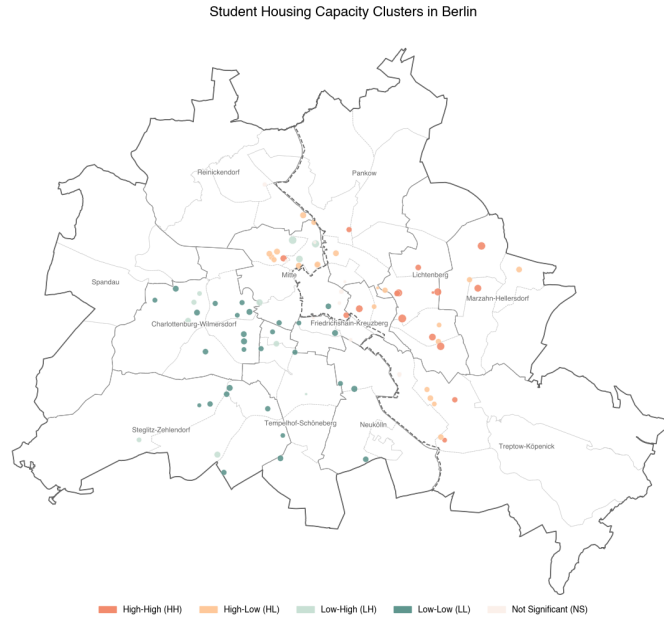


Figure 25. Student Accommodation Capacity Clusters in Berlin

3.9.7 Data Export

The student accommodation data obtained from global spatial autocorrelation was stored for further market analysis.

```
ef = r'../outputs/2.4_global_spatial_autocorrelation/pbsa_capacity_analysis.gpkg'
pbsa_c.to_file(ef, driver="GPKG")
```

3.10 Higher Education Universities International Demand

Similarly, an analysis of the spatial distribution of higher education university student demand across the city was conducted.

3.10.1 Data Import

```
uf = r'../inputs/2.4_global_spatial_autocorrelation/uni_c.gpkg'
uni_c = gpd.read_file(uf)
```

3.10.2 Standardisation

Since the research focused on international students, global autocorrelation will focus on number of international students (!s_aus), which measures the market demand.

```
uni_c["!s_aus_std"] = uni_c["!s_aus"] - uni_c["!s_aus"].mean()
```

3.10.3 Moran's I Scatterplots Test & K-Nearest Neighbor Test

Moran's I Scatterplots (Figure 26) demonstrate how the spatial structure of international student demand changes as the number of nearest neighbours increases (`k_values = list(range(1, 21))`) (Table 10). The main conclusion is that, unlike the student housing market, demand from universities does not form stable spatial clusters. Most points in all graphs are concentrated near zero on the standardised international demand axis

(`!s_aus_std`), indicating that neighbouring universities do not have a strong influence on demand values.

Q1 – High–High (HH)

Unlike student dormitories, there are virtually no high-density clusters because large universities are located far from each other, and their demand values do not correlate with each other, i.e., universities attract foreign students regardless of their location.

Q2 – Low–High (LH)

These are local pockets near universities with a large number of foreign students, where a particular university has low demand compared to its saturated surroundings.

Q3 – Low–Low (LL)

For most values of the number of nearest neighbours (k), there is a high concentration of universities with a low number of foreign students surrounded by similar institutions. This indicates a generally uniform low density of international student demand across the city.

Q4 – High–Low (HL)

Several universities with very high numbers of international students are isolated “peaks of demand”, not forming full-fledged spatial clusters.

In conclusion, Moran’s I scatterplots illustrate that the demand generated by international students is structurally dispersed and does not form clusters across the city.

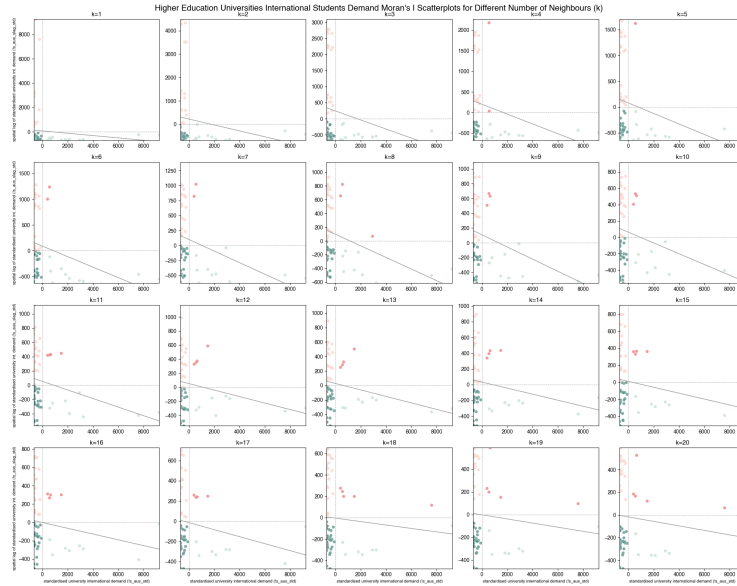


Figure 26. Higher Education Universities International Students Demand Moran’s I Scatterplots for Different Number of Neighbors (k)

```
np.random.seed(55)

k_values = [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
morans_I = []
p_values = []

for k in k_values:
    w = libpysal.weights.KNN.from_dataframe(uni_c, k=k)
    w.transform = "R"
    moran = Moran(uni_c['!s_aus'], w)
```

```

1820     morans_I.append(moran.I)
1821     p_values.append(moran.p_sim)
1822
1823 # Create DataFrame
1824 uni_k_test = pd.DataFrame({
1825     'k': k_values,
1826     "Moran's I": morans_I,
1827     'P -value': p_values
1828 })
1829
1830 uni_k_test

```

	k	Moran's I	P-value
0	2	-0.126627	0.022
1	3	-0.140827	0.003
2	4	-0.119096	0.003
3	5	-0.101951	0.009
4	6	-0.102231	0.004
5	7	-0.096559	0.004
6	8	-0.084318	0.014
7	9	-0.082463	0.005
8	10	-0.065725	0.039
9	11	-0.060102	0.061
10	12	-0.046249	0.138
11	13	-0.044825	0.134
12	14	-0.038477	0.221
13	15	-0.034638	0.255
14	16	-0.031578	0.280
15	17	-0.034663	0.224
16	18	-0.016068	0.472
17	19	-0.018201	0.477
18	20	-0.019882	0.448

1832 *Table 10. Higher Education Universities International Students Demand Moran's*
1833 *I index and P-value for Different Number of Neighbors (k)*

1834 The Moran's I distribution graph and P-value (Figure 27) show that the P-value
1835 takes acceptable values ($P < 0.05$) when k is between 2 and 10. Since we are talking about
1836 global spatial autocorrelation, the purpose of which is to identify patterns throughout
1837 Berlin, it is necessary to choose the maximum value of k. At the same time, the P-value
1838 when choosing 10 nearest neighbours is critically close to the significance threshold. There-
1839 fore, the previous value **k=9** was chosen.

Figure 27. Higher Education Universities International Students Demand Moran's I index and P-value for Different Number of Neighbors (k)

3.10.4 Significance Threshold Test

By analogy with student accommodation capacity, the significance threshold was calculated as half of the median of the absolute value (`median_abs * 0.5`).

Option 1 - Percentile -based threshold: 679.990
(75th percentile of absolute values)
Option 2 - Standard deviation -based threshold: 828.251
(0.5 * standard deviation)
Option 3 - Fixed threshold: 550.000
(Fixed value for your data range)
Option 4 - Median -based threshold: 304.745
(0.5 * median of absolute values)

Test 1: threshold = 679.990
NS: 54 (76.1%)
HH: 2 (2.8%)
LH: 6 (8.5%)
LL: 5 (7.0%)
HL: 4 (5.6%)

Test 2: threshold = 828.251
NS: 64 (90.1%)
HH: 2 (2.8%)
LH: 5 (7.0%)
LL: 0 (0.0%)
HL: 0 (0.0%)

Test 3: threshold = 550.000
NS: 23 (32.4%)
HH: 4 (5.6%)
LH: 7 (9.9%)
LL: 23 (32.4%)
HL: 14 (19.7%)

Test 4: threshold = 304.745
NS: 5 (7.0%)
HH: 5 (7.0%)
LH: 7 (9.9%)
LL: 34 (47.9%)
HL: 20 (28.2%)

3.10.5 Spatial Weights & Moran's I & Spatial Lag

```
# Generate spatial weights from the GeoDataFrame
w = KNN.from_dataframe(uni_c, k=9)
w.transform = "R" # Row -standardize weights

values = uni_c['!s_au$']

moran = Moran(values, w)
print(f"Moran's I: {moran.I:.3f}")
```

```
print(f"P -value: {moran.p_sim:.3f}")
```

```
Moran's I: -0.082
```

```
P -value: 0.009
```

Moran's I (Moran's I: -0.082) indicates a weak negative spatial autocorrelation. In the context of urban planning, this indicates that higher education universities do not create spatial clusters; in other words, they are unevenly distributed throughout the city.

P-value (P-value: 0.007) indicates that the probability of obtaining such a result by chance is 0.7%, i.e., the result obtained is statistically significant.

3.10.6 Spatial Lag & Standardised Spatial Lag

```
# Add spatial lag and standardized columns
uni_c["!s_aus_slag"] = lag_spatial(w, uni_c["!s_aus"])
uni_c["!s_aus_std"] = uni_c["!s_aus"] - uni_c["!s_aus"].mean()
uni_c["!s_aus_slag_std"] = lag_spatial(w, uni_c["!s_aus_std"])
```

The maps (Figure 28) visualise the different stages of preparing international demand data for spatial analysis.

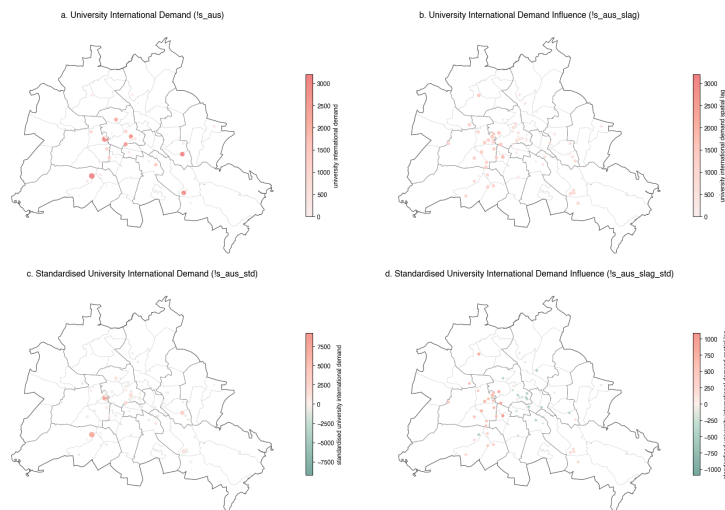


Figure 28. Stages of Preparing Higher Education Universities International Students Demand Data

Moran's graph (Figure 29) visualises the spatial relationships between standardised international university demand (!s_aus_std) and its spatial lag (!s_aus_slag_std). The division into four quadrants allows us to identify characteristic spatial patterns of demand distribution and understand where areas of high competition or, conversely, a lack of educational attractiveness are forming.

Q1 – High–High (HH)

These are clusters of high international demand, where universities with a large proportion of foreign students are located next to similarly significant neighbours. Such areas form stable 'magnets' for international students, for example, the College of Business and Law (HWR – 1,336 foreign students), the University of the Arts Berlin (UKB – 1,244 foreign students), and ESCP Business School Berlin (ESCP – 1,092 foreign students). At the same time, two of the three universities in the HH quadrant are publicly owned (HWR and UKB) and located in Willensdorf, while the private ESCP University is lo-

cated in Charlottenburg. These three universities form a powerful spatial core of global education in West Berlin.

Q2 – High–Low (HL)

Spatial anomalies, where a university with high international demand is located among neighbouring universities with low values. This is the largest cluster among all universities. It mainly consists of individual private universities that have a localised impact and signal the beginning of the formation of new educational centres. Among the universities belonging to the HL cluster but funded by the state are the Technical University of Berlin, the Protestant University of Applied Sciences Berlin, and the Charité – University Medicine Berlin campuses in the south-west of the city.

Q3 – Low–Low (LL)

Clusters of low demand, where both the university itself and its surroundings attract relatively few foreign students. LL zones indicate areas where international educational activity is weak, such as Pankow, Lichtenberg, Friedrichshain-Kreuzberg, and Neukölln. Among the reasons for unrealised educational potential, one can assume a lack of infrastructure, limited academic activity, or weak integration into the city's structure.

Q4 – Low–High (LH)

These are local failures, where a university demonstrates low international demand, but its environment demonstrates high demand. At the same time, seven of the nine universities in the LH quadrant are publicly owned. Universities such as the Free University of Berlin and Humboldt University of Berlin have the potential to participate in existing educational clusters, but are not yet strategically integrated into them.

This distribution indicates competition between universities, the existence of ‘hot spots’ of international activity, and, at the same time, structural territorial gaps. All these patterns are important for studying the student housing crisis, as international demand directly correlates with increased pressure on the rental market, especially in highly attractive local clusters.

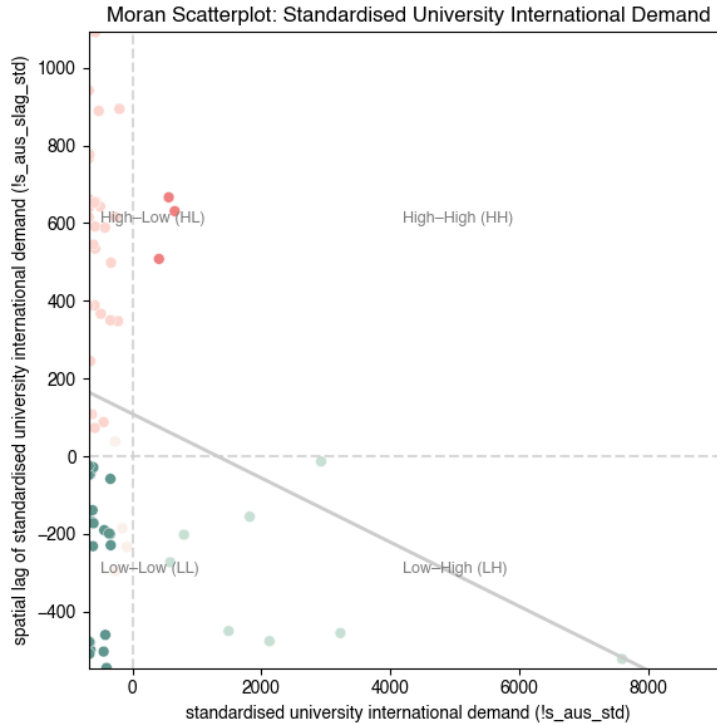


Figure 29. Higher Education Universities International Students Demand Moran Scatterplot

The map showing the spatial distribution of international student demand in Berlin (Figure 30) reveals that, unlike the supply of student housing, demand does not form stable clusters of high concentration. Instead of the expected spatial agglomeration around large public universities, there is a dispersion (scattering) of values. In other words, universities with high numbers of international students are not grouped geographically, but rather distributed in a fragmented manner and often separated from similar neighbours.

The east-west divide highlights Berlin's historical asymmetry: the West traditionally concentrates international academic demand, while the East remains unattractive in an international context.

The demand for international students is distributed more chaotically and does not correlate with the spatial patterns of student housing supply, which exacerbates the structural imbalance between the housing market and the city's academic infrastructure.

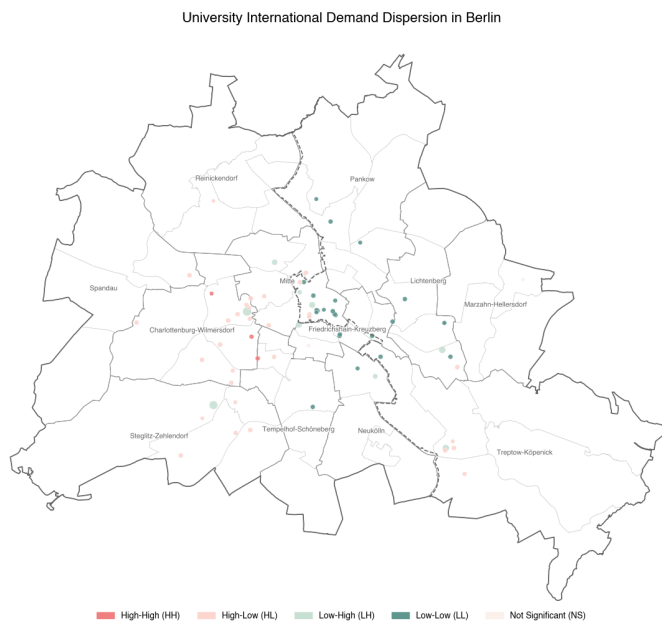


Figure 30. Higher Education Universities International Demand Dispersion in Berlin

3.10.7 Data Export

The data on higher education universities' international students demand obtained as a result of global spatial autocorrelation was collected for further market analysis.

```
hf = r'../outputs/2.4_global_spatial_autocorrelation/uni_int_demand_analysis.gpkg'
uni_c.to_file(hf, driver="GPKG")
```

4 3. Student Housing Market Imbalance

4.1 3.1 Overview

Chapter Student Housing Market Imbalance analyses the imbalance between university demand and student housing capacity.

4.1.1 *What? Student Housing Market Imbalance*

Student Housing Market Imbalance can be understood as a set of statistically significant patterns that constitute mismatches between student housing capacity supply and university demand for higher education.

Among these patterns, one of the most important is pressure on the student housing market. Based on the conclusions drawn in the Capacities, Demand & Spatial Distribution Chapter, it can be assumed that pressure on the market is not evenly distributed.

4.1.2 *Why? Assess Spatial Demand–Supply Imbalances*

The aim of this chapter is to assess the student housing market imbalances, identify areas where demand exceeds supply by answering the following research questions:

How do student university demand compare to student housing market housing capacities?

Where do spatial patterns of demand and supply generate elevated market pressure?

The answers to these questions are necessary to understand how and where pressure on the student housing market is formed in the city.

4.1.3 *How? Spatial Data Analysis*

Unlike the Capacities, Demand & Spatial Distribution Chapter, where the analysis of the two systems was conducted sequentially, this chapter aimed to combine the data using spatial join and conduct a comparative analysis of the student housing market according to the Central Place Theory outlined in the Theoretical Framework chapter.

Spatial Join

A spatial join combines represented geographic objects and their associated attributes based on a spatial relationship (University of West Florida and Morgan, 2023). In other words, spatial joins allow you to combine multiple data sets based on spatial characteristics. For example, you can calculate the total student housing capacity and university demand for the Mitte district.

However, such a rough approach is not suitable for a detailed analysis of the student housing market, as Berlin's districts vary greatly in size, so simple spatial intersection and subsequent aggregation distort the real picture of market pressure. In order to increase the accuracy of the spatial join, an isochrone accessibility analysis was performed.

Isochrone Analysis

An isochrone in geography and urban planning is a polygon that depicts the area accessible from a point within a certain time threshold, i.e., isochrones calculate accessibility in terms of time (OpenStreetMap Wiki, 2025). This allows administrative district boundaries to be replaced with more functional zones - zones of student mobility. Thus, isochrone analysis eliminates the problem of mismatched district areas and focuses on the actual patterns of student movement.

Based on isochrones, it is possible to perform a spatial join between university demand and student housing capacity. However, such isochrones often overlap with each other, which leads to the problem of overlap. One solution to the problem of overlap is rasterisation.

Rasterisation

Rasterisation converts vectors into rasters. In Geospatial Science, it usually means converting vector geometry to raster geometry. It can be used to summarise data from several objects. Since accessibility isochrones intersect with each other, rasterisation — the conversion of geometry into a regular grid — is performed to match spatially heterogeneous supply and demand data.

These research methods allow for a parallel assessment of market areas and show how spatial patterns exacerbate or mitigate the shortage of student housing, forming or neutralising market pressure.

Thus, the chapter demonstrates the geography of the student housing market, identifying the most problematic areas.

4.2 3.2 Central Place Analysis

The chapter Student Housing Market: Capacities, Demand, and Spatial Distribution provides a consistent analysis of two systems: student accommodations and higher education universities.

To proceed to market analysis, it is necessary to examine how these two systems interact with each other, forming the student accommodation market. Central Place Theory, outlined in the chapter Theoretical Framework, assists in navigating the analysis process.

4.2.0.1 Set Up Import Libraries

```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib as mpl
import geopandas as gpd
from shapely.geometry import Point, MultiPoint
from pyproj import CRS
from scipy.stats import norm
from matplotlib.colors import LinearSegmentedColormap
import matplotlib.path_effects as path_effects
import string
import requests
import time
from shapely.geometry import Polygon
import seaborn as sns
from shapely.geometry import MultiPolygon
```

Berlin Spatial Boundaries

```
pf = r"../inputs/3.2_central_place_analysis/lor_pgr.gpkg"
pgr = gpd.read_file(pf)
pgr.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

bf = r"../inputs/3.2_central_place_analysis/adm_bz.gpkg"
bz = gpd.read_file(bf)
bz.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

of = r"../inputs/3.2_central_place_analysis/adm_out.gpkg"
out = gpd.read_file(of)
out.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)
```

The analysis of student accommodation capacity and higher education universities' international student demand data, obtained as a result of Global Spatial Autocorrelation, served as the basis for student housing market research.

```
ef = r'../inputs/3.2_central_place_analysis/pbsa_capacity_analysis.gpkg'
pbsa_cap = gpd.read_file(ef)

mf = r'../inputs/3.2_central_place_analysis/uni_int_demand_analysis.gpkg'
uni_dem = gpd.read_file(mf, driver="GPKG")
```

4.3 Central Places Definition

In Central Place Theory King (1986) argue that the idea that there is a smooth and regular progression in urban centres from the smallest up to the largest center - is not a very useful one, and that greater insight can be gained into the nature of an urban pattern by referring to discrete population-size classes.

Taking this into account, central universities should be defined not as the ‘largest’ in terms of numbers, but as institutions that perform central functions in the urban system. Their significance is determined not so much by the scale of student enrolment as by their functional role, their position in the urban and educational hierarchy, and their ability to serve wider areas of attraction.

In order to determine which universities are considered central, the following concerns must be taken into account.

Firstly, based on spatial economics analysis of university clusters, only High–High (HH), Low–High (LH) and High–Low (HL) clusters are capable of generating sustained high demand in the housing market, fulfilling the functional roles of centres. Meanwhile, Low–Low (LL) clusters do not form centres of demand.

Secondly, the number of central universities must be limited, as 26 out of 52 universities cannot be central by definition.

Only by taking this into account can a quantitative test be conducted using the international student demand (!s_aus) threshold.

```
Option 1 - Percentile -based threshold: 414.000
(75th percentile of absolute values)
Option 2 - Standard deviation -based threshold: 828.251
(0.5 * standard deviation)
Option 3 - Fixed threshold: 550.000
(Fixed value for your data range)
Option 4 - Median -based threshold: 51.000
(0.5 * median of absolute values)
```

```
Test 1: threshold = 414.000
Total features: 15
Total demand: 40958.7
Features with cluster = 'NS': 0
Features with cluster = 'HH': 3
Features with cluster = 'LH': 9
Features with cluster = 'HL': 3
Features with cluster = 'LL': 0
```

```
Test 2: threshold = 828.251
Total features: 12
Total demand: 39607.76
Features with cluster = 'NS': 0
Features with cluster = 'HH': 3
Features with cluster = 'LH': 9
Features with cluster = 'HL': 0
Features with cluster = 'LL': 0
```

```
Test 3: threshold = 550.000
Total features: 12
Total demand: 39607.76
```

```

2124 Features with cluster = 'NS': 0
2125 Features with cluster = 'HH': 3
2126 Features with cluster = 'LH': 9
2127 Features with cluster = 'HL': 0
2128 Features with cluster = 'LL': 0
2129
2130 Test 4: threshold = 51.000
2131 Total features: 45
2132 Total demand: 46240.259999999995
2133 Features with cluster = 'NS': 0
2134 Features with cluster = 'HH': 3
2135 Features with cluster = 'LH': 9
2136 Features with cluster = 'HL': 19
2137 Features with cluster = 'LL': 14

```

2138 In order to correctly interpret the Threshold Test, it is necessary to pay attention
 2139 to the number of universities and their distribution across four spatial clusters.

2140 **Option 1 - Percentile-based threshold: 414,000**

2141 With a percentile-based threshold, the upper quarter of the distribution is selected, tak-
 2142 ing into account three spatial clusters - HH, LH and HL. It also significantly reduces the
 2143 sample from 52 to 15 universities, fixing the central nodes of the pair. Therefore, this
 2144 threshold creates a threshold that reflects the real structure of the student housing mar-
 2145 ket. (threshold = threshold_percentile).

2146 **Option 2 - Standard deviation-based threshold: 828.251**

2147 **Option 3 - Fixed threshold: 550.000**

2148 Standard deviation-based and fixed thresholds exclude HL clusters that are early indi-
 2149 cators of the development of new student attraction zones.

2150 **Option 4 - Median-based threshold: 51.000**

2151 With a median-based threshold, almost all universities become 'central,' which deval-
 2152 ues the category itself.

2153 Thus, using the spatial clusters and percentile-based threshold filter, 15 universi-
 2154 ties (Table XX) were identified as central.

2155 Thus, using the spatial clusters and percentile-based threshold filter, 15 universi-
 2156 ties (Table 11) were identified as central.

```

2157 central_places = uni_dem[
2158     (uni_dem['cluster_aus'].isin(['HL', 'HH', 'LH'])) &
2159     (uni_dem['!s_aus'] >= threshold_percentile)
2160 ]

```

```

2161 central_places[[
2162     "code",
2163     "abr",
2164     "name",
2165     "own",
2166     "cluster",
2167     "!s_aus"
2168 ]]

```


	code	abr	name	own	cluster	!s_ aus
2	HS0180	CU	Charite - Univer- sitätsmedi- zin Berlin	Public University	LH	1266.50
3	HS0180	CU	Charite - Univer- sitätsmedi- zin Berlin	Public University	HL	455.94
6	HS0200	HU	Humboldt- Universität Berlin	Public University	LH	3608.46
7	HS0200	HU	Humboldt- Universität Berlin	Public University	LH	2806.58
16	HS1380	FU	Freie Uni- versität Berlin	Public University	LH	8280.02
21	HS1690	TU	Technische Univer- sität Berlin	Public University	LH	9914.01
23	HS1700	ESCP	ESCP- Europe Wirtschafts- hochschule Berlin	Privat University	HH	1092.00
30	HS2290	BIU	Berlin Int. Univ. of Ap- plied Sciences	Privat University	HL	417.00
36	HS2790	UAB	Universität der Kün- ste Berlin	Public University	HH	1243.34
39	HS3270	GIU	German Internat- ional University Berlin	Privat University	HL	478.00
45	HS5802	UEAS	U of Eu- rope for Appl.Sc.Potsdam i.Berlin	Privat University	LH	2501.00
52	HS7690	HTW	Hochschule für Tech- nik und Wirtschaft Berlin	Public University	LH	3908.01
55	HS7710	BTS	Berliner Hochschule für Tech- nik	Public University	LH	2171.96
56	HS7721	HWR_BA	HWR Berlin, FB Wirtschaft	Public University	HH	1335.88

2170 *Table 11. Central Higher Education Universities*2171 **4.4 Market Areas Definition**

2172 Once the central universities have been identified, the next step is to define mar-
 2173 ket areas that jointly reflect data on higher education universities' international demand
 2174 and student accommodation capacity. As noted in the Overview chapter, there are sev-
 2175 eral approaches to combining these data. Here are three primary approaches.

2176 The first approach is to use Berlin's existing administrative boundaries, such as Life-
 2177 Oriented Spaces (LOR - Lebensweltlich Orientierte Räume) or their sub-level Neighbour-
 2178 hood Districts (BZR - Bezirksregionen). However, this method creates significant gaps:
 2179 many areas end up "empty" because they have neither universities nor student accom-
 2180 modation. This indicates that neighbourhood districts are too small for analysing the
 2181 urban supply and demand system. The next level, Prognosis Areas (PGR), on the con-
 2182 trary, is too large and aggregates too heterogeneous areas. Thus, the existing life-oriented
 2183 spaces do not reflect the real geography of the student housing market.

2184 The second approach is to build buffer zones around central universities, creating
 2185 new geometries for spatial data integration. However, due to Berlin's heterogeneous mor-
 2186 phology, it is impossible to correctly determine a universal buffer distance. In addition,
 2187 this method results in all market areas being the same size, which contradicts the ac-
 2188 tual structure of urban accessibility and central place theory.

2189 The third approach is to use accessibility isochrones, i.e., zones formed on the ba-
 2190 sis of actual travel time. Unlike administrative boundaries or fixed buffers, isochrones
 2191 reflect the actual spatial relationship and student mobility. This allows for the creation
 2192 of flexible and functionally relevant market areas in which supply and demand can be
 2193 matched more accurately.

2194 To construct accessibility isochrones, the study used the Travel Time API service,
 2195 which allows modelling reachability zones based on real-time travel. A distinctive fea-
 2196 ture of the Travel Time service is the ability to generate isochrones based on public trans-
 2197 port estimates (``type': `public_transport'`).

2198 Below is an example of using the Travel Time API to create 15-minute accessibil-
 2199 ity zones (`travel_time = 900`).

```
2200 # url = 'https://api.traveltimeapp.com/v4/time -map'
2201
2202 # tt_app_id = 'my_app_id'
2203 # tt_api_key = 'my_api_key'
2204
2205 # headers = {
2206 #     'content -type': 'application/json',
2207 #     'x -application -id': tt_app_id,
2208 #     'x -api -key': tt_api_key
2209 # }
2210
2211 # travel_time = 900 # 15 minutes
2212
2213 # for idx, row in central_places.iterrows():
2214 #     lon = float(row['lon'])
2215 #     lat = float(row['lat'])
2216
2217 #     payload = {
2218 #         "arrival_searches": [
```

```

2219 #         {
2220 #             "id": f"iso_{idx}",
2221 #             "coords": {"lat": lat, "lng": lon},
2222 #             "arrival_time": "2025 -08 -13T09:00:00+01:00",
2223 #             "travel_time": travel_time,
2224 #             "transportation": {"type": "public_transport", "disable_border_crossing": True}
2225 #         }
2226 #     ]
2227 # }
2228
2229 #     resp = requests.post(url, json=payload, headers=headers)
2230 #     print(f"request {idx}: {resp.status_code} {resp.reason} for location {[lon, lat]}")
2231
2232 #     if resp.ok:
2233 #         data = resp.json()
2234 #         for res in data.get('results', []):
2235 #             search_id = res.get('search_id')
2236 #             for shp in res.get('shapes', []):
2237 #                 shell = shp.get('shell', [])
2238 #                 if not shell:
2239 #                     continue
2240 #                 coords = [(pt['lng'], pt['lat']) for pt in shell]
2241 #                 if coords and coords[0] != coords[-1]:
2242 #                     coords.append(coords[0])
2243 #                 poly = Polygon(coords)
2244 #                 g = gpd.GeoDataFrame([{'geometry': poly, 'tt_id': search_id,
2245 #                                         'origin_lon': lon, 'origin_lat': lat}],
2246 #                                     crs='EPSG:4326')
2247 #                 iso_gdf = gpd.GeoDataFrame(pd.concat([iso_gdf, g],
2248 #                                                         ignore_index=True,
2249 #                                                         sort=False))
2250
2251 #                 time.sleep(15)
2252
2253 #     print(f"got {len(iso_gdf)} isochrone polygons.")
2254
2255 # iso_gdf["geometry"] = iso_gdf["geometry"].buffer(0)
2256 # iso_gdf.to_crs(epsg=25833)
2257
2258 # ef = r"../outputs/3.2_central_place_analysis/iso_without_data_15_min.gpkg"
2259 # iso_gdf.to_file(ef, driver="GPKG")

```

2259 4.5 Travel Time Test

2260 Since isochrones are sensitive to travel time, a travel time test was conducted (Fig-
2261 ure 31) to determine market areas where travel time varied from 10 to 35 minutes with
2262 5-minute intervals (isochrone_times = [10, 15, 20, 25, 30, 35]).

```

2263 # Define isochrone time ranges
2264 isochrone_times = [10, 15, 20, 25, 30, 35]
2265 base_path = "../outputs/3.2_central_place_analysis"
2266
2267 # Initialize results storage
2268 results_summary = []
2269 pbsa_market_summary = None

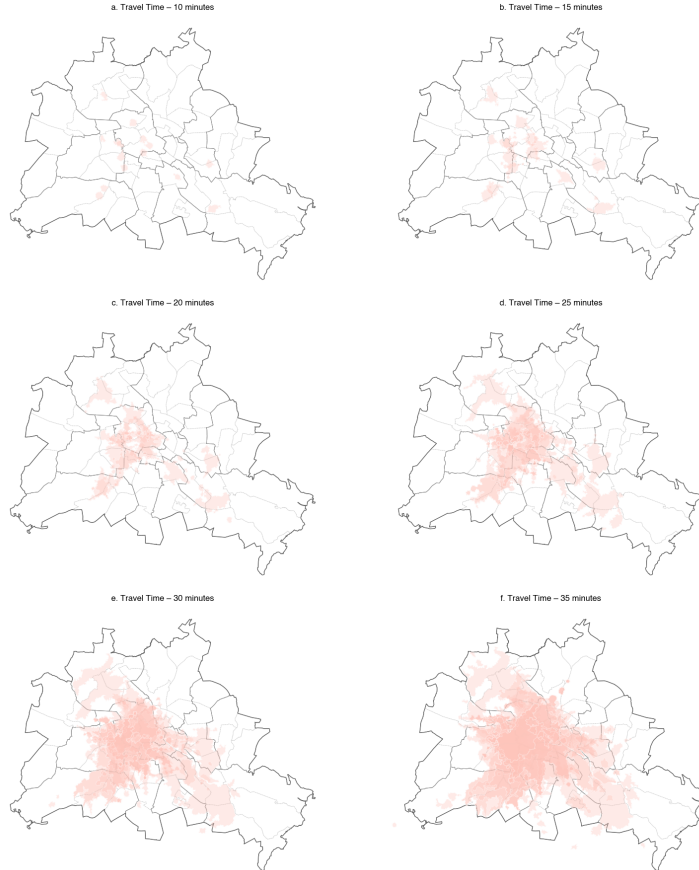
```

```

2270 iso_dem_ready = None
2271 pbsa_cap_ready = None
2272 summary_df = None

2273 total_capacity_complete = pbsa_cap["!cap"].sum()
2274 total_demand_complete = uni_dem["!s_aus"].sum()
2275
2276 pbsa_cap = pbsa_cap.to_crs("EPSG:25833")
2277 central_places = central_places.to_crs("EPSG:25833")

```



2278

2279 *Figure 31. Travel-Time API Test*

2280 When constructing isochrones, it is important to correctly interpret the allocation
 2281 of student housing capacity and higher education universities demand.

2282 The allocation of student accommodations is performed using a spatial join between
 2283 points and isochrone polygons. Since one student accommodations facility may be lo-
 2284 cated within the catchment area of several universities, directly assigning its full capac-
 2285 ity to each area would lead to a significant overestimation of supply. To avoid double-
 2286 counting, each student accommodation facility first receives information about the num-
 2287 ber of isochrones in which it is located, after which its capacity is divided equally among
 2288 all relevant areas. The adjusted values are then aggregated at the market area level. This
 2289 approach allows for the correct distribution of available housing in accordance with the
 2290 actual spatial structure of accessibility, taking into account the overlap of the areas of
 2291 influence of different universities.

2292 Similar to the distribution of dormitories, university demand (number of German
 2293 students (!s_de), number of foreign students (!s_aus) and total number of students (!s_all))

is assigned to isochrones based on the principle of spatial intersection. Universities may also lie within several isochrones, especially when accessibility times are long (30–35 minutes). Therefore, the demand for each university is divided among all the zones within which it falls. This method eliminates artificial doubling of demand and creates a balanced distribution that reflects the real impact of universities on the housing market within their accessibility zones. The final demand values are aggregated at the market area level and can be compared with the distributed student accommodation capacity.

```

2294 for time_min in isochrone_times:
2295     # Load isochrone data
2296     iso_file = f"{base_path}/iso_without_data_{time_min}_min.gpkg"
2297     iso_gdf = gpd.read_file(iso_file)
2298
2299     # Reproject immediately
2300     if iso_gdf.crs != "EPSG:25833":
2301         iso_gdf = iso_gdf.to_crs("EPSG:25833")
2302
2303     # Build multipolygons per tt_id
2304     iso_multi = (
2305         iso_gdf.groupby(["tt_id", "origin_lon", "origin_lat"])
2306         .geometry.apply(lambda g: MultiPolygon([geom for geom in g]))
2307         .reset_index()
2308     )
2309     iso_multi_gdf = gpd.GeoDataFrame(iso_multi, geometry="geometry", crs="EPSG:25833")
2310
2311     # Prepare central places (no geometry needed for merge)
2312     central_places_no_geom = central_places.drop(columns="geometry")
2313     central_places_no_geom["lon"] = central_places_no_geom["lon"].astype(float)
2314
2315     # Merge on origin_lon/lon
2316     iso_multi_gdf["origin_lon"] = iso_multi_gdf["origin_lon"].astype(float)
2317     iso_dem = iso_multi_gdf.merge(
2318         central_places_no_geom,
2319         left_on="origin_lon",
2320         right_on="lon"
2321     )
2322     iso_dem = gpd.GeoDataFrame(iso_dem, geometry="geometry", crs="EPSG:25833")
2323     iso_dem = iso_dem.drop(columns=["origin_lon", "url"])
2324     iso_dem["geometry"] = iso_dem["geometry"].buffer(0) # fix invalids
2325
2326     # From here everything is already in EPSG:25833
2327     pbsa_cap_ready = pbsa_cap.copy()
2328     if "pbsa_id" not in pbsa_cap_ready.columns:
2329         pbsa_cap_ready["pbsa_id"] = pbsa_cap_ready.index
2330
2331     iso_dem_ready = iso_dem.copy()
2332     if "iso_id" not in iso_dem_ready.columns:
2333         iso_dem_ready["iso_id"] = iso_dem_ready.index
2334
2335     uni_dem = uni_dem.copy()
2336     if 'uni_id' not in uni_dem.columns:
2337         uni_dem['uni_id'] = uni_dem.index
2338
2339     # PBSA allocation
2340     join_pbsa = gpd.sjoin(pbsa_cap_ready, iso_dem_ready, how="inner", predicate="intersects")

```

```

2348 n_iso_pbsa = join_pbsa.groupby("pbsa_id")["iso_id"].nunique().rename("n_iso_pbsa")
2349 join_pbsa = join_pbsa.merge(n_iso_pbsa, left_on="pbsa_id", right_index=True, how="left")
2350 cap_cols = ["!cap"]
2351 for c in cap_cols:
2352     join_pbsa[f"{c}_sum"] = join_pbsa[c] / join_pbsa["n_iso_pbsa"]
2353 iso_alloc_pbsa = (
2354     join_pbsa.groupby("iso_id")[[f"{c}_sum" for c in cap_cols]]
2355     .sum().reset_index()
2356 )
2357
2358 # - - - UNI allocation - - -
2359 join_uni = gpd.sjoin(uni_dem, iso_dem_ready, how='inner', predicate='intersects')
2360 n_iso_uni = join_uni.groupby('uni_id')['iso_id'].nunique().rename('n_iso_uni')
2361 join_uni = join_uni.merge(n_iso_uni, left_on='uni_id', right_index=True, how='left')
2362 dem_cols = ['!s_aus_left', '!s_de_left', '!s_all_left']
2363 for c in dem_cols:
2364     join_uni[f'{c}_sum'] = join_uni[c] / join_uni['n_iso_uni']
2365 iso_alloc_uni = (join_uni.groupby('iso_id')[[f'{c}_sum' for c in dem_cols]]
2366                 .sum()
2367                 .reset_index())
2368
2369 # Merge results
2370 market_area = iso_dem_ready.merge(iso_alloc_pbsa, on="iso_id", how="left")
2371 market_area = market_area.merge(iso_alloc_uni, on="iso_id", how="left")
2372
2373 market_area = market_area.rename(columns={
2374     "!s_aus_left_sum": "!s_aus_sum",
2375     "!s_de_left_sum": "!s_de_sum",
2376     "!s_all_left_sum": "!s_all_sum",
2377 })
2378
2379 # Ratios
2380 market_area["capacity_demand_ratio"] = (
2381     market_area["!cap_sum"] / market_area["!s_aus_sum"].replace(0, np.inf)
2382 )
2383 market_area["demand_capacity_pressure"] = (
2384     market_area["!s_aus_sum"] / market_area["!cap_sum"].replace(0, np.inf)
2385 )
2386 market_area["area"] = market_area.geometry.area / 1e6 # in sq km
2387
2388 # PBSA count
2389 pbsa_pts = pbsa_cap_ready.copy()
2390 wb = gpd.sjoin(pbsa_pts, market_area[["iso_id", "geometry"]], how="inner", predicate="within")
2391 pbsa_counts = wb.groupby("iso_id").size()
2392 market_area["pbsa_count"] = market_area["iso_id"].map(pbsa_counts).fillna(0).astype(int)
2393
2394 # Uni Count
2395 uni_pts = uni_dem.copy() # again, geometry is already active
2396 # Spatial join: UNI point within isochrone polygon
2397 wb = gpd.sjoin(uni_pts, market_area[["iso_id", "geometry"]], how='inner', predicate='within')
2398 # Count UNI per iso_id
2399 uni_counts = wb.groupby('iso_id').size()
2400 # Attach to market_area - fill missing with 0
2401 market_area['uni_count'] = market_area['iso_id'].map(uni_counts).fillna(0).astype(int)
2402

```

```

2403     # Travel time
2404     market_area["travel_time_min"] = time_min
2405
2406     #Cleaning
2407     market_area['cluster'] = market_area['cluster'].map({
2408         'HH': 'High -High (HH)',
2409         'HL': 'High -Low (HL)',
2410         'LH': 'Low -High (LH)',
2411         'LL': 'Low -Low (LL)',
2412         'NS': 'Not Significant (NS)'
2413     })
2414     columns_to_drop = ['origin_lat', 'origin_lon', 'cluster_aus', 'iso_id', 'lon']
2415     market_area = market_area.drop(columns=columns_to_drop, errors='ignore')
2416
2417     # Save
2418     output_file = f"{base_path}/market_areas_{time_min}_min.gpkg"
2419     market_area.to_file(output_file, driver="GPKG")
2420
2421     # Summary stats
2422     total_capacity_inside = market_area["!cap_sum"].sum()
2423     total_demand_inside = market_area["!s_aus_sum"].sum()
2424
2425     uncovered_demand = total_demand_complete - total_demand_inside
2426     unused_capacity = total_capacity_complete - total_capacity_inside
2427
2428     results_summary.append({
2429         "travel_time_min": time_min,
2430         "total_capacity_inside": total_capacity_inside,
2431         "total_demand_inside": total_demand_inside,
2432         "uncovered_demand": uncovered_demand,
2433         "unused_capacity": unused_capacity,
2434     })

```

2435 By changing the time thresholds, it is possible to check the stability of market ar-
2436 eas: whether the structure of supply and demand zones changes or remains stable. To
2437 do this, at the final stage, a summary of key indicators is formed for each tested time
2438 of availability, allowing us to assess how effectively the formed market areas reflect the
2439 real structure of supply and demand. The table (Table 12) contains: the total capacity
2440 of dormitories that fall within the isochrones (`total_capacity_inside`) and the total
2441 university demand covered by these same areas (`total_demand_inside`).

```

2442 # Create summary DataFrame
2443 summary_df = []
2444
2445 summary_df = pd.DataFrame(results_summary).round(1)
2446
2447 summary_df.sort_values("travel_time_min", inplace=True)
2448
2449 summary_df

```

	travel_time_min	total_capacity	total_demand	inside_demand	inside_capacity
0	10	3001.0	32358.5	16098.3	22816.0
1	15	6191.0	44461.9	3994.9	19626.0
2	20	9089.0	45344.2	3112.6	16728.0
3	25	15195.0	46592.4	1864.4	10622.0
4	30	17841.0	46976.1	1480.6	7976.0
5	35	20635.0	47224.8	1232.0	5182.0

2450

2451

Table 12. Travel Time Summary Statistics

2452

2453

2454

```

output_path = r"../outputs/3.2_central_place_analysis/market_area_travel_time.xlsx"
with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
    summary_df.to_excel(writer, sheet_name='base')

```

2455

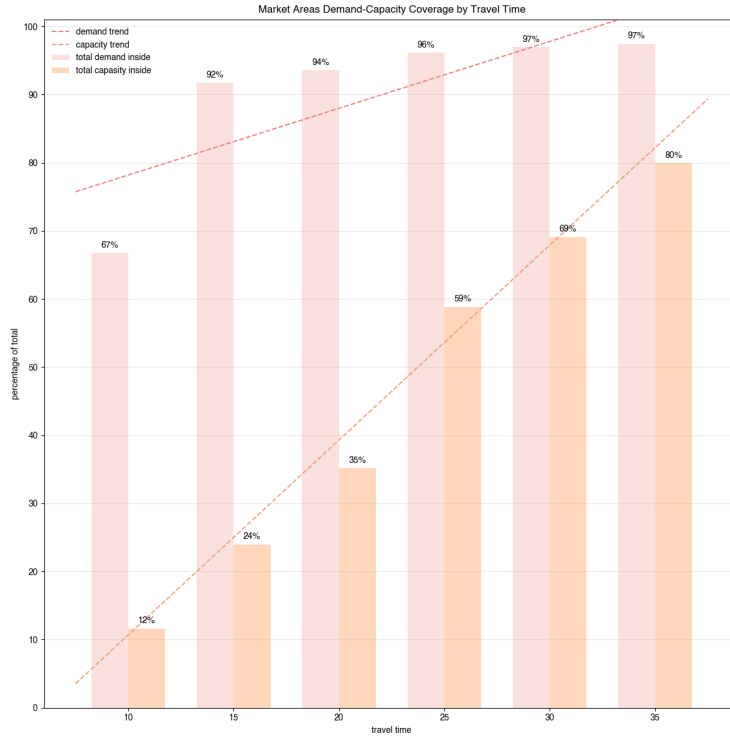
2456

2457

2458

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Comparing these indicators between different time thresholds (10–35 minutes) (Figure 32) allows us to determine the optimal radius of market influence of universities, as well as to identify structural imbalances between supply and demand associated with spatial constraints. This approach provides a quantitative basis for selecting the isochrone threshold and correctly forming market areas.



2460

2461

Figure 32. Market Areas Demand-Capacity Coverage by Travel Time

2462

2463

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2465

The results of the travel time test demonstrate that increasing the time threshold for accessibility significantly affects the degree of coverage of university demand and the capacity of student housing included in market areas. However, the effect grows unevenly, and it is precisely this dynamic that allows us to determine the optimal travel time value.

First, at thresholds of 10–25 minutes, there is a significant discrepancy between the share of demand covered and the share of capacity included. For example, at 25 minutes, the market already covers 96% of total demand, but only 59% of housing capacity. This imbalance indicates that with short isochrones, a significant portion of dormitories remain formally ‘outside the market,’ which leads to artificially inflated estimates of the deficit and local market pressure.

Secondly, starting at 30 minutes, the volume of included capacity increases sharply, reaching 69%, and by 35 minutes, 80%, which reflects a significant improvement in the spatial correspondence between university demand and student accommodation capacity. At the same time, there is no further noticeable growth in demand coverage: even at 20 minutes, more than 94% of student demand is covered, and at 30–35 minutes, 97%. This means that an increase in travel time in this range mainly affects the housing supply rather than changing the demand pattern.

Thus, the 35-minute threshold represents an optimal compromise.

Moreover, according to a study by Menz & Mandl (2024), students studying mostly in person spend 5 hours more per week on study-related activities compared to students studying mostly online. Thus, it can be concluded that full-time students spend an additional 5 hours per week commuting, which is equivalent to approximately 1 hour per day or 30 minutes each way.

4.6 Market Areas Analysis

Spatial economics and central place metrics were used to analyse market areas (Table 13).

	code	name	own	cluster	!cap_sum	!s_aus_sum	!n_de_sum	!s_all_sum
0	HS0180	Charite - Uni- ver- sitätsmedi- zin Berlin	Public Uni- versity	Low- High (LH)	1455.4	3072.6	8126.0	11198.7
1	HS0180	Charite - Uni- ver- sitätsmedi- zin Berlin	Public Uni- versity	High- Low (HL)	511.5	2563.9	10281.7	12845.5
2	HS5802	U of Europe for Appl.Sc.Potsdam i.Berlin	Privat Uni- versity	Low- High (LH)	1594.8	3520.2	9143.1	12663.4
3	HS7690	Hochschul für Tech- nik und Wirtschaft Berlin	Public Uni- versity	Low- High (LH)	3808.8	4446.4	19321.6	23768.1
4	HS7710	Berliner Hochschul für Tech- nik	Public Uni- versity	Low- High (LH)	1409.9	3023.1	7929.0	10952.1
5	HS7721	HWR Berlin, FB Wirtschaft	Public Uni- versity	High- High (HH)	884.2	5449.5	16643.3	22092.8
6	HS7971	SRH Berlin U of Appl. Sci- ences in Berlin	Privat Uni- versity	Low- High (LH)	1687.0	1244.6	1773.3	3017.9
7	HS0200	Humboldt UniversitÄt Berlin	Public Uni- versity	Low- High (LH)	1245.5	4352.1	18945.9	23297.9
8	HS0200	Humboldt UniversitÄt Berlin	Public Uni- versity	Low- High (LH)	2000.0	3190.8	9539.3	12730.1
9	HS1380	Freie Uni- ver- sität Berlin	Public Uni- versity	Low- High (LH)	885.8	2739.1	10922.5	13661.6
10	HS1690	Technische Uni- ver- sität Berlin	Public Uni- versity	Low- High (LH)	831.0	2567.9	6318.9	8886.8

2489 *Table 13. Market Areas*2490 **4.6.1 Spatial Economics Metrics**

2491 Spatial Economics indicators are important for further calculation of capacity–demand
2492 balance and central place metrics.

2493 **Capacity–Demand Ratio**

2494 This indicator reflects the ratio between the supply of student housing and university
2495 demand within each market area. It is calculated as the share of the total capacity of
2496 all student accommodation in the market area relative to the total demand from inter-
2497 national students (`market_area[‘capacity_demand_ratio’] = !cap_sum / !s_aus_sum`).
2498 High ratio values indicate areas with a relative oversupply, while low values indicate a
2499 shortage of housing relative to demand. This indicator helps to identify areas where the
2500 market is potentially balanced or, conversely, under the pressure.

2501 **Demand–Capacity Pressure**

2502 This indicator is the inverse ratio and is calculated as the share of total university de-
2503 mand relative to student housing capacity (`market_area[‘demand_capacity_pressure’]
2504 = !s_aus_sum / !cap_sum`). It is interpreted as the degree of market pressure: the higher
2505 the value, the greater the load on the available supply and the more acute the shortage.

2506 **Number of Student Accommodations**

2507 This indicator determines how many student accommodation facilities are located within
2508 a specific market area (`pbsa_counts = wb.groupby(‘iso_id’).size()`).

2509 **Number of Higher Education Universities**

2510 This indicator reflects how many higher education institutions fall within the boundaries
2511 of the market area (`uni_counts = wb.groupby(‘iso_id’).size()`).

2512 **Market Area Size**

2513 The area of each market area is calculated as the area of its geometry in square kilome-
2514 tres (`market_area[‘area’] = market_area.geometry.area / 1e6`).

2515 The main result of the analysis of spatial economic metrics is the capacity–demand
2516 balance (Figure 33), which allows for a direct comparison of the aggregate demand of
2517 international students within each market area with the available supply of places in stu-
2518 dent accommodation.

2519 As can be seen in the graph, in the vast majority of university market areas, uni-
2520 versity international demand significantly exceeds available student accommodation ca-
2521 pacity. Almost all points are below the ideal equilibrium line (45°), indicating a struc-
2522 tural supply shortage and systemic pressure on the student housing market. The size of
2523 the points reflects the degree of pressure on a particular university. The smallest pres-
2524 sure is on the market area formed by SRH Berlin University of Applied Sciences, where
2525 capacity exceeds demand. The greatest pressure is on the market area formed by the Uni-
2526 versity of Business and Law, where demand exceeds capacity by 6.2 times.

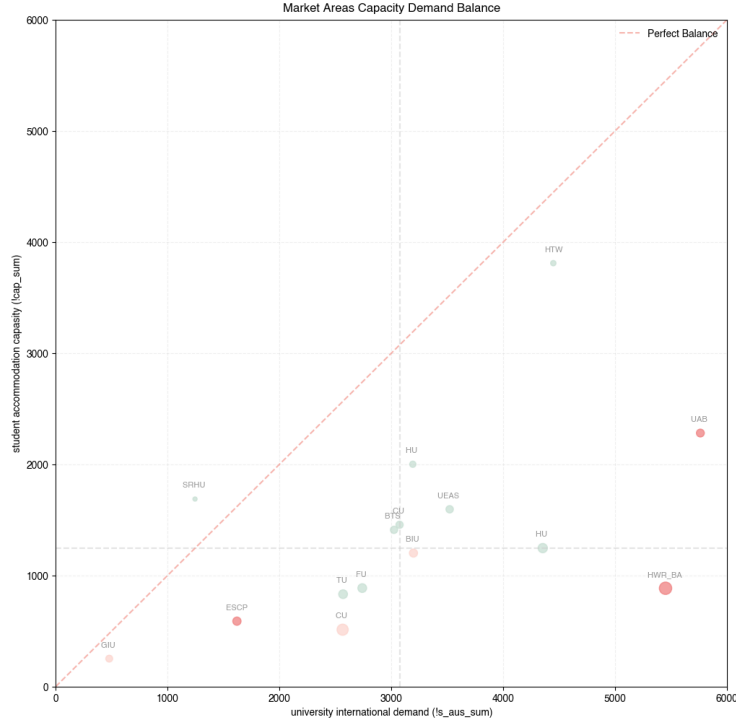


Figure 33. Market Areas Capacity Demand Balance

4.6.2 Central Place Metrics

Centrality

The first challenge was to find some way of measuring the centrality of an urban place (King, 1986). To calculate centrality Godlund (1956) used the following formula:

$$C = \sqrt{(B)(m) - (P)(k)} \quad (5)$$

where:

B — a count of the number of shops in the place
m — a weighting factor that took into account the comparative average size of the shops

P — the population of the town

k — was an index of the average accessibility to retail trade services enjoyed by any one inhabitant of the larger region in which the central place was located

In a study of the student housing market, centrality can be determined using a modified version of Godlund's formula.

$$C = \sqrt{(Csa + Cuni)(m) - (D)(Ci/Di)} \quad (6)$$

where:

Csa + Cuni — a sum of the number of student accommodations and higher education universities in the market area

m — is a constant calculated during manual tests (m=50)

D — the total international demand

Ci/Di — a capacity-demand ratio

```

2547 m = 50
2548
2549 market_area["centrality"] = np.sqrt(((market_area["pbsa_count"] + market_area["uni_count"]) * m
2550                                     (market_area["!s_aus_sum"] * market_area["capacity_demand_r
2551
2552 market_area[["fname", "pbsa_count", "uni_count", "capacity_demand_ratio",
2553             "!s_aus_sum", "centrality"]].head(5).sort_values("centrality", ascending=False)

```

	fname	pbsa_count	uni_count	capacity_denormalisation	centralisation	centrality
2	University of Europe for Applied Sciences	42	44	0.453028	3520.246	52.012
13	University of the Arts Berlin	50	48	0.395898	5760.527	51.180
5	Hochschule für Wirtschaft und Recht (Berlin, F...	32	37	0.162263	5449.490	50.653
0	Charité - Universitätsmedizin Berlin	37	38	0.473682	3072.597	47.902
8	Humboldt University of Berlin	44	41	0.626782	3190.825	47.435
12	Berlin International University of Applied Sci...	33	35	0.375696	3196.726	46.894
4	Berlin School of Technology (Technical Univers...	36	34	0.466377	3023.068	45.718
10	Technical University of Berlin	25	18	0.323625	2567.892	36.318
11	ESCP Business School Berlin	17	13	0.363232	1619.219	30.197
9	Free University of Berlin	19	16	0.323402	2739.128	29.397
1	Charité - Universitätsmedizin Berlin	11	12	0.199499	2563.897	25.269
6	SRH Berlin University of Applied Sciences	20	22	1.355402	1244.649	20.322

2555 *Table 14. Centrality Central Place Metrics*

2556 The market areas formed by the Humboldt University of Berlin and the Univer-
 2557 sity of Applied Sciences for Technology and Economics Berlin have zero centrality (Ta-
 2558 ble 14), which can be explained by their spatial location in Treptow-Köpenick and Licht-
 2559 enberg, respectively.

2560 **Service Range Adequacy**

2561 It is based on the key principle of central place theory: service range determines the size
 2562 of the market territory necessary for the sustainable operation of the facility. As King
 2563 (1986) notes, the higher the ‘service range’ of a product or service, the larger the area
 2564 that must provide a sufficient flow of users, and the higher the probability that such a
 2565 facility will be located in a larger urban centre. Service Range Adequacy (Table 15) al-
 2566 lows us to assess how well each territory meets the expected level of service and contributes
 2567 to the sustainable functioning of the student housing market. It is calculated as the rel-
 2568 ative share of demand attributable to each market area (`market_area[``cntr_service_range_adequacy']`
 2569 `= market_area[``!s_aus_sum'] / total_demand_complete)`)

```
2570 total_demand_complete = uni_dem['!s_aus'].sum()
2571 market_area['cntr_service_range'] = (
2572     market_area['!s_aus_sum'] / total_demand_complete).round(3)
2573
2574 market_area[["fname", "!s_aus_sum", "centrality", "cntr_service_range"]].head(5).sort_values("cn
```

2575

	fname	!s_aus_sum	centrality	cntr_service_range
13	University of the Arts Berlin	5760.527	51.180	0.119
5	Hochschule für Wirtschaft und Recht (Berlin, F...	5449.490	50.653	0.112
3	University of Applied Sciences for Technology ...	4446.418	0.000	0.092
7	Humboldt University of Berlin	4352.098	0.000	0.090
2	University of Europe for Applied Sciences	3520.246	52.012	0.073
8	Humboldt University of Berlin	3190.825	47.435	0.066
12	Berlin International University of Applied Sci...	3196.726	46.894	0.066
0	Charité - Universitätsmedizin Berlin	3072.597	47.902	0.063
4	Berlin School of Technology (Technical Univers...	3023.068	45.718	0.062
9	Free University of Berlin	2739.128	29.397	0.057
1	Charité - Universitätsmedizin Berlin	2563.897	25.269	0.053
10	Technical University of Berlin	2567.892	36.318	0.053
11	ESCP Business School Berlin	1619.219	30.197	0.033
6	SRH Berlin University of Applied Sciences	1244.649	20.322	0.026
14	German International University Berlin	478.000	9.936	0.010

2576 *Table 15. Service Range Adequacy Central Place Metrics*2577 **Hierarchy Rank**

2578 According to King's approach, the hierarchy of urban places is defined not only by their
 2579 size, but primarily by the number and order of the functions they perform (King, 1986,
 2580 p. 22).

2581 Following this logic, the ranking of market areas in this study is based on their func-
 2582 tional capacity, expressed through two key indicators: centrality and service range ad-
 2583 equacy. These values are combined into a composite index (`market_area[['cntr_hierarchy']`
 2584 `= ((market_area[['centrality']]/100) + (market_ara[['cntr_service_range']])).round(3)`).
 2585 The hierarchical ranking (Table 16) allows us to see not only the current configuration
 2586 of supply and demand, but also the structural role of each market area in the urban sys-
 2587 tem, identifying higher-order centres and lower-level areas.

```

2588 market_area['cntr_hierarchy'] = (
2589     (market_area['centrality']/100) + (market_area['cntr_service_range'])).round(3)
2590
2591 market_area['cntr_hierarchy_rank'] = pd.qcut(
2592     market_area['cntr_hierarchy'],
2593     q=5, labels=range(1, 6)    # 1-5 classes
2594 )
2595
2596 market_area[['fname', '!s_aus_sum', 'centrality', 'cntr_service_range',
2597             'cntr_hierarchy', 'cntr_hierarchy_rank']].head(5).sort_values("cntr_hierarchy", as

```

	fname	ls_aus_sum	centrality	cntr_service_change	cntr_hierarchy	cntr_hierarchy_rank
13	University of the Arts Berlin	5760.527	51.180	0.119	0.631	5
5	Hochschule für Wirtschaft und Recht (Berlin, F...	5449.490	50.653	0.112	0.619	5
2	University of Europe for Applied Sciences	3520.246	52.012	0.073	0.593	5
0	Charité - Universitätsmedizin Berlin	3072.597	47.902	0.063	0.542	4
8	Humboldt University of Berlin	3190.825	47.435	0.066	0.540	4
12	Berlin International University of Applied Sci...	3196.726	46.894	0.066	0.535	4
4	Berlin School of Technology (Technical Univers...	3023.068	45.718	0.062	0.519	3
10	Technical University of Berlin	2567.892	36.318	0.053	0.416	3
9	Free University of Berlin	2739.128	29.397	0.057	0.351	3
11	ESCP Business School Berlin	1619.219	30.197	0.033	0.335	2
1	Charité - Universitätsmedizin Berlin	2563.897	25.269	0.053	0.306	2
6	SRH Berlin University of Applied Sciences	1244.649	20.322	0.026	0.229	2

Table 16. *Hierarchy Rank Central Place Metrics*

Market Areas Hierarchy Rank demonstrates the distribution of central places by hierarchical classes in accordance with Central Place Theory.

Hierarchy Rank - 5. Primary Central Places

Primary Central Places (Figure 34, Table 17) are formed around the University of the Arts Berlin (UAB), the University of Europe for Applied Sciences (UEAS) and the School of Economics and Law (HWR_BA). Although according to King (1986), centrality is not determined by the physical size of the territory, these market areas have some of the largest areas, more than 90 km² each. This is because top-level zones must have a sufficiently wide coverage to serve a large number of functions and significant volumes of demand.

Geographically, these market areas are located mainly in the districts of Mitte, Charlottenburg-Wilmersdorf and Friedrichshain-Kreuzberg, as well as partly in Tempelhof-Schöneberg, Steglitz-Zehlendorf and Neukölln, forming a distinct axis in a south-westerly direction. Together, they account for about 32% of total international student demand, which highlights their systemic role in the city's university infrastructure.

It is worth noting that two of the three primary centres belong to the High-High cluster, meaning they are stable magnets for international students, around which areas of high demand and growing market pressure are forming. However, the greatest pressure is observed in the School of Economics and Law market zone, where demand exceeds available capacity by 6.2 times, making this zone the most vulnerable in the hierarchy of central locations. This imbalance highlights the critical shortage of infrastructure and the need to prioritise the expansion of the housing stock near this educational cluster.

Together, the primary central places form the core of Berlin's student urbanisation, where key educational functions, large volumes of demand and the most significant infrastructure imbalances are concentrated. Their high hierarchical position confirms the central role of these areas in the urban structure, but at the same time demonstrates the vulnerability of the system: even higher-order centres face a significant housing shortage and growing market pressure.

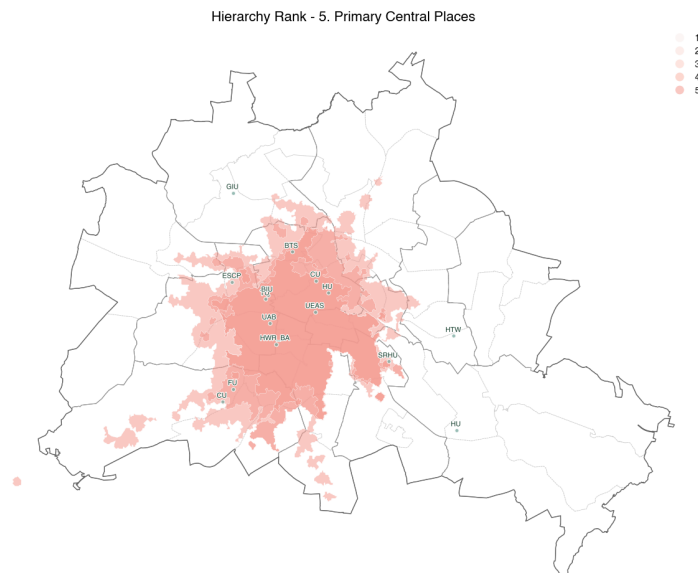


Figure 34. *Primary Central Places*

	fname	ls_aus_sum	centrality	cntr_service_change	cntr_hierarchy	cntr_hierarchy_rank
13	University of the Arts Berlin	5760.527	51.180	0.119	0.631	5
5	Hochschule für Wirtschaft und Recht (Berlin, F...	5449.490	50.653	0.112	0.619	5
2	University of Europe for Applied Sciences	3520.246	52.012	0.073	0.593	5

Table 17. Primary Central Places

Hierarchy Rank - 4. Secondary Central Places

Secondary Central Places Centres (Figure 35, Table 18) are formed by Humboldt University of (HU), International University of Applied Sciences (BIU) and Charité University Hospital (CU). These market areas are secondary central places, i.e., high-order centres that have significant functional saturation but are lower than primary centres in terms of scale of attraction and infrastructure concentration.

Geographically, secondary central places are located east of primary central places, covering parts of the Lichtenberg and Pankow districts and forming a distinct west-east axis in the spatial structure of the student housing market. This configuration reflects the expansion of the city's education system in an easterly direction, which is historically less saturated with higher education institutions. Together, these areas account for about 20% of the total demand from international students, confirming their significant but not dominant role in the urban hierarchy.

The average demand–capacity pressure value of 2.1 confirms that demand exceeds current supply by more than double. This creates moderate but stable market pressure, indicating the need for strategic expansion of student infrastructure and further development of public-private initiatives in the field of student accommodation.

Thus, secondary central places function as sub-regional centres in the urban hierarchy, playing a key role in rebalancing demand, ensuring access to higher education and shaping future growth trends in Berlin's student housing market.

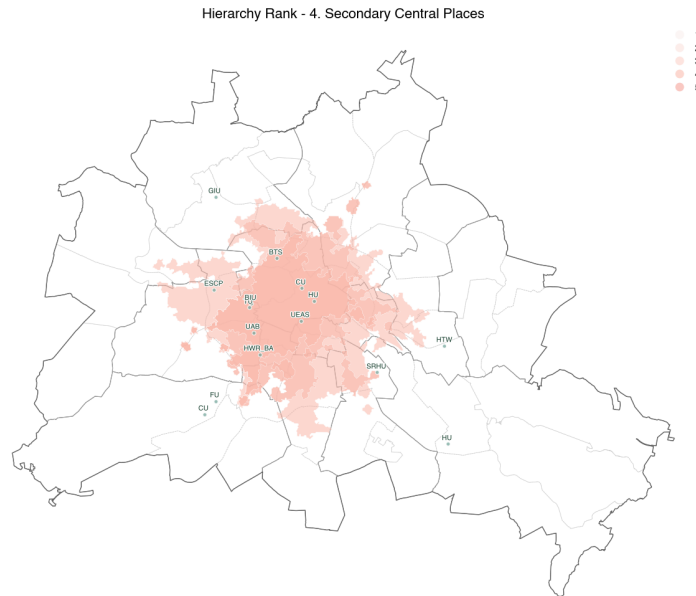


Figure 35. Secondary Central Places

	fname	ls_aus_sum	centrality	cntr_service_change	cntr_hierarchy	cntr_hierarchy_rank
0	Charité - Univer- sitätsmedi- zin Berlin	3072.597	47.902	0.063	0.542	4
8	Humboldt University of Berlin	3190.825	47.435	0.066	0.540	4
12	Berlin Internat- ional Univer- sity of Applied Sci...	3196.726	46.894	0.066	0.535	4

Table 18. Secondary Central Places

Hierarchy Rank - 3. Tertiary Central Places

Tertiary Central Places (Figure 36, Table 19) are formed around the Technical University of Berlin (TUB), Free University (FU) and Berlin School of Technology (BTS). These universities represent significant but less dominant centres of demand, located on the periphery of the primary and secondary central places.

Geographically, tertiary central places occupy areas to the north and south of secondary central places, covering a significant part of Steglitz-Zehlendorf and Reinickendorf. In this way, they support the established south-western educational axis extending from the primary central places and, at the same time, form a new north-south axis in the spatial structure of the student housing market. This configuration reflects the complex multipolar organisation of demand characteristic of a large polycentric city.

Together, these market areas account for about 17% of total international student demand, making them significant but not dominant elements in the structure of central places.

It is noteworthy that all three universities forming tertiary central places belong to the Low-High (LH) cluster and are public. This combination indicates the high educational potential of these areas and their ability to be included in existing clusters, but at the same time demonstrates the lack of full strategic integration into established centres of student urbanisation.

The average demand-capacity pressure value is 2.8, which is higher than that of secondary central places. This indicates that demand is almost three times higher than the available supply. Consequently, tertiary central places are critically important areas of hidden deficit, where a growing gap is forming between the educational potential of universities and the lack of adequate housing infrastructure.

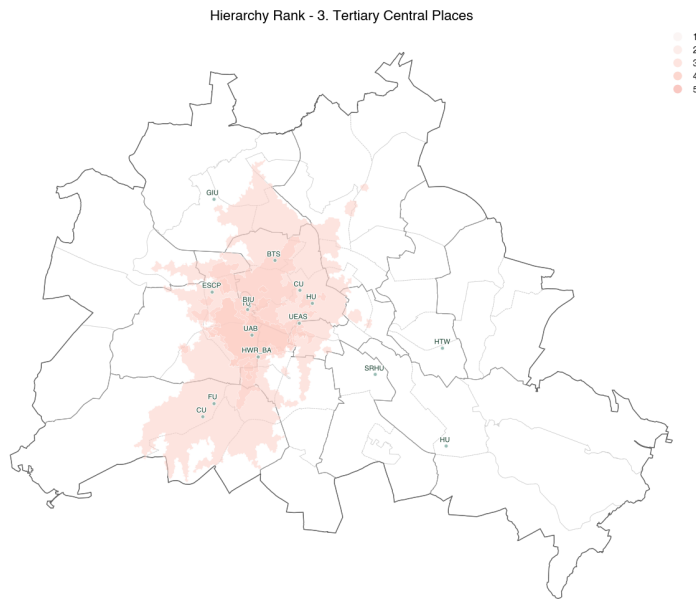


Figure 36. Tertiary Central Places

	fname	!s_aus_sum	centrality	cntr_service_change	cntr_hierarchy	cntr_hierarchy_rank
4	Berlin School of Technology (Technical Univers...	3023.068	45.718	0.062	0.519	3
10	Technical University of Berlin	2567.892	36.318	0.053	0.416	3
9	Free University of Berlin	2739.128	29.397	0.057	0.351	3

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Table 19. Tertiary Central Places

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Hierarchy Rank - 2. Local Central Places

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Local Central Places (Figure 37, Table 20) are formed around SRH Berlin University of Applied Sciences (SRHU), Charité University Hospital (CU) and ESCP Business School Berlin (ESCP). These universities act as local centres, primarily serving the surrounding areas and acting as a link between larger secondary and tertiary central places.

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Geographically, these market areas are mainly located in the districts of Friedrichshain-Kreuzberg and Neukölln, as well as partly in Mitte, Charlottenburg-Wilmersdorf, Tempelhof-Schöneberg, Treptow-Köpenick and Lichtenberg. Spatially, they occupy positions south of secondary central places and simultaneously support two key axes of the urban structure: the west-east axis formed by secondary central places and the north-south axis formed by tertiary central places. In this way, local central places reinforce the polycentric organisation of the student housing market, ensuring connectivity between areas of different hierarchical levels and serving 11% of student demand.

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What is particularly interesting is that the three universities that form this hierarchical level belong to different types of spatial clusters: ESCP Business School Berlin is characterised by a High-High (HH) pattern, Charité University Hospital by a High-Low (HL) pattern, and SRH Berlin University of Applied Sciences by a Low-High (LH) pattern. This highlights the heterogeneity of these areas and their varying degrees of integration into existing educational clusters.

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The intensity of market pressure within local central places also varies. Market areas formed by private universities ESCP and SRHU show relatively moderate indicators: 2.7 and 0.7, respectively. At the same time, the area formed by the state-owned Charité University Hospital experiences a pronounced imbalance between supply and demand: demand exceeds available capacity by five times. This indicates a structural shortage of student housing in one of the city's most important medical and scientific centres.

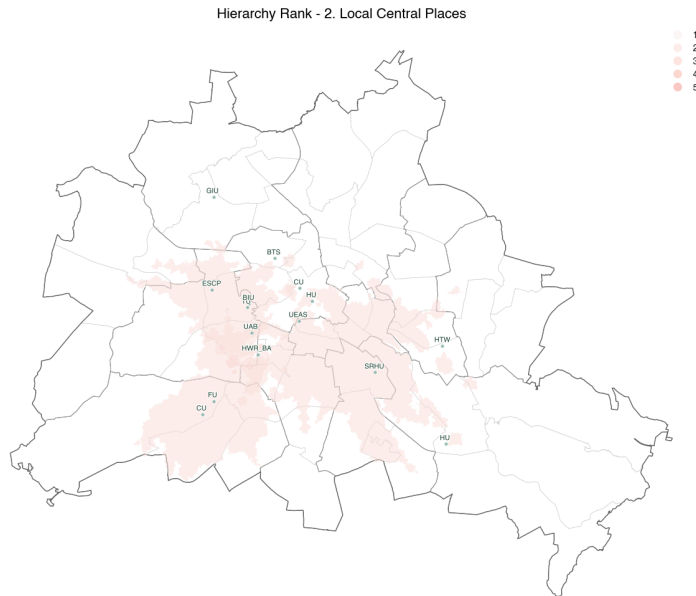
2710

2711

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2713

Thus, local central places reflect an important intermediate level of the city's educational and housing structure. They combine areas with stable patterns of demand and supply, adjacent to local shortages, and form a significant component of Berlin's polycentric student space.



2714

Figure 37. Local Central Places

Table 20. Local Central Places

Hierarchy Rank - 1. Basic Service Areas

Basic Service Areas (Figure 3, Table 21) are formed around the German International University Berlin (GIU), University of Applied Sciences for Technology, Economics Berlin (HTW) and Humboldt University of Berlin (HU). These universities are basic service centres whose role in the structure of central locations is primarily to provide fundamental educational functions and support peripheral parts of the urban system.

Geographically, these market areas are located mainly in the districts of Reinickendorf, Treptow-Köpenick and Lichtenberg. They are situated to the south-east and north-west of the secondary central places. This location emphasises their functional role as areas serving peripheral or poorly integrated parts of the city's educational space, acting as 'edge' centres in Berlin's polycentric structure.

Collectively, Basic Service Areas account for about 19% of international student demand — more than tertiary central places. This indicates that even the most basic centres in the hierarchy play a significant role in the distribution of student mobility and the formation of market pressure, serving significant shares of both educational and housing demand in the periphery.

It is worth noting that two of the three universities forming this hierarchical level belong to the Low-High (LH) cluster, demonstrating low supply with higher regional demand. The third university is private and belongs to the High-Low (HL) cluster, reflecting the opposite dynamic: a high value at the facility level in an environment with relatively low values. This distribution of clusters shows that the basic market areas are characterised by strong heterogeneity in demand and supply, as well as varying levels of integration into existing educational clusters.

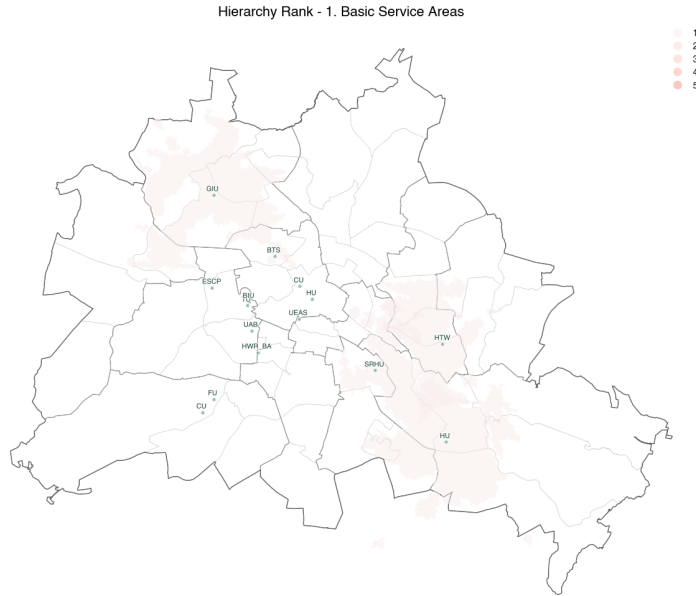


Figure 38. Basic Service Areas

Table 21. Basic Service Areas

The constructed hierarchy of market areas revealed a pronounced polycentricity of the city's educational space: central locations at the highest levels concentrate the great-

2745 est student demand and form the main clusters of attraction, while lower-level zones per-
2746 form supporting functions. At the same time, demand significantly exceeds supply at
2747 all levels.

2748 Thus, the analysis shows that the student housing crisis in Berlin has a clearly de-
2749 fined spatial structure. The uneven distribution of both universities and accommoda-
2750 tion infrastructure exacerbates market pressure, creating persistent areas of overload and
2751 local shortages, which require further strategic intervention and planning.

2752 `market_area.to_crs(epsg=25833)`

2753

2754 `ef = r"../outputs/3.2_central_place_analysis/market_area_central_place_analysis.gpkg"`
2755 `market_area.to_file(ef, driver="GPKG")`

4.7 3.3 Market Pressure Analysis

In the chapter Central Place Analysis, spatial economics metrics were calculated, including demand–capacity pressure for each market area. The task of the chapter Market Pressure Analysis is to move from numerical characteristics to spatial ones and to calculate the local pressure on the student housing market and particular student accommodations.

4.7.1 Set Up

Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import geopandas as gpd
from matplotlib.colors import LinearSegmentedColormap
import seaborn as sns
import matplotlib.path_effects as path_effects
from matplotlib.colors import ListedColormap
from shapely.ops import unary_union
import rasterio
from rasterio import features
from affine import Affine
from scipy.ndimage import gaussian_filter
```

Berlin Spatial Boundaries

```
pf = r"../inputs/3.3_market_pressure_analysis/lor_pgr.gpkg"
pgr = gpd.read_file(pf)
pgr.replace({'-': np.nan, 0: np.nan, ': np.nan}, inplace=True)

bf = r"../inputs/3.3_market_pressure_analysis/adm_bz.gpkg"
bz = gpd.read_file(bf)
bz.replace({'-': np.nan, 0: np.nan, ': np.nan}, inplace=True)

of = r"../inputs/3.3_market_pressure_analysis/adm_out.gpkg"
out = gpd.read_file(of)
out.replace({'-': np.nan, 0: np.nan, ': np.nan}, inplace=True)
```

Data Import

```
ff = r"../inputs/3.3_market_pressure_analysis/market_area_central_place_analysis.gpkg"
market_area = gpd.read_file(ff)

cf = r"../inputs/3.3_market_pressure_analysis/central_places.gpkg"
central_places = gpd.read_file(cf)

pf = r"../inputs/3.3_market_pressure_analysis/pbsa_capacity_analysis.gpkg"
pbsa_cap = gpd.read_file(pf)
```

4.8 Overlap Problem

Whereas in the original Central Place, theory market areas do not overlap, in order to analyse the aggregate market pressure on the student housing market, it is nec-

essary to move from the overlapping market areas obtained as a result of the Central Place Analysis (Figure 39) to a more accurate local structure that reflects the aggregate pressure on a particular part of the city.

```
market_area['demand_capacity_pressure_rank'] = pd.qcut(
    market_area['demand_capacity_pressure'],
    q=5, labels=range(1, 6)    # 1-5 classes
)
```

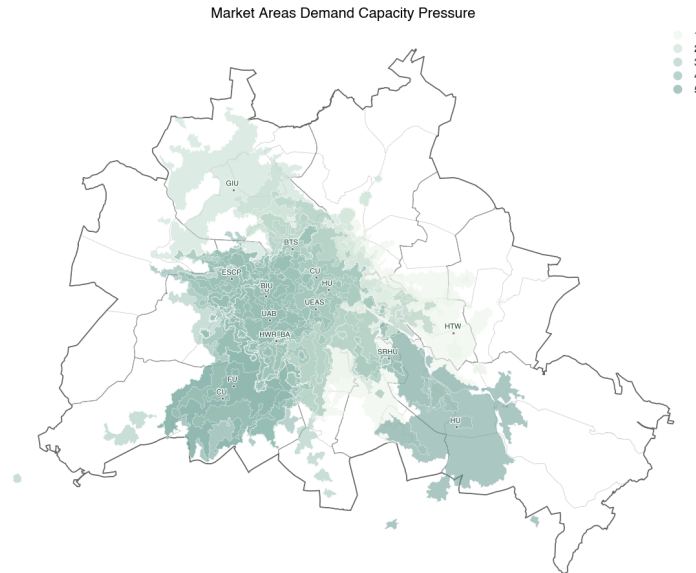


Figure 39. Market Areas Demand Capacity Pressure Rank

Market Areas Demand Capacity Pressure Rank (Figure XX) illustrates that each university forms its own accessibility zone, and these zones naturally overlap. On the one hand, this overlap reflects the real competition for student flows for the same housing stock. On the other hand, it creates analytical complexity: some territories are part of several market areas at the same time, which complicates the interpretation of local pressure.

This is partly due to the fact that, initially, isochrones were calculated using multipolygons, which form complex geometries and consist of several polygon parts. These fragments overlap with each other and with the polygons of other universities.

According to the results of spatial analysis, individual market areas of overlap and are part of at least five different market areas, while particularly central areas, such as parts of the Mitte district, are part of 14 market areas simultaneously. In order to simplify such a dense network of intersecting zones of influence, it is necessary to eliminate overlaps.

```
market_area['er_overlap'] = 0

# Calculate overlaps
for i, current_poly in market_area.iterrows():
    overlap_count = 0
    for j, other_poly in market_area.iterrows():
        if i != j: # Don't compare with itself
            if current_poly.geometry.intersects(other_poly.geometry):
```

```

2832         overlap_count += 1
2833     market_area.at[i, 'er_overlap'] = overlap_count
2834
2835     print(f"Overlap Counts Range for Market Areas: {market_area['er_overlap'].min()} to {market_area['er_overlap'].max()}")
2836
2837     ff = r'../outputs/3.3_market_pressure_analysis/market_areas_not_explode_overlap.gpkg'
2838     market_area.to_file(ff, driver="GPKG")

```

```

2839     Overlap Counts Range for Market Areas: 5 to 14

```

One way to simplify geometry is to divide complex multipolygon geometry into simple polygons `market_area.explode(ignore_index=True)`. In this case, polygons intersect between 0 and 51 times.

```

2843     market_area_exp = market_area.explode(ignore_index=True)
2844
2845     market_area_exp['er_overlap'] = 0
2846
2847     for i, current_poly in market_area_exp.iterrows():
2848         overlap_count = 0
2849         for j, other_poly in market_area_exp.iterrows():
2850             if i != j:
2851                 if current_poly.geometry.intersects(other_poly.geometry):
2852                     overlap_count += 1
2853             market_area_exp.at[i, 'er_overlap'] = overlap_count
2854
2855     print(f"Overlap Counts Range for Exploded Market Areas: {market_area_exp['er_overlap'].min()} to {market_area_exp['er_overlap'].max()}")

```

```

2856     Overlap Counts Range for Exploded Market Areas: 0 to 51

```

This means that in order to switch to simple geometry, it will be necessary to apply the division 51 more times. Such a task is not only difficult in terms of computing power, but will also result in the resulting non-intersecting polygons being extremely small.

2860 4.9 Rasterisation

2861 Another way to eliminate the problem of multiple market area overlaps is to switch
 2862 from vector geometry to raster representation, i.e., to a regular grid of fixed size. Ras-
 2863 terisation allows complex, overlapping polygons to be replaced with a set of uniform pixel
 2864 cells. Unlike a vector model, where a single territory can belong to several market ar-
 2865 eas at once, a raster grid forces a unique spatial correspondence: one cell equals one pres-
 2866 sure value.

2867 The use of a regular 25-meter grid provides a sufficient level of spatial detail to iden-
 2868 tify micro-zones of local congestion, especially in areas where the density of universities
 2869 and student accommodation is highest.

```

2870     value_col = "demand_capacity_pressure"
2871
2872     # Define raster resolution (in meters)
2873     resolution = 25
2874
2875     # Bounding box
2876     minx, miny, maxx, maxy = market_area.total_bounds
2877

```

```

2878 # Create transform (affine)
2879 transform = Affine.translation(minx, maxy) * Affine.scale(resolution, -resolution)
2880
2881 # Calculate raster shape
2882 width = int((maxx - minx) / resolution)
2883 height = int((maxy - miny) / resolution)
2884
2885 raster = np.zeros((height, width), dtype=np.float32)
2886
2887 # For each polygon, add pressure values
2888 for _, row in market_area.iterrows():
2889     shapes = [(row.geometry, row[value_col])]
2890     layer = features.rasterize(
2891         shapes=shapes,
2892         out_shape=(height, width),
2893         transform=transform,
2894         fill=0,
2895         all_touched=True,
2896         dtype=np.float32
2897     )
2898     raster += layer

```

2898 4.10 Demand Capacity Pressure

2899 Thus, rasterisation can be used to assess Market Areas Local Demand Capacity Pres-
2900 sure.

2901 The Local Demand–Capacity Pressure (Figure 40) shows the spatial distribution
2902 of the total market pressure arising from the overlap of the catchment areas of all uni-
2903 versities. Unlike capacity–demand pressure, which is calculated at the level of individ-
2904 ual market areas, this visualisation shows the ‘accumulated’ pressure, reflecting the cu-
2905 mulative effect of student competition for housing resources at each point in the urban
2906 area.

2907 The most intense pressure zones are marked in saturated red shades and are con-
2908 centrated in the central part of the city — primarily in the districts of Mitte, Charlottenburg-
2909 Wilmersdorf, and Friedrichshain-Kreuzberg. This core coincides with the spatial loca-
2910 tion of the largest educational clusters: the University of the Arts Berlin (UAB), Uni-
2911 versity of Europe for Applied Sciences (UEAS), Berlin International University (BIU),
2912 Berlin School of Technology (BTS), and School of Economics and Law (HWR_BA). The
2913 catchment areas of these universities overlap most densely, which greatly increases the
2914 pressure on the limited housing stock in the central districts. It is here that local pres-
2915 sure reaches its maximum values — over 30, which indicates extremely high competi-
2916 tion among international students for available places.

2917 As move away from the central core, the intensity of the pressure gradually decreases,
2918 transitioning into moderate greenish hues. However, the decrease is asymmetrical: in the
2919 south-west (Steglitz-Zehlendorf), south-east (Treptow-Köpenick), and partly in the north-
2920 east (Pankow), there are still local areas of increased local pressure caused by the influ-
2921 ence of the large universities FU, HU, and HTW. At the same time, the districts of Reinick-
2922 endorf, Spandau, and Marzahn-Hellersdorf show the lowest local pressure values, indi-
2923 cating weak involvement in competition for housing stock and relatively low attractive-
2924 ness for students.

2925 Thus, the local demand–capacity pressure map clearly shows that market tension
2926 in the student housing market in Berlin is concentrated and polycentric, reaching its peak
2927 in the central ‘education corridor’ and gradually weakening towards the periphery.

Importantly, this visualisation emphasises that the pressure is not generated by individual market areas, but by the overlap of multiple accessibility zones, which explains why it is the concentration of university functions — and not just the local shortage of student accommodation — that is the key driver of market pressure in the central areas of the city.

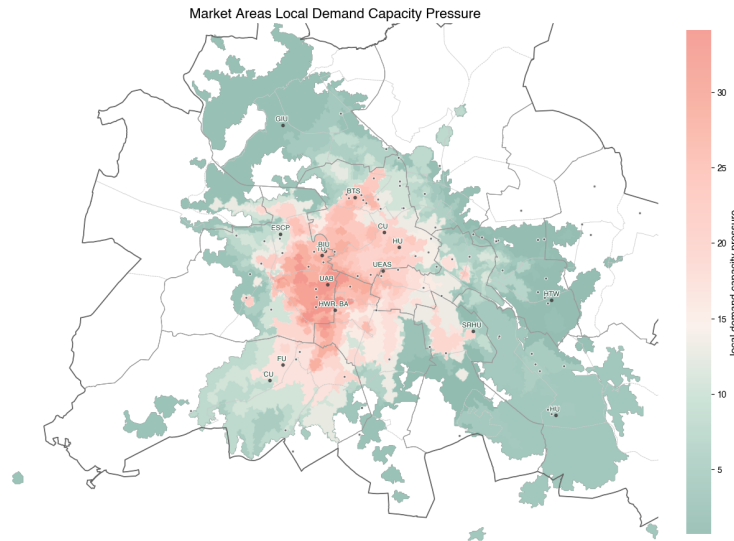


Figure 40. Market Areas Local Demand Capacity Pressure

```
out_raw = r'../outputs/3.3_market_pressure_analysis/market_pressure_sum_raw.tif'
```

4.11 Student Accommodation Pressure

Data Import

```
out_raw = r'../outputs/3.3_market_pressure_analysis/market_pressure_sum_raw.tif'
```

```
with rasterio.open(out_raw) as src:
```

```
    raster = src.read(1) # read first band as numpy array
```

```
    transform = src.transform
```

```
    crs = src.crs
```

```
ef = r"../inputs/3.3_market_pressure_analysis/pbsa_capacity_analysis.gpkg"
```

```
pbsa_cap = gpd.read_file(ef)
```

```
pbsa_cap.drop(columns=['link', '!cap_slag', '!cap_std', '!cap_slag_std', 'cluster', 'model', 'c
```

```
coords = [(geom.x, geom.y) for geom in pbsa_cap.geometry]
```

```
with rasterio.open(out_raw) as src:
```

```
    pbsa_cap["local_demand_capacity_pressure"] = [val[0] for val in src.sample(coords)]
```

```
pbsa_cap["local_demand_capacity_pressure"] = pbsa_cap["local_demand_capacity_pressure"].round(3)
```

Student accommodations with the lowest level of local market pressure (Table 22) form a separate and segment of the Berlin student housing market. This group primarily includes properties located in the eastern districts of the city — Pankow, Lichtenberg and Friedrichsfelde — and are mainly state-owned (Berlinovo and Studierendenwerk Berlin). The lowest local demand–capacity pressure values (≈ 0.74 – 1.17) are recorded, for exam-

2959 ple, at the Walter-Friedländer-Straße (Pankow) and Storckower Straße (Lichtenberg)
 2960 halls of residence, which means that supply here is either comparable to demand or even
 2961 exceeds it.

```
2962 pbsa_cap[pbsa_cap["local_demand_capacity_pressure"] > 0][[
2963     "name",
2964     "plz",
2965     "ownership",
2966     "!cap",
2967     "ownership_model",
2968     "cluster_cap",
2969     "local_demand_capacity_pressure",
2970 ]].sort_values("local_demand_capacity_pressure").head(10)
```

	name	plz	ownership	!cap	ownership	chotel	local_demand_capacity_pressure
86	Walter-Friedländer-Straße Pankow	10249	Berlinovo	285.0	State-Owned	HL	0.738
37	Storckower Straße Lichtenberg I	10369	Berlinovo	141.0	State-Owned	HL	0.738
31	Rhinstraße Lichtenberg	10315	Berlinovo	800.0	State-Owned	HH	1.167
55	WH Aris- totelessteig	10318	Studierende Berlin	913.0	State-Owned	HH	1.167
54	WH Allee der Kosmo- nauten	10315	Studierende Berlin	298.0	State-Owned	HH	1.167
42	Studentenwohnheim Alfred-Jung- Straße	10369	Home in Berlin GmbH	800.0	Privat-Owned	HH	1.167
8	Elfriede- Tygör-Straße	10319	Berlinovo	202.0	State-Owned	HL	1.167
11	Friedenhorst Straße Friedrichs- felde	10319	Berlinovo	138.0	State-Owned	HL	1.167
80	WH Sewanstraße	10319	Studierende Berlin	700.0	State-Owned	HH	1.167
83	WH Storkower Straße (Ferdinand Thomas)	10367	Studierende Berlin	508.0	State-Owned	HH	1.595

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Table 22. Less Pressured Student Accommodations

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Low market pressure in these locations is explained by a combination of several factors. Firstly, they are located on the periphery or semi-periphery of key university centres and educational clusters, which reduces competition from students focused on central areas. Secondly, these dormitories tend to have a relatively large capacity (up to 800 places), which helps to smooth out local peaks in demand. Thirdly, the state manage-

ment model is focused not on maximising profits, but on providing stable and affordable housing, which reduces selectivity in occupancy and lessens structural pressure.

It is important to emphasise that low demand–capacity pressure in this case does not mean that there are no problems, but rather indicates a spatial imbalance in demand: students are willing to accept less central and less prestigious locations in exchange for affordability and relative stability. Thus, halls of residence with minimal pressure perform a compensatory function in Berlin’s student housing system, redistributing demand from congested central areas to the eastern and peripheral areas of the city.

Student accommodation with the highest level of local market pressure (Table 23) forms the opposite pole of the Berlin student housing market and clearly demonstrates the depth of the structural deficit in the central areas of the city. Extremely high local demand–capacity pressure values (from ~27 to over 34) were recorded for properties located mainly in Charlottenburg-Wilmersdorf, Schöneberg, Tiergarten and Mitte — areas with a high concentration of universities, private educational institutions and international student demand.

```
pbsa_cap[pbsa_cap["local_demand_capacity_pressure"] > 0][[
  "name",
  "plz",
  "ownership",
  "!cap",
  "ownership_model",
  "cluster_cap",
  "local_demand_capacity_pressure",
]] .sort_values("local_demand_capacity_pressure", ascending=False).head(10)
```

	name	plz	ownership	!cap	ownership	cluster	local_demand_capacity_pressure
2	Am Mühlen- berg	10825	Gewobag	158.0	State- Owned	LL	34.094002
15	Insider Apart- ments Berliner Straße	10715	Insider Apart- ment Com- munity Living GmbH	200.0	Privat- Owned	LL	33.355999
63	WH Düssel- dorfer Straße	10719	Studierende Berlin	73.0	State- Owned	LL	31.245001
78	WH Pfalzbürgerstraße	10717	Studierende Berlin	28.0	State- Owned	LL	31.245001
62	WH Derfflingerstraße (Haus Unger)	10785	Studierende Berlin	43.0	State- Owned	LL	28.344000
81	WH Sieg- munds Hof	10555	Studierende Berlin	59.0	State- Owned	LH	28.344000
17	Insider Apart- ments Leibniz Straße	10625	Insider Apart- ment Com- munity Living GmbH	200.0	Privat- Owned	LL	28.344000
76	WH Nollendorfsstraße	10777	Studierende Berlin	18.0	State- Owned	LL	28.344000
70	WH Hardenbergstraße	10623	Studierende Berlin	46.0	State- Owned	LL	28.344000
47	Studentenwohnheim Walter- May- Haus	13353	Home in Berlin GmbH	276.0	Privat- Owned	NS	27.153999

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Table 23. Most Pressured Student Accommodations

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The greatest pressure is observed (Figure 41) in the Am Mühlenberg (Schöneberg) and Insider Apartments Berliner Straße halls of residence, where demand exceeds available capacity by more than 30 times. It is noteworthy that most of the properties in this group belong to the Low-Low (LL) cluster, i.e. they are characterised by both low local capacity and extremely high demand. In some cases, the capacity of such halls of residence is only 28–70 places (e.g. WH Pfalzbürgerstraße or WH Hardenbergstraße), which

makes them particularly vulnerable to overcrowding in conditions of high spatial competition.

It is important to note that high market pressure is observed in both private and public halls of residence, including those run by Studierendenwerk Berlin. This indicates that the reason for overcrowding lies not so much in the form of ownership as in the spatial location and integration into dense university clusters. Even relatively large halls of residence, such as WH Siegmunds Hof (590 places), are under extreme pressure due to their overlap with several university catchment areas and the concentration of international demand.

Thus, student residences with the highest level of demand–capacity pressure serve as indicators of bottlenecks in Berlin’s student housing market. They are located in the most prestigious and functionally rich areas of the city, where educational centrality is combined with limited opportunities for expanding supply. These areas form the core of the crisis, where the housing shortage is becoming chronic and cannot be compensated for by local increases in capacity without systemic urban planning and housing policy interventions.

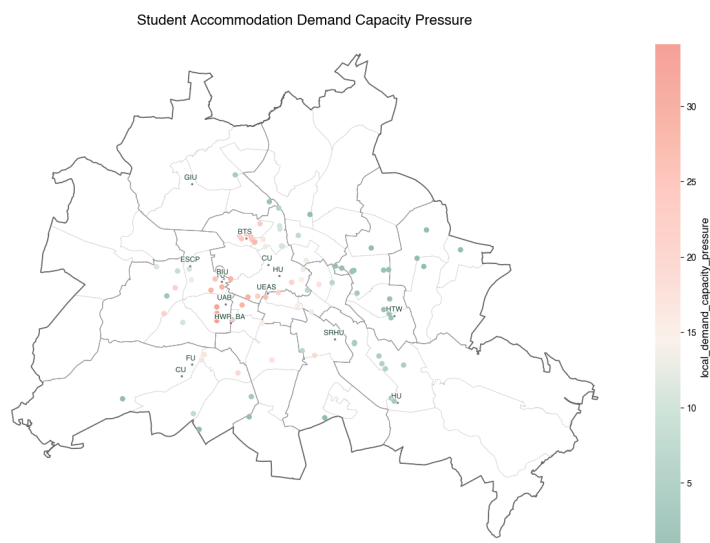


Figure 41. Student Accommodation Demand Capacity Pressure

5 4. Students Behind the Numbers

5.1 4.1 Overview

While the previous chapters described and examined the components of student housing and the market relationships between them, this chapter focuses on the users of student accommodations.

5.1.1 *What? Students Behind the Numbers*

Behind the student housing crisis numbers there are actual users. In order to rethink the housing crisis, it is necessary not only to assess the student housing market, but also to address users—customers—directly.

The first attempt to understand users was made in the chapter on defining **demand**. It is worth remembering that for this research, the target group is international students enrolled in Berlin's higher education institutions, at any level, regardless of current residency. However, **international students** themselves do not represent a homogeneous group, but rather a multitude of disparate qualitative groups that differ from each other in terms of housing needs and preferences.

This chapter helps to get to know **international students** as users in more detail.

5.1.2 *Why? Bridge Demand and Supply*

The aim of this chapter is to define users and assess user experience by answering the following research questions:

Who are the users of student accommodations?

What experience they have in ongoing housing crisis? What are common needs, preferences, difficulties and problems?

The answers to these questions can be used in developing strategies to improve the situation on the student housing market and bridge demand and supply.

5.1.3 *How? Behavioral Studies*

This chapter was written in collaboration with Joseph Kampfmann as part of the preparations for the ENHANCE workshop Perspectives on the European Student Accommodation Crisis, which was organised by Hannah Berner and Jesco Lippke.

Surveys, Questionnaires, Interviews

Surveys, questionnaires, and interviews are traditional behavioral study methods widely used to collect data on people's opinions, attitudes, and behavior. To improve the accuracy and relevance of the results, it is important not to generalise students but to divide them into qualitative groups. This takes into account the diversity of their needs and preferences, which contributes to a deeper understanding of the phenomena under study.

Storytelling & Narrative Analysis

Storytelling & Narrative Analysis can be seen as an alternative method of behavioral studies that allows the exploration of participants' personal stories and experiences, revealing the emotional and social aspects of their behavior (Riessman, 2008). Although this method differs from an overall data-driven approach, it makes it difficult to automate data analysis. Nevertheless, individual experiences are important for understanding hidden barriers to finding student housing, cultural sensitivity, and adaptation in a new city.

3071 Future-proof urban solutions require collaborative planning, where city authorities,
3072 public and private investors, and citizens act together.

5.2 4.2 Student Needs and Preferences

As part of behavioural studies, an online survey of student dormitory users was conducted, in which more than 100 people took part.

The survey of Berlin students partially reproduces and refines the study by Von Wittenhorst Zu Sonsfeld and Beusker conducted in 2024 in the German city of Aachen.

As part of the study, the survey was published in thematic groups on housing search, and was also distributed in the form of QR codes on the campus of the Technical University of Berlin (TU) and Studierendenwerk Berlin accommodations.

Affordability Theory was chosen as the theoretical basis for behavioural studies, which assumes that affordability is not limited to financial indicators, but also includes education, transport and social indicators.

5.2.1 Set Up

Import Libraries

```
import pandas as pd
import geopandas as gpd
import chardet
import gspread
import seaborn as sns
import matplotlib.pyplot as plt
import textwrap
from pywaffle import Waffle
import numpy as np
from matplotlib.patches import Patch
from math import pi
```

5.2.2 Data Mining

The input data for the analysis was tabular data downloaded from Google Forms responses.

```
fp = r"../inputs/4.2_students_needs/student_experience_090126.xlsx"
df = pd.read_excel(fp)
```

The survey consists of 34 questions, designed to take a total of 10-15 minutes to complete, and consists of three thematic sections:

Personal Data

The first section is aimed at collecting basic socio-demographic and contextual information necessary to understand the profile of respondents and correctly interpret behavioural patterns in the student housing market.

Socio-demographic characteristics

- gender
- age group
- origin (`origin`)
- study status (`stat`)

These questions allow us to analyse whether housing problems vary between different social groups and help to determine whether international students face a higher degree of market pressure than local students.

Financial and housing information

- residential address (**adr**)
 - postcode (**plz**)
 - current rent (**price**),
 - sources of funding (**fin**)
 - housing model (**model**)
- These questions provide insight into budget constraints and types of accommodation.

This block forms the basis for analysis and helps answer the question:

Who are the users of student accommodation?

Needs and Preferences

The second section is devoted to identifying the factors that determine the choice of housing, subjective assessment of needs and living conditions. The structure of the section reflects the key parameters of housing affordability, location choice, mobility behaviour and social environment.

Affordability The questions assess the financial component of current housing:

- monthly rent within budget (**a_price**) — whether the rent is in proportion to personal budget.
 - price matches apartment condition (**a_cond**) — subjective perception of the fairness of the cost.
 - hidden costs (**a_hidd**) — the presence of additional financial barriers (deposits, commissions).
- This block allows us to assess the degree of financial stress and identify the gap between expectations and market reality.

Location & Proximity

This measures the significance of spatial factors:

- proximity to the university (**l_uni**);
 - accessibility of the ring-bahn as a key transport axis (**l_ring**);
 - availability of inexpensive food (Mensa) (**l_mensa**);
 - availability of green areas (**l_green**);
 - proximity to shops and essential services (**l_grocer**).
- This data allows us to assess which amenities have the greatest influence on the choice of place of residence.

Transportation

The questions are aimed at assessing the quality of the transport infrastructure:

- quality and frequency of public transport (**t_connect**);
 - cycling infrastructure (**t_bike**);
 - availability of night transport (**t_night**).
- This block is critical for understanding how students commute to and from university and how important this is to them.

Social & Living Conditions

The questions clarify the quality of the social environment:

- neighbourhood safety (`s_safe`);
- access to cultural/student events (`s_cultur`);
- proximity to student communities (`s_common`);
- friendliness towards international students (`s_env`);
- inclusiveness (`s_incl`).

This section identifies how the social atmosphere affects satisfaction with accommodation and choice of neighbourhood.

Difficulties and Satisfaction

The third section explores the subjective experience of finding accommodation, stress levels and potential barriers, including discrimination.

The housing search process

- method of finding housing (`way`);
- duration of the search (`term`);
- existence of a long-term contract (`longtermagree`);
- number of applications submitted (`applic_count`).

These questions provide insight into the time and effort required to integrate into the rental market.

Emotional and social impact

- level of stress when searching for housing (`stress`);
 - level of satisfaction with current housing (`satisfied`).
- This helps to identify the degree of individual market pressure.

Problems and discrimination

- the existence of difficulties (`diff`);
- types of discrimination (`discrimination`).

These questions allow us to analyse the vulnerability of different groups of students and assess the social justice of the market.

The interpretation of the third section is beyond the scope of this chapter and will be discussed in more detail in the next chapter, Students' Experiences.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 103 entries, 0 to 102
Data columns (total 34 columns):
```

#	Column	Non -Null Count	D
0	Gender	103 non -null	o
1	Age Group	103 non -null	o
2	Origin	103 non -null	o
3	University	103 non -null	o
4	Study Status	103 non -null	o
5	Address (street & house number)	59 non -null	o
6	Postal Code	103 non -null	o

```

3200 7 Current Rental Price 103 non -null ol
3201 8 Sources of Financing 103 non -null ol
3202 9 Housing Model 103 non -null ol
3203 10 Monthly Rent Within Budget 103 non -null in
3204 11 Price Matches Apartment Condition 103 non -null in
3205 12 Hidden Costs (deposit, agency fees) 103 non -null in
3206 13 Proximity to the University 103 non -null in
3207 14 Proximity to the Ring -Bahn 103 non -null
3208 15 Proximity to the Mensa (or affordable food options) 103 non -null in
3209 16 Proximity to Green Spaces (parks, canals) 103 non -null in
3210 17 Proximity to Grocery Stores/Essential Services 103 non -null in
3211 18 Quality of public transport connections (frequency, reliability) 103 non -null in
3212 19 Bike -friendly infrastructure (storage, parking places, bicycle paths) 103 non -null
3213 20 Night transport options (for late classes/part -time work) 103 non -null
3214 21 Safety in the Neighborhood 103 non -null in
3215 22 Access to Cultural/Student events 103 non -null in
3216 23 Proximity to Student Communities 103 non -null in
3217 24 International student -friendly environment 103 non -null
3218 25 Inclusivity 103 non -null in
3219 26 How did you find your accomodation in Berlin? 103 non -null ol
3220 27 How long have you searched for your long term accommodation in Berlin? 103 non -null ol
3221 28 Do you have a Long Term Rental Agreement? 103 non -null ol
3222 29 How stressful would you describe your housing search? 103 non -null in
3223 30 How many accommodations did you apply for? 103 non -null ol
3224 31 How satisfied are you with your current housing situation? 103 non -null in
3225 32 Did you experience any kinds of difficulties? 103 non -null ol
3226 33 What kind of discrimination did you have to face with? 103 non -null ol
3227 dtypes: int64(18), object(16)
3228 memory usage: 27.5+ KB

```

3229 When working with survey responses, standard data preparation operations used
3230 in data science were performed.

3231 First, the columns and response categories were renamed to shorter and more uni-
3232 form designations. This simplified data access made the code more compact and ensured
3233 uniformity in the structure of the dataset.

```

3234 # Clean up any hidden characters
3235 df.columns = df.columns.str.strip().str.replace('\xa0', ' ', regex=False)
3236
3237 # Renaming cols for data clarity
3238 df = df.rename(columns={
3239     'Gender': 'gender',
3240     'Age Group': 'age',
3241     'Origin': 'origin',
3242     'University': 'uni',
3243     'Study Status': 'stat',
3244     'Address (street & house number)': 'adr',
3245     'Postal Code': 'plz',
3246     'Current Rental Price': 'price',
3247     'Sources of Financing': 'fin',
3248     'Housing Model': 'model',
3249     'Monthly Rent Within Budget': 'a_price',
3250     'Price Matches Apartment Condition': 'a_cond',
3251     'Hidden Costs (deposit, agency fees)': 'a_hidd',
3252     'Proximity to the University': 'l_uni',

```

```

3253     'Proximity to the Ring -Bahn': 'l_ring',
3254     'Proximity to the Mensa (or affordable food options)': 'l_mensa',
3255     'Proximity to Green Spaces (parks, canals)': 'l_green',
3256     'Proximity to Grocery Stores/Essential Services': 'l_grocer',
3257     'Quality of public transport connections (frequency, reliability)': 't_connect',
3258     'Bike -friendly infrastructure (storage, parking places, bicycle paths)': 't_bike',
3259     'Night transport options (for late classes/part -time work)': 't_night',
3260     'Safety in the Neighborhood': 's_safe',
3261     'Access to Cultural/Student events': 's_cultur',
3262     'Proximity to Student Communities': 's_common',
3263     'International student -friendly environment': 's_env',
3264     'Inclusivity': 's_incl',
3265     'How did you find your accomodation in Berlin?': 'way',
3266     'How long have you searched for your long term accommodation in Berlin?': 'term',
3267     'Do you have a Long Term Rental Agreement?': 'longtermagree',
3268     'How stressful would you describe your housing search?': 'stress',
3269     'How many accommodations did you apply for?': 'applic_count',
3270     'How satisfied are you with your current housing situation?': 'satisfied',
3271     'Did you experience any kinds of difficulties?': 'diff',
3272     'What kind of discrimination did you have to face with?': 'discrimination'
3273 })

3274 #Renaming long answers
3275 model_mapping = {
3276     "Studierendenwerk Dorms (State -subsidised student housing)": "PBSA",
3277     "PBSA (e.g., Neon Wood - private investor housing with high amenities)": "PBSA",
3278     "Wohngemeinschaften (WGs - shared flats, informal sublets)": "Wohngemeinschaften (WG)",
3279     "Living at Home (with family, commuting)": "Living at Home",
3280     "Commons / Co -ops (e.g., STUK - self -managed, solidarity -based housing)": "Co -ops / Commons",
3281     "PPPs (e.g., Camplus - public -private university housing models)": "PPP Housing",
3282     "Private Apartment / Studio": "Private Apartment"
3283 }
3284
3285 #Apply the renaming
3286 df["model_short"] = df["model"].replace(model_mapping)

```

3287 Secondly, since the correct type of variables is critical for further analysis, numerical indicators were converted from text format to numerical format. This made it possible to perform statistical calculations, construct distributions, and apply quantitative analysis methods.

```

3291 int_columns = [
3292     'a_price', 'a_cond', 'a_hidd', 'l_uni', 'l_ring', 'l_mensa', 'l_green',
3293     'l_grocer', 't_connect', 't_bike', 't_night', 's_safe', 's_cultur',
3294     's_common', 's_env', 's_incl', 'stress', 'satisfied'
3295 ]
3296
3297
3298 for col in int_columns:
3299     df[col] = pd.to_numeric(df[col], errors='coerce').astype('Int64')

```

3300 Thirdly, multiple-choice questions were standardised: answers were broken down into binary indicators (dummy variables), which made it possible to correctly account for each selected option when modelling and analysing frequencies. This approach ensures data interpretability and prevents the loss of information that is characteristic of

aggregated text fields. In addition, gaps were eliminated, and categories were brought into a uniform format.

```

3306 fin_categories = ['Public Funding', 'Private Support', 'Part -Time Job', 'Scholarship']
3307
3308 # Step 1: Clean and split the values
3309 df['fin_e'] = (
3310     df['fin']
3311     .str.split(',') # Split by comma
3312     .apply(lambda x: [item.strip() for item in x] if isinstance(x, list) else [])
3313     .apply(lambda x: [item for item in x if item]) # Remove empty strings
3314 )

```

Together, these steps provided a clean, consistent, and analysable dataset that meets the methodological requirements of quantitative behavioral research.

5.3 Users Personal Data

5.3.1 Student Status

The majority of survey participants (98.1%) were university students, including undergraduates, postgraduates and college students (Figure 42). This group constitutes the core user base of student accommodation and most accurately reflects the typical demand for long-term, affordable housing near universities.

However, it is important to emphasise that student accommodation in Berlin serves a much wider range of educational tracks. A significant proportion of users are applicants from other countries who are studying at a preparatory college (Studienkolleg) and preparing to enrol at universities in the German capital. These students have specific needs related to their short stay and high dependence on the availability of housing when taking their first steps in a new country.

Exchange students, including participants in Erasmus and Double-Degree programs, form a separate category. Unlike regular university students, they are mainly interested in short-term rentals of 3 to 12 months, which creates additional seasonal and structural demand for flexible, temporary forms of accommodation.

In addition, the urban student ecosystem includes students at higher levels of education — doctoral students, graduates seeking additional qualifications, interns, and young research and teaching staff. Their housing needs vary greatly, ranging from short-term stays during internships to long-term accommodation near research institutes.

All of these groups represent a significant research gap, as they fall outside the scope of both the original study by Von Wittenhorst Zu Sonsfeld and Beusker (2024) and the current work. Their inclusion requires separate analysis, as they create additional hidden demand, which in practice increases the burden on the student housing market but is rarely taken into account in regulatory and statistical assessments.

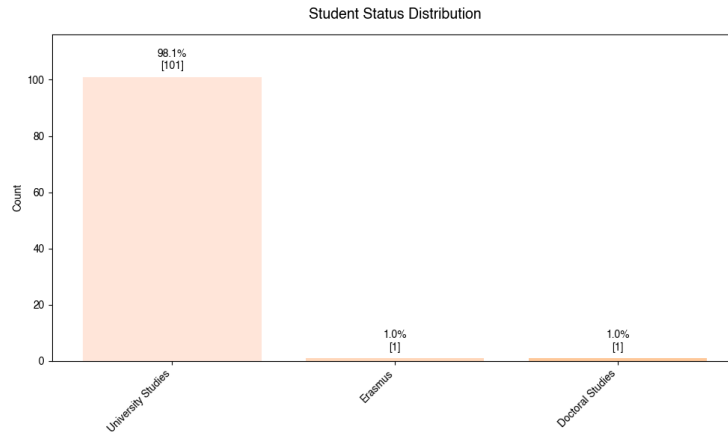


Figure 42. Student Status Distribution

5.3.2 Age

According to the survey results, the largest groups were students aged 24–26 (30.1%) and 21–23 (29.1%), who together make up almost two-thirds of the entire sample. These age categories reflect the typical structure of Berlin’s student population, which is dominated by senior undergraduate and master’s students.

The younger group — students under the age of 21 — accounts for 17.5%, which corresponds to the proportion of first-year students and students in preparatory programs. The remaining categories — 27–29 years old and over 30 years old — include late starters, doctoral students, participants in international programs, and young professionals who combine work with study; each of these groups accounts for 11.7%.

The data on student age obtained from the survey is consistent with the official statistics from Statistik Berlin Brandenburg (2025), where 57.8% of students are over 25 years old.

Thus, the age distribution (Figure 43) shows a predominance of mature students in the later stages of their studies, which influences their housing preferences, level of independence, and requirements for the quality and affordability of housing.

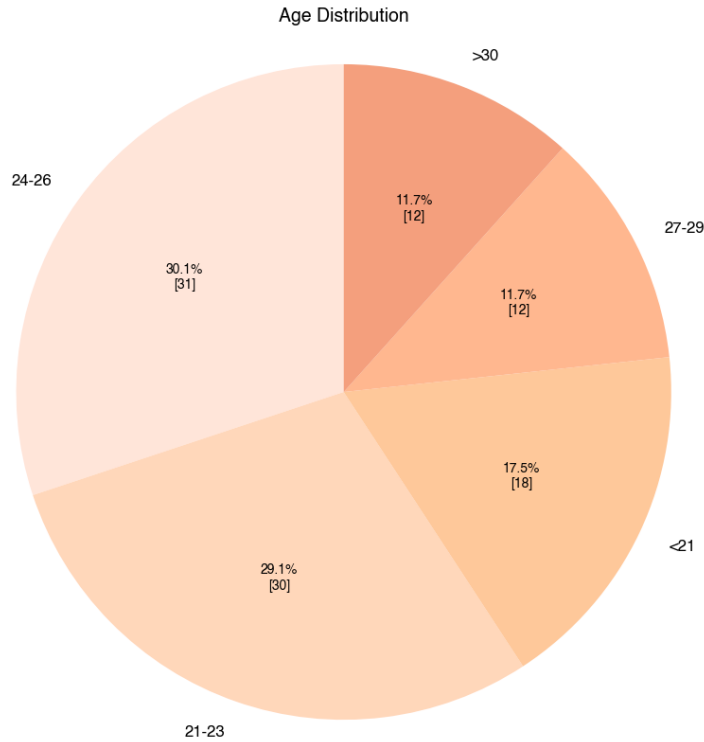


Figure 43. Age Distribution

5.3.3 Origin

Most of the survey participants are international students, with the majority coming from non-European countries (Non-EU) — 75.7%, while students from European Union countries make up another 13.6% of the sample. Only 10.7% of participants are German citizens.

This distribution (Figure 44) is fully consistent with the target audience of the study, which focuses on analysing the living conditions and barriers faced by international students — the group experiencing the most significant difficulties in the Berlin rental market.

The high proportion of non-EU students further reflects Berlin's growing international appeal as an educational center and highlights the relevance of studying the pressure on the student housing market.

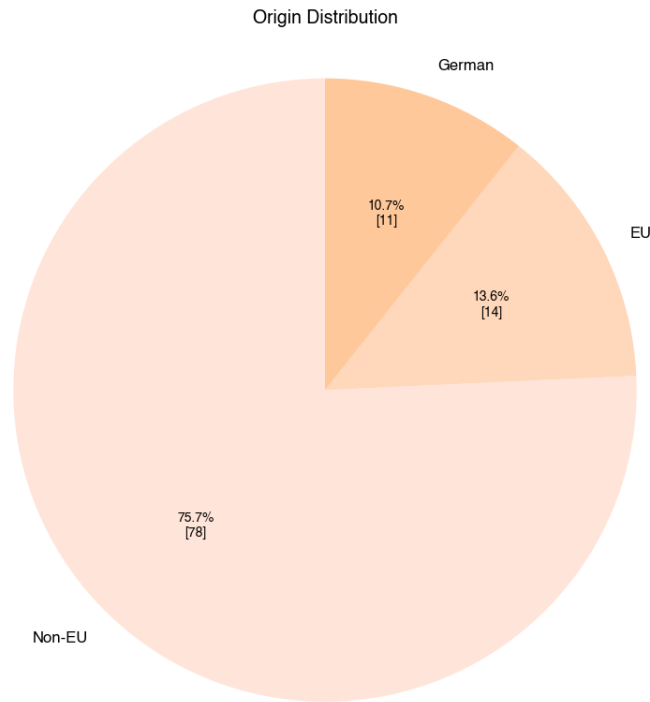


Figure 44. Origin Distribution

5.3.4 Housing Model

Students in Berlin use a wide range of accommodation models (Figure 45), each with its own structural features, advantages, risks and local context. These models form a heterogeneous student housing market, where public mechanisms coexist with the private sector, cooperatives and informal solutions. Let us consider three of them that are most frequently mentioned in the literature.

Purpose-Built Student Accommodation (PBSA) Purpose-Built Student Accommodation can be divided into two categories: public and private.

State-owned purpose-built student accommodation is funded and managed by public agencies and is the most affordable form of student housing. Its key advantages are low cost and social diversity among residents. However, limited capacity and long waiting lists significantly reduce their availability. This makes them a scarce resource and increases pressure on other types of housing.

Private purpose-built student accommodation is an investment-oriented segment of student housing, offering modern infrastructure, additional services and the possibility of immediate occupancy. At the same time, such properties are very expensive, which makes them inaccessible to many students and contributes to spatial segregation.

Although 53.4% of users indicated that they live in PBSAs, in reality, only 5-6% of students in Berlin live in PBSAs. The majority of students live in shared flats or other corporate models due to the predominantly rental housing situation in Germany (Janet Merkel, 2025).

Wohngemeinschaften (WGs) One of the most common forms of accommodation for students in Berlin is shared rented flats with an informal system of renting and subletting. The most important advantages of WGs are flexibility, affordability and a high level of social interaction. However, such conditions are often accompanied by in-

stability, short-term contracts and discrimination when moving in, which particularly affects foreign students.

Living at Home

This option involves living in the parental home with daily commutes to the place of study. It ensures no or minimal rental costs, but increases the risk of social isolation and imposes time costs associated with long commutes. Unlike in Eastern European countries, this model is much less common in Berlin (Janet Merkel, 2025).

With regard to Housing Model Distribution (Figure XX), the survey results are heavily skewed towards users of purpose-built student accommodation, as the main target group of the study.

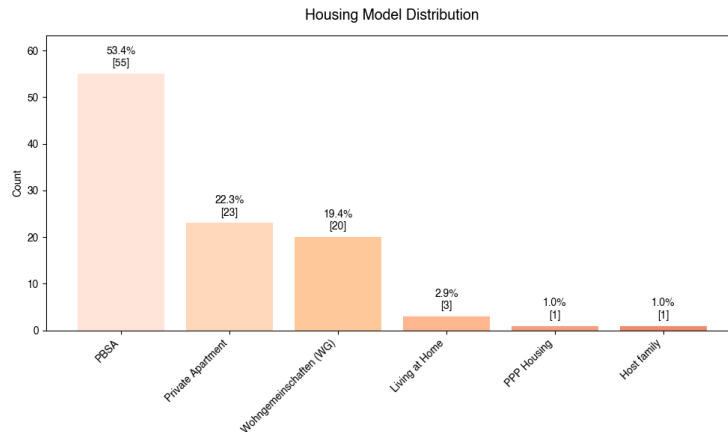


Figure 45. Housing Model Distribution

5.3.5 Sources of Financing

According to the survey results (Figure 46), the most common source of financing for students' educational and personal expenses is part-time employment: almost half of the respondents (47.3%) work alongside their studies to cover their daily expenses and living costs.

A significant proportion of respondents (36.6%) receive financial support from private individuals, most often from parents or relatives. However, formal support mechanisms, such as government funds (5.3%) and academic scholarships (6.9%), are used relatively rarely, which highlights the limited official channels of assistance for foreign students.

It is particularly telling that 81.3% of students identify rent as their largest expense. This confirms the key finding of the study: the cost of living in Berlin has a decisive influence on students' economic behavior and constitutes the main financial pressure affecting both their choice of place of residence and the need for additional employment.

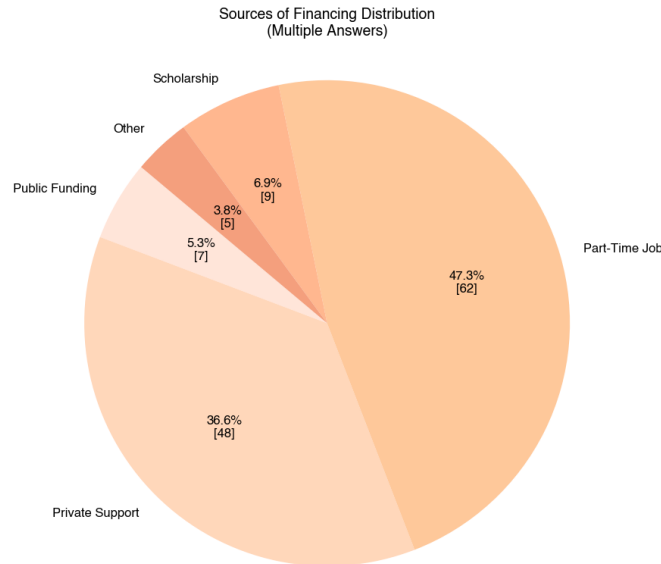


Figure 46. Sources of Financing Distribution

5.4 Needs and Preferences

In the second part of the survey, the target audience was asked to answer the question: what is most important when looking for housing? This helps to identify the needs and preferences of international students across the city.

The **Affordability** category covers issues such as whether the rent fits the budget, whether the price is commensurate with the quality of the accommodation, and whether there are any hidden costs. The **Location & Proximity** block records students' spatial preferences, including proximity to the university, everyday infrastructure and leisure areas. **Transportation** reflects accessibility and quality of mobility, while **Social & Living Conditions** describes the social environment, sense of security, inclusiveness, and access to student communities.

To ensure comparability of results, all responses were converted to a numerical format and aggregated within the relevant groups. For each question, respondents were asked to rate the importance of a particular factor on a scale from 1 (not important at all) to 10 (extremely important). In addition, a uniform order of display of indicators was established, which made it possible to maintain the logical structure of the analysis and ensure consistent interpretation of the results both at the level of individual indicators and at the level of thematic blocks.

Thus, this stage of the analysis serves as a diagnostic cross-section, allowing us to identify key areas of tension and satisfaction in students' perceptions of the Berlin housing market and to create a basis for further research of subjective assessments.

5.4.1 General Statistics

As a preliminary analysis of the data, similar to the Descriptive Market Statistics chapter, a generalised summary statistics assessment of respondents' answers was conducted across four key thematic topics reflecting the main dimensions of the student housing experience: affordability, location and proximity, transportation, and social and living conditions.

```
group_columns = {
```

```

3457     'Affordability': ['a_price', 'a_cond', 'a_hidd'],
3458     'Location & Proximity': ['l_uni', 'l_ring', 'l_mensa', 'l_green', 'l_grocer'],
3459     'Transportation': ['t_connect', 't_bike', 't_night'],
3460     'Social & Living Conditions': ['s_safe', 's_cultur', 's_common', 's_env']
3461 }
3462
3463 order = [
3464     'a_price', 'a_cond', 'a_hidd', 'Affordability',
3465     'l_uni', 'l_ring', 'l_mensa', 'l_green', 'l_grocer', 'Location & Proximity',
3466     't_connect', 't_bike', 't_night', 'Transportation',
3467     's_safe', 's_cultur', 's_common', 's_env', 's_incl', 'Social & Living Conditions'
3468 ]
3469 # 'stress', 'satisfied'
3470
3471 item_rename = {
3472     'a_price': 'Monthly Rent Within Budget',
3473     'a_cond': 'Price Matches Apartment Condition',
3474     'a_hidd': 'Hidden Costs',
3475     'l_uni': 'Proximity to the University',
3476     'l_ring': 'Proximity to the Ring -Bahn',
3477     'l_mensa': 'Proximity to the Mensa',
3478     'l_green': 'Proximity to Green Spaces (parks, canals)',
3479     'l_grocer': 'Proximity to Essential Services',
3480     't_connect': 'Quality of Public Transport Connections',
3481     't_bike': 'Bike -Friendly Infrastructure',
3482     't_night': 'Night Transport Options',
3483     's_safe': 'Safety in the Neighborhood',
3484     's_cultur': 'Access to Cultural/Student Events',
3485     's_common': 'Proximity to Student Communities',
3486     's_env': 'International Student -Friendly Environment',
3487     's_incl': 'Inclusivity',
3488     'stress': 'Level of Stress',
3489     'satisfied': 'Level of Satisfaction',
3490     'Affordability': 'Affordability',
3491     'Location & Proximity': 'Location & Proximity',
3492     'Transportation': 'Transportation',
3493     'Social & Living Conditions': 'Social & Living Conditions'
3494 }
3495
3496 int_columns = [
3497     'a_price', 'a_cond', 'a_hidd', 'l_uni', 'l_ring', 'l_mensa', 'l_green',
3498     'l_grocer', 't_connect', 't_bike', 't_night', 's_safe', 's_cultur',
3499     's_common', 's_env', 's_incl'
3500 ]
3501 # 'stress', 'satisfied'
3502 group_columns = {
3503     'Affordability': ['a_price', 'a_cond', 'a_hidd'],
3504     'Location & Proximity': ['l_uni', 'l_ring', 'l_mensa', 'l_green', 'l_grocer'],
3505     'Transportation': ['t_connect', 't_bike', 't_night'],
3506     'Social & Living Conditions': ['s_safe', 's_cultur', 's_common', 's_env']
3507 }

```

Based on the descriptive statistics presented (Table XX), several key patterns can be identified in students' preferences for living conditions.

The **Affordability** category shows the highest and most consistent ratings among all the blocks analysed. The average value for the group is 8.29 with a low standard deviation, which indicates relative uniformity of perception. The affordability of rent is particularly highly rated (mean = 9.12, median = 10), reflecting students' adaptation to financial constraints or forced compromises when choosing accommodation. At the same time, issues of hidden costs and value for money have a higher variance, indicating heterogeneous experiences and the presence of problematic cases.

The **Location & Proximity** block is characterised by a significantly lower average value (5.74) and greater variability in ratings. The most important aspect is proximity to canteens and affordable places to eat (mean = 3.52), as well as to universities and the Ring-Bahn, reflecting the spatial gap between places of study and actual areas of residence for students. At the same time, access to everyday services is rated relatively highly, indicating the compensatory role of urban infrastructure outside university clusters.

In the **Transportation** category, there is a marked contrast between individual indicators. The quality of public transport receives one of the highest ratings in the sample (mean = 8.53), confirming the important role of Berlin's transport system as a compensatory mechanism for long-distance commuting. At the same time, cycling infrastructure is rated significantly lower (mean = 4.42) and shows high dispersion, indicating a spatially uneven distribution of bicycle-friendly conditions.

The most vulnerable group of indicators is **Social & Living Conditions**, with an average score of 5.19. Despite a relatively high sense of security in the neighbourhood, access to student communities, cultural events and a sense of an inclusive environment receive low ratings and are characterised by high variability. This indicates social fragmentation in student housing and weak integration of some students, especially international students, into local social structures.

Overall, the results of the initial analysis show that economic affordability and transport connectivity are priority factors in a stressed market.

	Item	mean	std	min	median	max
0	Monthly Rent Within Budget	9.12	1.13	5.0	10.0	10.0
1	Price Matches Apartment Condition	8.06	1.82	3.0	8.0	10.0
2	Hidden Costs	7.7	2.06	2.0	8.0	10.0
3	Affordability	8.29	1.17	<NA>	8.33	<NA>
4	Proximity to the University	6.1	2.18	1.0	6.0	10.0
5	Proximity to the Ring-Bahn	5.67	2.47	1.0	6.0	10.0
6	Proximity to the Mensa	3.52	2.4	1.0	3.0	10.0
7	Proximity to Green Spaces (parks, canals)	5.67	2.39	1.0	6.0	10.0
8	Proximity to Essential Services	7.72	1.91	1.0	8.0	10.0
9	Location & Proximity	5.74	1.6	<NA>	5.8	<NA>
10	Quality of Public Transport Connections	8.53	1.6	4.0	9.0	10.0
11	Bike-Friendly Infrastructure	4.42	2.84	1.0	4.0	10.0
12	Night Transport Options	6.94	2.38	1.0	8.0	10.0
13	Transportation	6.63	1.57	<NA>	6.67	<NA>
14	Safety in the Neighborhood	7.95	2.24	1.0	8.0	10.0
15	Access to Cultural/Student Events	4.39	2.56	1.0	4.0	10.0

Table XX. General Student Needs and Preferences

Correlation analysis of student needs and preferences (Figure 47) reveals stable relationships between individual aspects of housing choice, thereby deepening the interpretation of descriptive statistics results. Overall, the correlation matrix shows moderate but structurally meaningful relationships, indicating the multidimensional nature of student housing capacity, where no single factor is completely isolated.

Affordability indicators show relatively weak correlations with spatial and social characteristics. Rent affordability and hidden costs are more closely related to each other than to other blocks, confirming their autonomous role in the decision-making process. This indicates that financial constraints are seen by students as a separate structural condition rather than an element of the integrated quality of the living environment.

Blocks associated with **Location & Proximity** form a dense cluster of interrelationships. Proximity to the university, the ring-bahn, the mensa, green areas, and basic services shows consistent positive correlations with each other (in the range of approximately 0.27–0.50). This indicates that students tend to view accessibility not in terms of individual objects, but as a comprehensive characteristic of the living environment. A particularly strong link is observed between proximity to the Mensa and student communities, as well as between green areas and cultural access, which emphasises the role of everyday routes and infrastructure in creating a comfortable environment.

Transportation indicators act as a link between spatial and social characteristics. The quality of public transport correlates moderately with proximity to key urban functions and night-time transport, reflecting the importance of the transport system as a mechanism for commuters. At the same time, cycling infrastructure shows more selective links, including a negative correlation with the sense of safety in the area, which may indicate local conflicts between different types of mobility or spatial heterogeneity of cycling conditions in the city.

Social & Living Conditions form a separate but interrelated cluster. Access to cultural events, proximity to student communities, an internationally friendly environment, and inclusiveness demonstrate moderate positive correlations. The strongest link is observed between student communities and an international-friendly environment, which highlights the importance of social networks and “social density” for international students. At the same time, the feeling of safety is weakly correlated with most social indicators, which may indicate a gap between the physical and social comfort of the environment.

Overall, correlation analysis confirms that student housing choices are shaped by the intersection of three factors: economic adaptation, spatial accessibility, and social integration. At the same time, it is precisely spatial and social connections that are the most structured, while financial parameters act more as a strict limitation than as an active factor in shaping preferences.

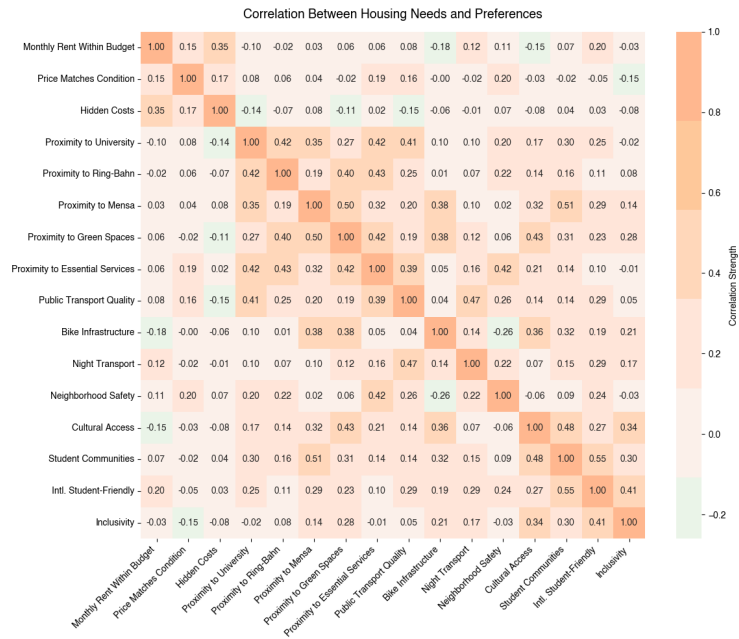


Figure 47. Correlation Between Housing Needs and Preferences

5.4.2 Statistics by Origin

In order to avoid generalising students, it is necessary to conduct a comparative analysis of students' housing needs and preferences (Table 24) depending on their country of origin (German, EU, Non-EU) and identify both common patterns and structural differences between groups.

origin_cat	German			EU			Non-EU		
stat	mean	median	std	mean	median	std	mean	median	std
Item									
Monthly Rent Within Budget	9.09	5.0	0.94	9.29	5.0	0.99	9.09	5.0	1.19
Price Matches Apartment Condition	7.18	5.0	1.33	7.50	5.0	1.87	8.28	5.0	1.83
Hidden Costs	7.36	5.0	1.50	8.57	5.0	1.83	7.59	5.0	2.15
Affordability	7.88	5.0	0.98	8.45	5.0	0.96	8.32	5.0	1.23
Proximity to the University	5.00	5.0	2.28	5.64	5.0	2.27	6.33	5.0	2.11
Proximity to the Ring-Bahn	4.55	5.0	2.77	5.21	5.0	2.67	5.91	5.0	2.36
Proximity to the Mensa	4.00	5.0	2.49	4.14	5.0	2.88	3.35	5.0	2.30
Proximity to Green Spaces (parks, canals)	6.00	5.0	2.14	5.50	5.0	2.79	5.65	5.0	2.37
Proximity to Essential Services	6.09	5.0	2.30	7.50	5.0	2.68	7.99	5.0	1.57
Location & Proximity	5.13	5.0	1.91	5.60	5.0	1.98	5.85	5.0	1.48
Quality of Public Transport Connections	7.27	5.0	1.56	8.64	5.0	1.69	8.69	5.0	1.52
Bike-Friendly In-	6.09	5.0	2.34	3.86	5.0	2.85	4.28	5.0	2.85

Table 24. Student Needs and Preferences by Origin

In the **Affordability** section, the differences between the groups are relatively slight. For all categories of students, the affordability of rent is rated very highly (9), confirming the universal importance of the price factor. At the same time, students from the EU and especially non-EU countries rate the value for money of housing and the importance of hidden costs slightly higher, which may reflect their lower awareness of the local market and higher costs of searching for housing. In the aggregate indicator of affordability, it is foreign students who demonstrate the highest values, which emphasises their increased sensitivity to financial constraints.

The **Location & Proximity** section shows more noticeable differences. On average, non-EU students rate proximity to the university, the ring-bahn and, in particular, basic services higher than German students. This indicates that international students are more dependent on a compact and well-served urban environment where everyday functions are accessible without complicated navigation around the city. German students, on the other hand, give lower ratings for proximity, which may be due to their better knowledge of the city and greater willingness to make spatial compromises.

The differences are most noticeable in the Transportation category. Foreign students (EU and non-EU) rate the quality of public transport and night-time transport options significantly higher, while German students tend to value cycling infrastructure more. This reflects differences in mobility practices: for local students, bicycles are a familiar mode of transport, while international students rely more on public transport, especially at night.

The Social & Living Conditions section reveals the most contrasting results. Non-EU students rate the importance of neighbourhood safety significantly higher, indicating increased vulnerability and a need for a subjective sense of security in an unfamiliar urban environment. At the same time, foreign students rate access to cultural events, student communities and an international-friendly environment lower than German students, which may indicate structural barriers to integration and limited inclusion in local social networks. German students, on the other hand, rate their proximity to student communities and cultural life higher, reflecting their more stable position in the urban and academic environment.

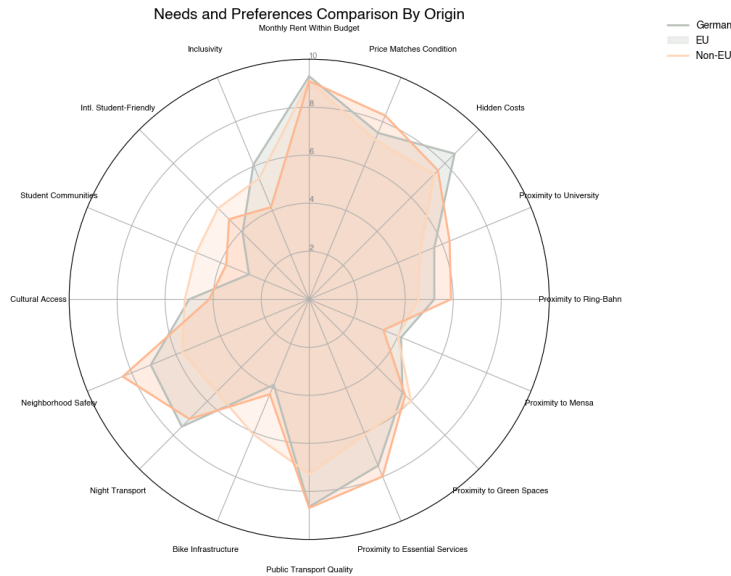


Figure 48. Student Needs and Preferences by Origin

Overall, the results (Figure 48) show that students' origins have a significant impact on the structure of their housing priorities. For international students, accessibility, transport and safety are key, while for German students, social and cultural aspects as well as alternative forms of mobility play a greater role. These differences highlight the need for a differentiated approach to analysing student housing capacity and confirm that a single 'student' model does not reflect the real diversity of user groups in the student housing market in Berlin.

```
df.to_excel(os.path.join(hf, "user_data.xlsx"), index=False)
```

5.5 4.3 Student Experiences

The student experience of finding accommodation clearly demonstrates the difficulties and problems faced by the most vulnerable groups during the ongoing housing crisis. In order to better understand the students, individual interviews were conducted, where open-ended questions were asked to guide the storytelling.

These problems and difficulties need to be analysed before proposing ways to resolve the student housing crisis, as they serve as barriers between student accommodation and users.

5.5.1 Set Up

Import Libraries

```
import pandas as pd
import geopandas as gpd
import chardet
import gspread
import seaborn as sns
import matplotlib.pyplot as plt
import textwrap
from pywaffle import Waffle
import numpy as np
from matplotlib.patches import Patch
from math import pi
```

Data Import

```
fp = r"../inputs/4.3_students_experiences/user_data.xlsx"
df = pd.read_excel(fp)
```

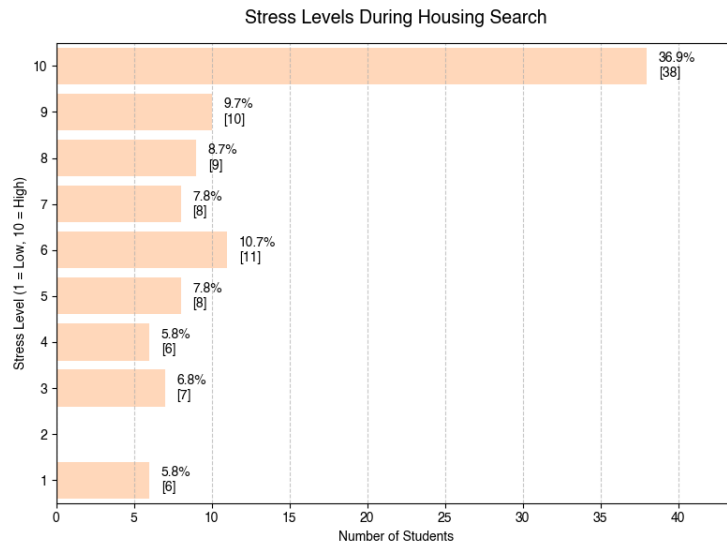
5.6 Difficulties and Problems

5.6.1 Too Much Competition

One of the most frequently mentioned problems when looking for accommodation is excessive competition, which permeates all segments of the student rental market. Virtually all respondents describe a situation in which the number of applicants significantly exceeds the available supply, especially in the budget segment. Even when actively submitting applications, students rarely receive responses, and scheduled viewings are often cancelled at the last minute. As one participant noted: **The worse problem is competition. I wrote people and even arranged meetings, but they did not respond and cancelled meetings because they rented the place already to another person.** In such a competitive environment, time becomes a critical factor — a delay of even a few hours can mean losing an opportunity: **You had to apply very quickly, or you'd be too late.**

5.6.2 Stress

Due to excessive competition, 55.3% of student housing users experience stress levels (Figure 49) above 8 points. Similar to the high demand–capacity pressure indicators recorded in the spatial analysis, high stress levels reflect not only a structural shortage of student housing, but also a combination of related difficulties, including uncertainty, temporary solutions, and emotional costs associated with the housing search process.



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Figure 49. Stress Levels During Housing Search

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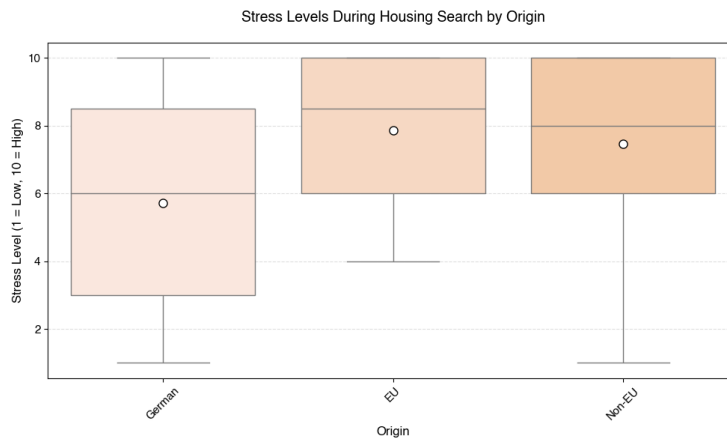
The stress levels diagram (Figure 50) demonstrates differences and similarities in stress levels among students with different backgrounds when searching for accommodation. For German students, the median stress level is around 6 points, with a relatively wide range of responses, indicating a heterogeneous experience — from a relatively comfortable search to an extremely stressful one. At the same time, for students from EU and non-EU countries, the median values are significantly higher (around 8–9 points), and the upper quartiles in both cases reach the maximum value on the scale, indicating a systematically high level of stress.

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It is particularly telling that the distributions for the EU and non-EU groups largely overlap: despite differences in legal status and access to resources, both groups face comparable levels of emotional stress.



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Figure 50. Stress Levels During Housing Search by Origin

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Uncertainty, constant rejections and fear of being left homeless directly affect mental health and academic adaptation. One participant directly links the housing crisis to depression: It was kind of difficult and, to be honest, scary because of the new country.

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Another student emphasises that the stress reached such a level that professional help was required: I was really very depressed, but I took therapy at the TU.

Thus, housing instability becomes not a temporary difficulty, but a factor that directly affects the well-being and ability of students to study and integrate.

5.6.3 Searching Time

The distribution of searching time (Figure 51) demonstrates the high heterogeneity of student experiences and highlights the structural tension in the market. For the largest share of respondents, the search took between 1 and 3 months (35.0%), which indicates a relatively ‘standard’ scenario, requiring, nevertheless, significant time and emotional resources. Another 27.2% of students were able to find accommodation in less than a month, which is probably due to a combination of favorable market conditions and an increased budget.

At the same time, almost one in five students (21.4%) searched for accommodation for more than six months, and another 16.5% searched for 3 to 6 months. Taken together, this means that more than a third of respondents faced a protracted and problematic search that went beyond one semester. Such long search periods directly correlate with increased stress and housing instability and point to a systemic shortage of affordable and long-term student housing in Berlin, as well as a general lack of transparency in the search process.

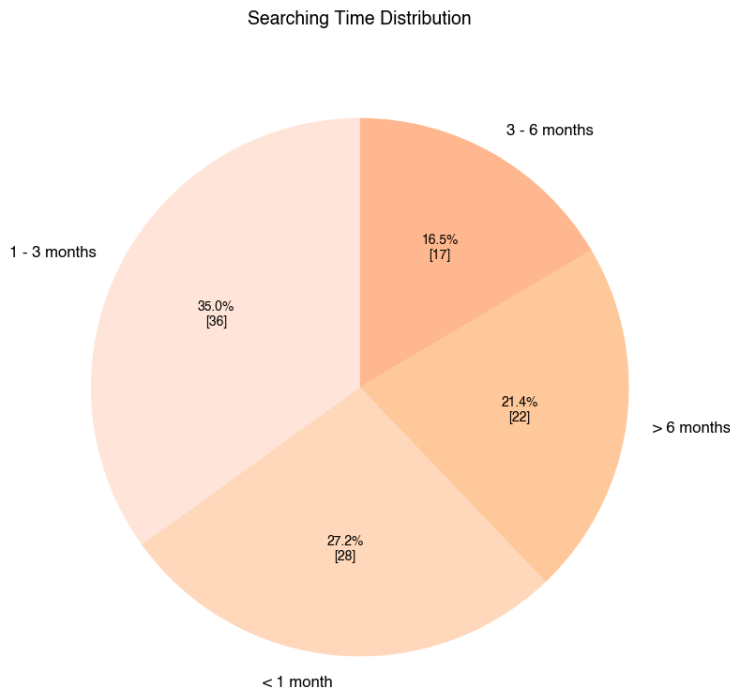


Figure 51. Searching Time Distribution

5.6.4 Applications Count

The application count distribution diagram (Figure 52) clearly reflects the scale of competition in the student housing market in Berlin and the efforts that students have to make in their search.

The largest group of respondents (35.9%) submitted between 1 and 5 applications, which is typically characteristic of students who managed to find accommodation relatively quickly — most often through acquaintances, university channels or state-run student halls of residence. Another 16.5% submitted between 6 and 15 applications, which

may also indicate relatively favorable search conditions or a lucky coincidence in terms of timing and budget.

At the same time, almost half of the survey participants reported a significantly higher level of search effort. About 14.6% of students submitted 16–30 applications, and the same number submitted 31–50 applications. It is particularly noteworthy that 18.5% of respondents submitted more than 50 applications, including 10.7% who submitted more than 100 applications. These figures indicate a high degree of frustration, a prolonged search and a systemic shortage of available housing options.

Thus, the distribution of applications highlights the polarisation of the search experience: some students find accommodation relatively quickly, while for a significant proportion, the search becomes a protracted and time-consuming process. Combined with high levels of stress, this confirms the existence of a structural imbalance between supply and demand in the student housing market in Berlin.

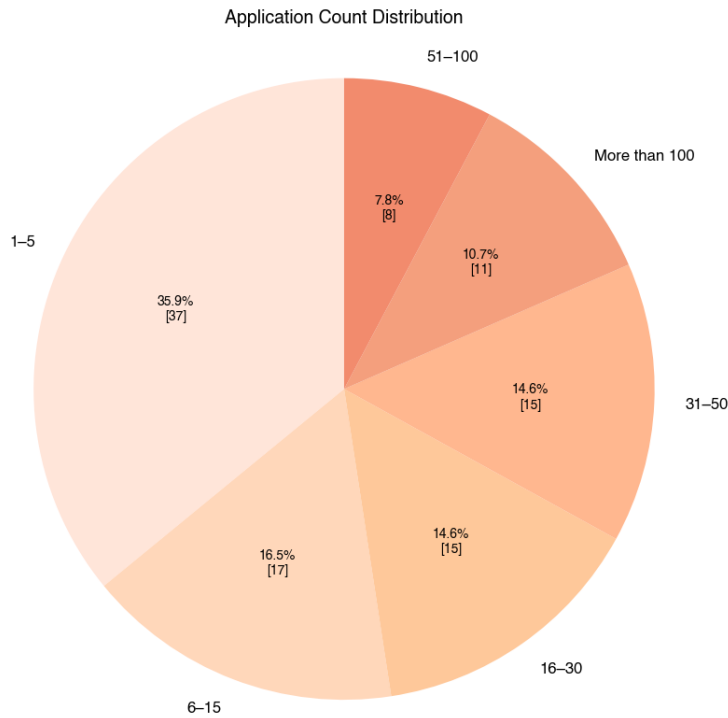


Figure 52. Application Count Distribution

5.6.5 Short Term Agreement

Almost a quarter of students (24.3%) do not have a long-term rental contract for a period of one year or more throughout their studies (Figure 53). Given that the duration of studies can range from 2 to 6 years, this means that a significant proportion of students have to repeatedly search for accommodation, facing recurring stress, uncertainty, and additional financial costs.

Against this backdrop, state-run student halls of residence remain the most stable and reliable landlords. They usually offer one-year contracts with the possibility of renewal, providing basic housing security and predictability of living conditions. However, the limited capacity of such accommodations significantly reduces their systemic role and does not compensate for the instability characteristic of the private segment of the student housing market.

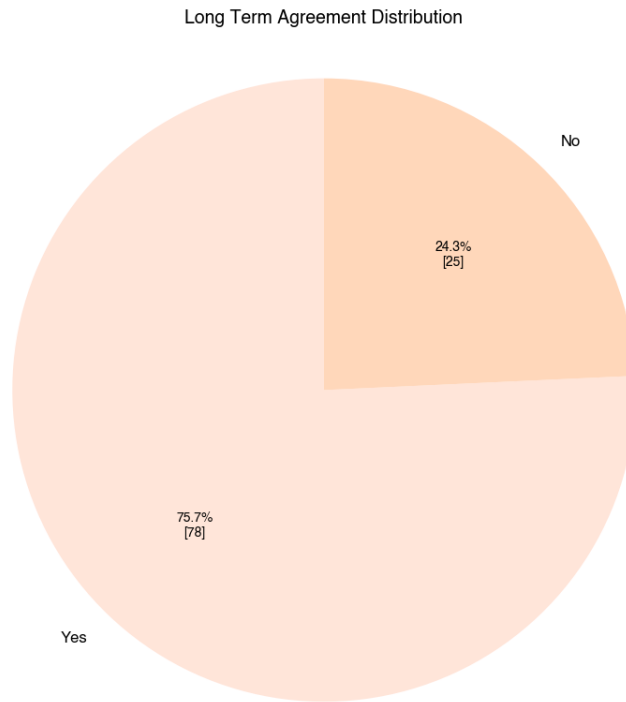


Figure 53. Long Term Agreement Distribution

5.6.6 No Anmeldung

The lack of a long-term rental agreement is directly linked to the issue of registering at the place of residence (Anmeldung), which, in turn, is a mandatory condition for extending visas and residence permits in Germany. Several survey participants who did not have a long-term contract indicated that they had to pay extra to register their residence or register at the address of acquaintances, which indicates the prevalence of informal and legally vulnerable practices in the student housing market.

In addition, the absence of a long-term contract often indicates subletting or re-renting from previous tenants. This practice is usually accompanied by higher rents and, at the same time, deprives students of the right to official registration. This creates a vicious circle of housing instability, in which short-term rentals increase financial pressure, legal insecurity and administrative risks, especially for international students.

Several participants indicated that the possibility of registration was promised before signing the contract, but later turned out to be impossible due to illegal subletting. This creates serious legal and administrative risks: **Before signing the contract, I was told that Anmeldung would be possible, and afterwards the landlord told us that it isn't.** In one interview, the situation is described as a vicious circle: **You need papers to get an Anmeldung, and you need an Anmeldung before you can apply for your residence permit.**

Many students are forced to live without official registration or use old addresses, which creates a chain of further problems — from renewing their residence permits to accessing public services. Anmeldung is transformed from a basic administrative procedure into a tool of pressure and control by landlords.

3771 **5.6.7 Hidden Costs**

3772 In addition to high rents, students regularly encounter hidden and non-transparent
3773 costs that significantly increase the actual cost of living and make housing less afford-
3774 able than advertised. These costs include high deposits, additional payments for util-
3775 ities, furniture, and internet, as well as administrative fees and agency services.

3776 The search for housing takes place in parallel with the start of studies, adaptation
3777 to a new country and, in some cases, a lack of social and financial support. One respon-
3778 dent sums up the general mood as follows: **The entire system is screwed. Students**
3779 **who don't have financial support from their families have it especially rough.**

3780 **5.6.8 Scams**

3781 Housing shortages and high competition make students particularly vulnerable to
3782 fraud in the rental market. Interviews show that some respondents lost significant amounts
3783 of money due to fake advertisements, fictitious landlords, or requests to transfer a de-
3784 posit before viewing the property.

3785 One of the most dangerous and financially painful types of fraud in the rental mar-
3786 ket involves fake offers and advance deposit payments, where the victim is told that a
3787 flat has been found and that a deposit (Kaution) must be transferred before meeting or
3788 signing the contract. This happened to one participant in the study, who noted: **That's**
3789 **how 300 other people lost 2000 euros of deposit...** The scheme involves the prospec-
3790 tive tenant transferring a significant amount of money as a deposit or advance payment,
3791 waiting for the keys and confirmation of the transaction, but after the transfer, contact
3792 with the 'landlord' is lost and they disappear. This is a typical sign of a rental scam —
3793 the scammer creates the appearance of legitimacy, but has no right to rent out the prop-
3794 erty, and the transferred funds become non-refundable.

3795 When time is short and there aren't many options, students often have to make
3796 quick decisions, which makes it harder for them to think critically about the risks. In-
3797 ternational students who are outside Germany or aren't familiar with local laws and mar-
3798 ket conditions are especially vulnerable. The fear of being left without accommodation
3799 leads to unsafe decisions, including temporary accommodation without a contract or in
3800 illegal conditions. In this context, fraud is not a separate problem, but a direct conse-
3801 quence of the structural crisis in student housing, in which the shortage and asymme-
3802 try of information systematically work against students.

3803 **5.6.9 Discrimination**

3804 Although more than half of respondents formally stated that they had not encoun-
3805 tered direct discrimination (Figure 54), qualitative interviews show that discrimination
3806 often takes hidden and institutionalised forms. The most common practices are ignor-
3807 ing applications, failing to respond, and rejecting applications without explanation. One
3808 student emphasises: **Haven't encountered direct discrimination, rather landlords**
3809 **ignore applications.** Such "invisible discrimination" is particularly characteristic of
3810 an overheated market, where refusal does not require justification but systematically af-
3811 fects students, foreigners and people with diverse social profiles.

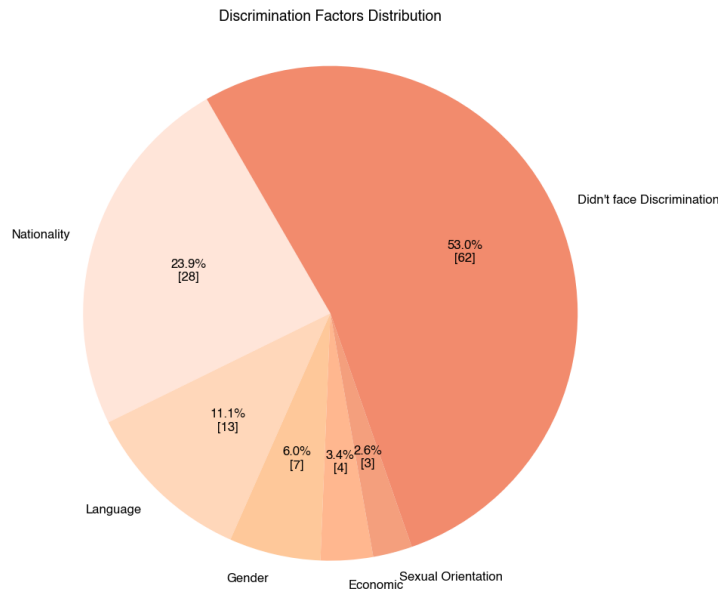


Figure 54. Discrimination Factors Distribution

Discrimination is most evident on the basis of nationality and origin, as confirmed by both quantitative data and interviews. Some respondents encountered outright refusals related to their citizenship or name: I don't get a lot of replies to be honest, I'm assuming that the reason behind it may be me being a student and my non-German name, while another participant notes: Actually, you can really see how important a German surname can sometimes be when competing for accommodation. In some cases, discrimination is explicit, such as refusing to accept students of a certain nationality: Due to today's situation, the landlord decided not take Russian students. These examples highlight the vulnerability of international students in the absence of transparent control mechanisms.

Financial discrimination is another key barrier and is closely linked to student status. Many landlords and housing companies require proof of income at a level that is unaffordable for most students: Many housing companies require proof of income and require you to make at least three times the monthly rent, which is entirely unattainable for most students. As a result, housing is effectively reserved for working professionals, while students are systematically excluded from the market: Apartments only for people with work contracts. This is particularly painful for foreign students who cannot provide German guarantors or credit history (Schufa).

A separate and particularly vulnerable group are students who face discrimination on the basis of gender, family status and queer identity. One interviewee describes being denied a rental because of their 'non-traditional' family configuration: Another agent denied our application because we are an alternative family configuration and not a straight couple. Others mention advertisements with strict restrictions: Women only, no LGBT, which directly excludes certain groups from the housing market. The inevitable consequence of such discrimination is a retreat into the grey area of renting, accompanied by risks and exploitation: I had to live in flats without contracts, hiding from the Hausverwaltung, creating a very stressful power struggle.

Taken together, this evidence shows that discrimination in the student housing sector is rarely an isolated or purely interpersonal phenomenon. Rather, it is embedded in the structure of a market where competition, scarcity and excessive requirements legitimise the exclusion of students, particularly international and socially vulnerable groups.

5.7 Key Findings

The housing crisis in Berlin has a systemic and multi-level impact on students' lives, significantly limiting their choices and undermining their sense of stability. High prices and intense competition dramatically narrow the range of available options, making even minimally acceptable housing difficult to obtain. Many students note that there are virtually no affordable options on the market that are specifically tailored to their needs: **nothing affordable for students or fewer flats, more competition.** In this situation, housing ceases to be a matter of preference and becomes a matter of survival.

As a result, students are forced to lower their expectations and settle for inadequate or temporary solutions. Some remain in dormitories with outdated facilities and a lack of privacy, as alternatives are financially unattainable: **I'm stuck with my affordable dormitory option with older amenities and a lack of privacy.** Others consider moving to neighbouring cities or the suburbs, which increases the time and cost of daily commutes: **Flats are expensive and/or in districts not suitable for my daily living.** Thus, the crisis directly affects students' spatial mobility and their daily organisation of life.

The crisis has a particularly acute impact on students' psychological and emotional state. The search for housing is described as an exhausting and demanding process that takes priority over studying and social life. Respondents emphasise that they had to spend several hours a day exclusively on searching, often without any results: **It was very stressful and time-consuming. I couldn't do much else besides search for an apartment.** For some, this situation becomes almost unbearable: **Not yet, but soon it will break me,** which indicates the risk of long-term emotional exhaustion.

Overall, the results show that the housing crisis creates an atmosphere of constant instability and forced gratitude for minimally acceptable conditions. Even those students who have managed to find accommodation realise that they were simply lucky, rather than that the system works effectively: **I was lucky, but just a couple of months later it was almost impossible to find a room.** Awareness of the shortage and lack of alternatives forces students to give up actively searching in advance and rely entirely on student dormitories as the only realistic option. This highlights the structural nature of the crisis and its long-term consequences for the student population, especially those who do not have financial support or social connections.

6 5. Data-Driven Decisions

6.1 5.1 Conclusion

I attempted to combine spatial, quantitative, and qualitative aspects to rethink the student housing crisis in Berlin and came up with the following conclusions:

1. Public student accommodations are more evenly distributed across the periphery, reflecting their role as affordable, welfare-oriented infrastructure. Private facilities are more concentrated in central areas and high-demand areas, particularly in Mitte, Friedrichshain-Kreuzberg and Charlottenburg-Wilmersdorf, reflecting market logic that prioritises profitable locations with high rents over equal access.
2. East Berlin shows a high concentration of large student complexes and compact clusters. Western Berlin, on the other hand, is characterised by a predominance of stable areas with low capacity and a shortage of supply.
3. Private universities are mainly concentrated in Mitte, Charlottenburg-Wilmersdorf and Friedrichshain-Kreuzberg – areas that are already characterised by high demand for housing – while public universities occupy both central areas and more peripheral areas such as Lichtenberg, Steglitz-Zehlendorf and Treptow-Köpenick.
4. Universities with high numbers of international students are not grouped geographically, but rather distributed in a fragmented manner and often separated from similar neighbors. The West Berlin traditionally concentrates international academic demand, while the East remains unattractive in an international context.
5. In the vast majority of university market areas, university international demand significantly exceeds available student accommodation capacity.
6. Market pressure is not generated by individual market areas, but by the overlap of multiple accessibility zones, which explains why it is the concentration of university functions — and not just the local shortage of student accommodation.
7. Berlin have pronounced polycentricity of educational space: central locations at the highest levels concentrate the greatest student demand and form the main clusters of attraction, while lower-level zones perform supporting functions. At the same time, demand significantly exceeds supply at all levels. Thus, the analysis shows that the student housing crisis in Berlin has a clearly defined spatial structure.
8. Student accommodation in Berlin serves a wider range of educational tracks. A significant proportion of users university students, including preparatory undergraduates, college students, postgraduates, exchange students, interns, young research workers and teaching staff.
9. The affordability of rent is highly rated, reflecting students' adaptation to financial constraints or forced compromises when choosing accommodation. The location and proximity to universities reflecting the spatial gap between places of study and actual areas of residence for students. In other words students priorities affordability over proximity to university.
10. The housing crisis in Berlin has a systemic and multi-level impact on students' lives, significantly limiting their choices and undermining their sense of stability. High prices and intense competition dramatically narrow the range of available options, making even minimally acceptable housing difficult to obtain. The crisis has a particularly acute impact on students' psychological and emotional state. The search for housing is described as an exhausting and demanding process that takes priority over studying and social life. The housing crisis creates an atmosphere of constant instability and forced gratitude for minimally acceptable conditions.

In a broader context, these findings imply that the student housing crisis in Berlin is not a temporary market failure or the result of individual student decisions. Rather, it is the result of institutional and political priorities that have placed social student housing secondary to the investment logic of urban development. The spatial structure of the

crisis points to a systemic mismatch between education policy, migration dynamics and urban housing planning.

The polycentric structure of Berlin's educational space, with demand concentrated in central areas and insufficient supply of affordable housing at all levels, demonstrates a lack of coordinated policy between universities, senates and housing operators. Universities continue to increase their international student enrolment without accompanying this growth with an equivalent expansion of public housing infrastructure. As a result, responsibility for student accommodation is effectively shifted to the private market, which operates on the logic of profit maximisation rather than equal access.

Particularly problematic is the fact that purpose-built student accommodation, despite being formally positioned as a solution to the crisis, reinforces social selection and spatial inequality. The concentration of private halls of residence in central and prestigious areas does not reduce overall market pressure but, on the contrary, contributes to rising rents and the displacement of socially vulnerable groups. Thus, private purpose-built student accommodation functions more as an investment asset than as an element of social housing policy, which contradicts Berlin's stated goals of providing affordable housing.

At the city policy level, this points to the need to rethink the role of student housing as critical social infrastructure rather than a niche segment of the real estate market. The research data shows that without active public sector involvement — through the expansion of state-owned student housing, regulation of private operators, and the integration of housing into educational and spatial planning — the crisis will continue and deepen. This particularly affects international students, for whom the lack of stable housing is directly linked to their legal status, access to registration and the ability to participate fully in city life.

In a political sense, the student housing crisis is becoming an indicator of a broader problem—the crisis of accessibility in the city as such. Limited choices, the normalisation of temporary and unstable forms of housing, increased psychological stress, and forced gratitude for minimally acceptable conditions all point to the gradual loss of the city as an inclusive space. This calls into question the implementation of the Right to the City principle and requires a shift from reactive measures to a long-term, structural strategy in which students are seen not as temporary users of the city, but as its rightful residents.

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6.2 5.3 Future Research

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here will be future actions and collaborations that can take place

7 6. Appendix

7.1 6.1 References

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7.2 Colour Palettes Design

Figure 7.2: Design of colour palettes. The figure shows a grid of 12 color swatches arranged in 3 rows and 4 columns. The rows represent different data types: sequential, diverging, and qualitative. The columns represent different data ranges: low to high, high to low, and mid-point. The colors are chosen to be visually distinct and easy to distinguish.

- sequential schemes - mapping orderable or numeric data that progress from low to high (or vice versa)
- diverging schemes - mapping quantitative data above and below an interesting mid-point
- qualitative schemes - mapping categorical differences in qualitative data

Figure 7.3: Comparison of ColorBrewer, MapColPal and CartoColors. The figure shows three columns of color swatches, each representing a different palette. The rows represent different data types: sequential, diverging, and qualitative. The colors are chosen to be visually distinct and easy to distinguish.

7.2.1 Colour Definitions

- sequential schemes - “Mint”, “Pi”, “Or”, “GrSc”
- diverging schemes - “MintPi”, “MintOr”, “PiOr”
- qualitative schemes - “MiPiOr”

```
color_palettes = {
    "Mint": ["#EAF4E8", "#C8E0D4", "#A9CCC1", "#8FB9AE", "#76A79D", "#5F968E", "#498581"],
    "Pi": ["#F8EDEB", "#FAE1DD", "#FCD5CE", "#FEC5BB", "#F8AD9D", "#F4978E", "#F08080"],
    "Or": ["#FFE5D9", "#FFD7BA", "#FEC89A", "#FFB78F", "#F49F7D", "#F28B6D", "#E1755CFF"],
    "GrSc": ['#EFEFEF', '#D9D9D9', '#CCCCCC', '#B7B7B7', '#999999', '#666666', '#434343'],
    "MintPi": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#FBF0EA", "#FCD5CE", "#FEC5BB", "#F8AD9D"],
    "MintOr": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#FBF0EA", "#FFD7BA", "#FEC89A", "#F8AD9D"],
    "PiOr": ["#F4978E", "#F8AD9D", "#FEC5BB", "#FCD5CE", "#FBF0EA", "#FFD7BA", "#FEC89A", "#FFB78F"],
    "MiPiOr": ['#EAF4E8', '#FAE1DD', '#C8E0D4', '#FFD7BA', '#F8EDEB', '#FCD5CE', '#FFE5D9']
}
```

```

    "MiPiOr": ['#EAF4E8', '#FEC89A', '#FAE1DD', '#C8E0D4', '#FFD7BA', '#8FB9AE', '#F8EDEB',
    '#FCD5CE', '#A9CCC1', '#FFE5D9']"#e1755cff"

    "GrSc": ["#434343", "#666666", "#999999", "#b7b7b7", "#cccccc", "#d9d9d9",
    "#efefef",

    colors = ['#FAE1DD', '#FEC89A', '#C8E0D4', '#FEC5BB', '#FFE5D9', '#EAF4E8',
    '#FBF0EA', '#FFB78F']

```



Figure XX. bla bla bla

7.2.2 Color Blindness Test

Differences in vision cover both blindness and color vision deficiency. These can be described by three concepts: accessibility, usability, and universal design. (Iwarson and Ståhl, 2003)

To ensure graphical materials are universal and enjoyable for everyone theories about color blindness are investigated alongside with accessibility and usability theories.

Color Blindes can be seen as differences in vision cover both blindness and color vision deficiency. Color perception in the human eye is build up by three different types of cones. Each type is sensitive to a certain wavelength of light (red, green, and blue) and every perceived color is therefore a mixture of stimuli of those three cone types. (<https://www.color-blindness.com/what-is-color-blindness/>)

Coblis — Color Blindness Simulator

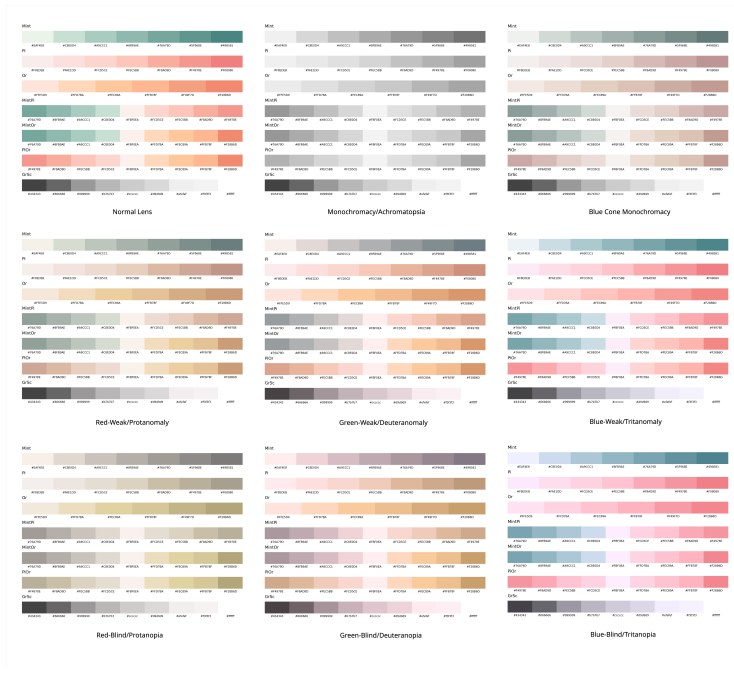


Figure XX. bla bla bla

! use Pi und Or together only with different color strenght !
! PiOr colour palete dont use !

TODO: MintPiOr check second time , do i really need it

```
color_palettes = {
    "Mint": ["#EAF4E8", "#C8E0D4", "#A9CCC1", "#8FB9AE", "#76A79D", "#5F968E", "#498581"],
    "Pi": ["#F8EDEB", "#FAE1DD", "#FCD5CE", "#FEC5BB", "#F8AD9D", "#F4978E", "#F08080"],
    "Or": ["#FFE5D9", "#FFD7BA", "#FEC89A", "#FFB78F", "#F49F7D", "#F28B6D", "#E1755CFF"],
    "GrSc": ['#EFEFEF', '#D9D9D9', '#CCCCCC', '#B7B7B7', '#999999', '#666666', '#434343'],
    "MintPi": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#F8E0E8", "#FCD5CE", "#FEC5BB", "#F8AD9D"],
    "MintOr": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#F8E0E8", "#FFD7BA", "#FEC89A", "#F8AD9D"],
    "MiPiOr": ['#EAF4E8', '#FAE1DD', '#C8E0D4', '#FFD7BA', '#F8EDEB', '#FCD5CE', '#FFE5D9']
}
```

7.2.3 Colour Palettes Export

```
from matplotlib.colors import LinearSegmentedColormap

cmap_mint = LinearSegmentedColormap.from_list("Mint", color_palettes["Mint"])
cmap_pi = LinearSegmentedColormap.from_list("Pi", color_palettes["Pi"])
cmap_or = LinearSegmentedColormap.from_list("Or", color_palettes["Or"])
cmap_mintpi = LinearSegmentedColormap.from_list("MintPi", color_palettes["MintPi"])
cmap_mintor = LinearSegmentedColormap.from_list("MintOr", color_palettes["MintOr"])
cmap_grsc = LinearSegmentedColormap.from_list("GrSc", color_palettes["GrSc"])
cmap_mipior = LinearSegmentedColormap.from_list("MiPiOr", color_palettes["MiPiOr"])
```

7.2.4 New color pallets

here will be a color pallets fro new visualisations and for apendix content

```
import matplotlib.pyplot as plt
```

```
4255 import numpy as np
4256
4257 colors = ['#FAE1DD', '#FFD7BA', '#A9CCC1', '#FEC5BB', '#FFE5D9', '#C8E0D4', '#FFB78F', '#F08080']
4258
4259 fig, ax = plt.subplots(figsize=(8, 1))
4260 ax.set_xlim(0, len(colors))
4261 ax.set_ylim(0, 1)
4262 for i, c in enumerate(colors):
4263     rect = plt.Rectangle((i, 0), 1, 1, facecolor=c)
4264     ax.add_patch(rect)
4265     ax.text(i + 0.5, 0.5, c, ha='center', va='center', fontsize=8, color='black')
4266 ax.axis('off')
4267 plt.tight_layout()
4268 plt.savefig('colors.jpg', dpi=300, bbox_inches='tight')
4269 plt.close(fig)
```

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